Fresh Start or Fresh Water: The impact of Environmental Lender Liability

by

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Abstract

I study the impact of lenders' environmental responsibility. The empirical setting exploits the U.S. Lender Liability Act of 1996, which reduced lenders' exposure to the environmental clean-up costs attached to some of their debtors' collateral, and employs difference-indifferences specifications estimated using EPA and U.S. Census microdata. Firms whose lenders face lower environmental liability risks increase pollution, reduce investment in abatement technologies by 14.7%, while experiencing small production and employment distortions. Lenders facing higher liability risks offer loans with less favorable pricing, thus financially incentivizing firms to become more environmentally responsible, and potentially monitor borrowers via shorter debt maturity.

Keyword: Environmental Lender Liability Risks; Pollution; Corporate Finance; Sustainable Finance; CERCLA; Secured Debt; Debt Structure; Environmental Regulation

JEL Classification: G21; Q58; K32; D22

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A classical view in finance is that corporations should focus solely on maximizing their shareholder returns or corporate profit (Friedman, 1970; Bénabou and Tirole, 2010). However, an alternative view argues that corporations, including lenders, should also play a role in correcting society's negative externalities (Bénabou and Tirole, 2010; Hart and Zingales, 2017). Consistent with this view, financial intermediaries are facing increasing pressure from stakeholders to play a pivotal role in making the economy more environmentally sustainable (Houston and Shan, 2022).

Yet, the nature of lenders' business raises questions about whether and how they can influence the environmental practices of their debtors in a cost-efficient way. Unlike equity holders, lenders grant loans, which, by definition, do not confer formal control rights over firms' decisions. Unlike environmental agencies, lenders may lack the expertise to effectively monitor firms' environmental behavior, potentially incurring excessive costs. If these costs are passed on through stricter credit terms, they could hinder borrowers' economic activity and employment. Thus, can lenders exert environmental influence, and if so, does it come at the cost of large distortions on firms' productive decisions?

In this paper, I provide evidence consistent with the idea that, when credibly incentivized, lenders can influence and monitor the environmental practices of their borrowers. They can do so by requiring higher interest payments and shorter debt maturities on firms that pollute. Higher interest payments create financial incentives for borrowers to reduce pollution. Shorter maturities enable lenders to renegotiate loan terms and subsequently increase the cost of capital if borrowers pollute more after receiving the loan. The lender-induced decline in pollution occurs with relatively limited economic distortions, reflected in modest reductions in firms' output and employment. These findings highlight a policy-relevant trade-off between achieving environmental improvements and preserving economic value for firms and their stakeholders.

Empirically identifying the effect of lenders' responsibility on their borrowers' environmental practices and economic activity presents a key identification problem. Both lenders and their borrowers are subject to external pressure from stakeholders to reduce pollution. As a result, observed

changes in borrower and lender behaviors may reflect a shared response to these unobserved external forces, rather than a causal influence exerted by lenders.

To mitigate this identification problem, the paper develops a strategy that isolates one specific source of external pressure on lenders: their legal exposure to the environmental cleanup costs tied to a borrower's collateral. Under the U.S. Comprehensive Environmental Response, Compensation, and Liability Act (CERCLA), lenders holding a secured lien on a contaminated facility may be held financially responsible for the costs associated with the environmental cleanup of their collateral. This legal exposure affects lenders' incentives to care for their debtors' pollution activities, because debtors that pollute are exposed to more frequent and severe environmental contamination, which leads to higher and more frequent cleanup costs.

The identification strategy relies on plausibly exogenous variation in the scope of lenders' legal exposure to the environmental cleanup costs of their debtors' collateral. In the United States, prior to 1996, lenders with a loan secured by a facility located within the jurisdiction of the U.S. Court of Appeals for the Eleventh Circuit (henceforth, the 11th Circuit) faced broader environmental liability than lenders with secured interests outside this circuit. Unlike other circuit courts, the 11th Circuit adopted a "capacity-to-control" standard in 1990, holding lenders liable for cleanup costs if they had the ability to prevent contamination. This interpretation significantly expanded lender liability risks (Madden, 1990). However, in 1996, the federal Lender Liability Act preempted the capacity-to-control doctrine by explicitly limiting lenders' liability, thereby narrowing their legal exposure and harmonizing liability standards across circuit courts. I exploit this legal shift using a difference-in-differences design that compares environmental and economic outcomes of establishments located inside the 11th Circuit (treated group) to those outside it (control group), before and after the 1996 policy change.

I estimate the difference-in-differences specifications on a unique and rich micro-administrative dataset. Specifically, I match environmental data from the U.S. Environmental Protection Agency (hereinafter, EPA) with de-identified firm-level U.S. Census micro-datasets and augment the dataset

with loan information from Dealscan. I use this granular dataset at the year-establishment-chemical-loan level to study a rich set of outcomes and refine the empirical identification by including detailed control and fixed effects.

In the first part of the paper, I document that several distinct environmental practices deteriorated in the treated group compared to the control group after 1996. First, treated firms increased overall normalized on-site pollution, with a 15% increase in the intensive margin and a 2% increase in the extensive margin of pollution. Second, they experienced a 14.7% reduction in abatement activities. Third, they were more likely to experience an unexpected environmental release by 25% of the baseline rate.

Establishments within the 11th circuit were affected by different intensities. I exploit variations in these different intensities to validate the results. First, the legal shock used in this identification strategy applies only to those firms that rely on secured debt. The results are significantly stronger for firms that use more secured debt or have most of their assets in the 11th circuit. Second, CERCLA liability does not apply to contamination caused by air pollution; consequently, I estimate an effect that is close to zero and statistically non-significant regarding air releases. Third, litigators strategically focus on solvent parties because they can pay for the liabilities or are less likely to file for bankruptcy after the judgment.[1] This implies that lenders are more exposed to environmental liability if the borrower is closer to financial distress. As expected, the effects of environmental lender liability on pollution are higher for firms that are closer to financial distress. Overall, the results are stronger for the firms and practices that are more greatly affected by the treatment. These differential effects lend additional credibility to the empirical design.

In the second part of the paper, I investigate the mechanism through which lenders can influence their debtors' environmental practices. To guide the empirical investigation, I formulate a stylized model that extends the insights of Pitchford (1995). Lowering total pollution is costly but reduces the probability of contamination. The main economic friction that causes excess pollution at the

[1] See for instance Hinerman and Taft (n.d.), CHAPTER 26

firm level is that once contamination occurs, the firm can avoid paying the full cleanup costs. Many reasons could explain such avoidance, such as the owner's lack of deep pockets, the discharge of these costs in bankruptcy, or the location of the headquarters outside of the contaminated plant's jurisdiction.

I next introduce lenders in this framework. Lenders can issue long-term and short-term debt. Rolling over debt is costly, so firms prefer to use long-term debt. Short-term debt can be renegotiated after lenders observe their debtors' environmental practices. If debtors exhibit poor environmental practices, then lenders can subsequently renegotiate with higher borrowing costs. Debtors anticipate that if they pollute, they will face higher interest payments, and therefore, in equilibrium, reduce pollution until the expected marginal reduction in borrowing costs equals the marginal abatement costs.

The model yields several key predictions. In equilibrium, when lenders are less legally responsible for cleanup costs, they issue more long-term debt, and borrowing costs become more independent of the firm's pollution practices, incentivizing firms to pollute more. Lenders implicitly monitor their debtors less when they face lower environmental liability risks.

A key prediction of the model is that firms should engage in more long-term loans if their lenders are less exposed to the environmental liability of their collateral. The alternative to a long-term debt contract is a short-term debt contract that is rolled over into a short-term debt contract. Rolling over debt is costly, so firms prefer to use long-term debt. However, the advantage of short-term debt is that new information on firms' environmental practices is factored into the borrowing costs. Firms are rewarded for their environmental efforts when they borrow short-term, and the reward is stronger when environmental lender liability is higher. This mechanism implies that the use of long-term debt over short-term debt increases when environmental lender liability is lower. I provide empirical support for this prediction. Specifically, the maturity of secured debt from firms in the treated group increases on average by 17% following the Lender Liability Act.

Firms with higher recent pollution should face lower borrowing costs when environmental liability

decreases. This is because lenders bear less risk when rolling over debt for more-polluting firms under weaker environmental lender liability rules. The data support this last prediction. Firms with on-site releases one standard deviation above the mean in 1995 experienced a 24.5% drop in their all-in-drawn spread and a 32.9% drop in their all-in-undrawn spread.

One alternative channel is that lenders rely more on secured debt instead of unsecured debt when lenders become less liable. I do not observe any economically and statistically meaningful change in the amount of secured lending following the shock. That said, I lack comprehensive measures of secured lending, which limits the strength of this conclusion. Even if some substitution occurred, I show in a model extension that as long as lenders can require more unsecured debt from borrowers than pollute more, and that unsecured debt is more costly for the borrowers, then lenders can influence debtors' practices to reduce pollution.

In the third part of the paper, I examine how reduced environmental lender liability distorts firms' real decisions. In the conceptual framework, pollution is treated as an input in a Cobb-Douglas production function, and firms operate under monopolistic competition, following the approach of Shapiro and Walker (2018). The effect of reduced liability on output and labor depends on two key parameters: the firm's market power and the output elasticity of pollution. Industrial firms in the US devote a relatively small share of total expenses to pollution abatement, suggesting that pollution, while important, is not a major input (Shapiro and Walker, 2018). The average firm is large and in industries that exhibit high entry costs, supporting the presence of significant market power.

Together, these facts predict that lower environmental liability induces only minimal changes in production, amount borrowed, and employment, but significantly benefits firms' profitability. When pollution is not a critical input, reducing its cost has a limited impact on input allocation or scale. And when firms have market power, they do not fully pass through cost savings into higher output, but experience instead improved profitability.

I validate empirically these predictions. Specifically, firms in the treated group experience a small

increase in employment of 2%. Moreover, production increases by 2.38%, and the coefficient is not statistically significant. At the same time, firms' return on assets increases significantly, supporting the view that the reduction in the implicit cost of pollution leads to higher profit margins.

This paper contributes to the literature on limited liability. It presents the first empirical evidence of lending theories on vicarious liability,[2] studying environmental safety decisions and production outcomes when a lender is liable for the actions of another party. This literature is mostly theoretical, with papers predicting either positive (Heyes, 1996; Shavell, 1997), negative (Pitchford, 1995), or ambiguous effects (Boyer and Laffont, 1997; Balkenborg, 2001) of increased liability on environmental safety.

This paper also relates to empirical studies that examine how environmental claims are treated in bankruptcy (Boomhower (2019), Wittry (2021)). Ohlrogge (2020) analyzes how the dischargeability of the Resource Conservation and Recovery Act (RCRA) cleanup claims in Chapter 11 bankruptcy affects environmental decision outcomes. The non-dischargeability of environmental cleanup claims differs from environmental lender liability. Using a different shock at a different time, Ohlrogge (2020) also finds that lenders influence debtors' environmental practices when incentivized by law, which provides external validity to this study. While Ohlrogge (2020) documents the importance of debt covenant, I study the complementary role of debt structure and pricing. Another departure is that I quantify the potential distortions on firms' production and employment decisions.

Similar to Ivanov, Kruttli, and Watugala (2023), I also document that lenders react to environmental risks by changing the debt maturity and pricing of the loan they grant to affected borrowers. While Ivanov, Kruttli, and Watugala (2023) study a change in environmental risk that affects both lenders and borrowers, I focus on a shock that plausibly only alters lenders' exposure. This approach enables the analysis of lenders' influence on borrowers' behavior, while ensuring that the

[2] Naaraayanan and Nielsen (2021) study the role of vicarious liabilities in the context of boards of directors, Akey and Appel (2021) examine how the exposure of parent companies to the environmental liabilities of their subsidiaries shapes their environmental and production outcomes. Chen (2022) studies the role of purchaser liability.

results are not confounded by borrowers' direct responses to changes in environmental risk exposure. Another departure is that I quantify the impact of this lender's influence on firms' pollution, production, and employment choices.

Finally, this paper contributes to the literature on sustainable finance. Recent papers have investigated how lenders that communicate their position as being environmentally responsible affect their debtors' environmental behaviors (Houston and Shan (2022), Kacperczyk and Peydró (2021), Green and Vallee (2023), Giannetti et al. (2023)).[3] These voluntary environmental engagements can subject lenders to a commitment problem: it is unclear whether lenders face punishment if they change their engagements over time. In contrast, this paper focuses on a setting wherein environmental lender liability commits lenders to consider their debtors' environmental behaviors. Another difference is that this paper takes a legal definition of environmental responsibility and exploits an identification strategy in which this responsibility changes for lenders. Focusing on a legal change mitigates the concern that lenders' environmental commitments and debtors' environmental outcomes are jointly affected by an omitted variable, such as future environmental regulations. Other works have studied, using international comparisons, the ways in which broad differences in legal systems affect corporate social responsibility and pollution (Liang and Renneboog (2017) and Shapiro (2022)). The results of this paper highlight the role of environmental liability rules in explaining the relationship between legal systems and corporate social responsibility.

[3]Other factors affecting corporate environmental behaviors include supply chains (Schiller (2018)), CEO preferences (Di Giuli and Kostovetsky (2014), Li, Xu, and Zhu (2021)), financial constraints (Bartram, Hou, and Kim (2022), Xu and Kim (2022), De Haas and Popov (2019), Levine et al. (2019), Bartram, Hou, and Kim (2022), and Cohn and Deryugina (2018)), and shareholders (Chu and Zhao (2019), Naaraayanan, Sachdeva, and Sharma (2019), Brandon et al. (2020), Bellon (2020), Krueger, Sautner, and Starks (2020)).

1 Data and summary statistics

1.1 Datasets

The primary sample comes from matching EPA datasets with confidential datasets from the U.S. Census Bureau from 1992 and 1999. The sample also relies on traditional financial datasets such as Dealscan and Compustat.

The EPA's Toxics Release Inventory (TRI) provides detailed data on firms' environmental practices. Originally established to support emergency planning by identifying facilities that handle hazardous chemicals, the TRI focuses on plants where contamination risks are both relevant and material. The dataset includes chemical-level measures of on-site releases to air, water, land, and underground injection, along with establishment-level identifiers that enable linkage to other administrative datasets.

All chemicals reported in the TRI dataset are hazardous but with different intensities. I use the EPA's Integrated Risk Information System to construct the variable toxic, a dummy variable taking the value one if the chemical threatens human health and zero otherwise. A component is toxic if people exposed to it consequently develop alterations in their biological system, such as the cardiovascular or dermal system, or if it causes cancer.

I augment the TRI data with information from the EPA's Pollution Prevention (P2) files to observe additional variables at the chemical level. The P2 files also contain the variable Production ratio$_{cit}$, which is the ratio of the output at time t over the output at time $t-1$ from which the chemical is used. The P2 files also report some indicators that explain whether the companies is doing some abatement effort for this chemical, and indicate accidental releases, which capture unplanned discharges due to equipment failure, spills, or other non-routine events.

I use the Longitudinal Business Dataset (LBD), a longitudinal database at the establishment level that tracks information on annual payrolls, employment, and linkages of establishments for multi-unit firms across years. It contains the population of firms with at least one employee in the United

States (Chow et al. (2021)). The LBD is built from administrative and mandatory survey data and information transmitted by the Internal Revenue Service (IRS).

Using both the LBD and the TRI offers several advantages.[4] The TRI is an establishment database. It is possible to reconstruct some linkages between establishments using the reported D&B numbers in the TRI. However, several data issues have been noticed regarding the D&B numbers (Khanna (2019)). The LBD is not exposed to these problems. The LBD records whether two different establishments belong to the same firm. Moreover, the LBD contains the precise number of employees in an establishment as reported to administrative authorities.

The TRI samples firms that are primarily industrial. The Census Bureau runs two surveys for industrial firms: the Census of Manufactures (CMF) and the Annual Survey of Manufactures (ASM). These surveys collect detailed information on establishments' quantity of inputs, costs, and the real output among the population of manufacturing firms. These surveys are mandatory and expose firms to fines if they misreport. The CMF samples all manufacturing establishments but is conducted for every year that ends in the number 2 or 7 (e.g., 1992, 1997). The ASM contains a fraction of the total manufacturing establishments for the years that do not end in 2 and 7. Specifically, all establishments with more than 250 employees are included, while the remaining ones are sampled with a probability that increases according to the number of employees. I don't use the CMF/ASM in all the tests because matching these datasets to the main sample causes a 16% drop in sample size.

I use Dealscan and Compustat. Dealscan contains debt information for companies that borrow in the syndicated market. It provides information on borrowing costs (variables All-In-Drawn and All-in-Undrawn), debt maturity, and the amount borrowed. One caveat is that Dealscan heavily favors large, syndicated loans, especially before 2000. Compustat is used to construct firm-level

[4]Researchers working with U.S. Census Bureau microdata are bound by strict confidentiality requirements under Title 13 of the U.S. Code. These rules prohibit the disclosure of any information that could directly or indirectly identify a business, establishment, or individual. As a result, all output must undergo a formal disclosure review before release, and results may be suppressed to protect confidentiality. This process can limit the level of detail reported.

controls and variables that are used in the cross-sectional tests, such as leverage. Table 2 describes how the variables are created. I use the Compustat-SSEL bridge developed by the Census Bureau to merge Compustat with the LBD.

The TRI datasets can be matched together at the establishment level using their administrative identifiers and at the chemical level using their CAS numbers. Similarly, the Census Bureau datasets can be matched using their administrative identifiers. Census Bureau provides a linkage table between Compustats and the LBD. I perform several fuzzy-matching steps to connect the environmental datasets from the EPA to the Business Register (BR) from the Census Bureau, which I describe in Section E in the online appendix. The algorithm exploits a notable advantage of the BR database: the ability to perform the match at the establishment level. Specifically, the matching uses the street name and establishment names; the physical address number; the ZIP, county, and state codes; and the two-digit NAICS industry code when this information is present. The accuracy of all final matches is manually verified.

In the online appendix section F, I explain how I constructed the database and report how I merged the datasets step-by-step and the number of observations in each step. I clean the raw files in a way that is consistent with previous works using Census Bureau data or the TRI.

Figure 2 replicates known relationships between pollution, production, and financial distress, helping validate the definitions of variable, data cleaning steps, and the LBD-TRI match. Panel A shows that establishments that are bigger, as proxied by their capital in the building and other structures used by the facility, pollute more. This result is consistent with Shapiro and Walker (2018), and more generally, with standard economic production functions that incorporate pollution (Copeland and Taylor, 2017). Panel B plots a binscatter that exhibits a negative relationship between pollution and the Altman Z-score. Firms close to bankruptcy are more likely to increase pollution, consistent with a large literature in economics and finance (Cohn and Deryugina, 2018; De Haas and Popov, 2019; Levine et al., 2019; Xu and Kim, 2022; Ohlrogge, 2023; Bellon and Boualam, 2024).

I perform two additional data consistency checks of the final link table. Participating in the TRI

implies that the firm met an employment threshold. I verify that this employment threshold is met using information from the LBD. Next, I confirm that the exit and entry patterns of facilities in the TRI are consistent with the LBD's exit and entry patterns, supporting the link table's quality.

1.2 Main variables construction

The construction of all variables used in the empirical analysis is summarized in Table 1. I next provide additional detail on selected variables to motivate their use and clarify their interpretation. This paper relies on various methods for calculating a measure of pollution intensity. The variable $\frac{\text{Pollution}_{cit}}{\text{Emp}_{it}}$ normalizes total on-site CERCLA toxic releases by the number of employees in the establishments. As shown in the section A of the Online Appendix, this measure arises naturally if we understand environmental lender liability as increasing the implicit cost of pollution for a firm with a Cobb-Douglas production function. This way of modeling pollution is standard in the economic literature (Copeland and Taylor, 2017; Shapiro and Walker, 2018). [5]

The variable $\frac{\text{Pollution}_{cit}}{\text{Emp}_{it=1994}}$ divides total on-site CERCLA toxic releases with respect to employment in 1994, thereby renormalizing the data using a scale variable that precedes the shock. This strategy ensures that variation in pollution is not confounded by changes in firm size that may be influenced by the treatment. As explained by Cohn, Liu, and Wardlaw (2022) and Chen and Roth (2024), normalizing the outcome variable using pre-determined values is an appropriate method to mitigate the influence of scale effects.

The paper also decomposes the intensive and extensive margins of pollution. The intensive margin is captured by Log(Pollution$_{cit}$), which is the logarithm of CERCLA toxic releases conditional on pollution being positive. The extensive margin is captured by $\mathbb{1}$(Pollution)$_{cit}$, which is a dummy variable that takes the value 100 if the firm reports strictly positive chemicals, and zero otherwise. Finally, the paper also uses the Log(Pollution+1$_{cit}$), primarily to compare the magnitudes with

[5] As put by Shapiro and Walker (2018) when describing the Cobb-Douglas function of pollution: "most of the relevant environmental literature has used this functional form."

other papers that have used the TRI. While many papers have raised concerns about the use of log Y+1, they are less likely to be first order in the context of the TRI. First, chemicals need to be reported in the TRI if they are released above a certain quantity.[6] As a result, the zeros are less likely to be true zeros. Second, Cohn, Liu, and Wardlaw (2022) shows that the bias of the log $Y+1$ is stronger if the conditional value of Y is low.[7] While the conditional value of Y is not observed, the unconditional expected value of Y is very large (equal to 10,500 in the baseline sample).

The variable $\mathbb{1}(\text{Accidental releases})_{cit}$ takes the value one if there is at least one accidental releases for chemical c at time t for establishment i. Accidental releases refer to unplanned chemical discharges resulting from events such as equipment failure, human error, or natural disasters, and is strongly linked to potential clean-up liabilities.

Similar to Akey and Appel (2021), I construct a variable $\mathbb{1}(\text{Process-related abatement})_{cit}$, which measures activities that modify how the product is made to reduce pollution. Examples include reusing chemicals, minimizing packaging, or altering the chemical content of products. According to the EPA's Waste Management Hierarchy (see Figure H.1 in the online appendix), this form of source reduction is the most preferred method, as it eliminates pollution before it is generated. As a robustness test, I define a variable $\mathbb{1}(\text{Abatement Activity})_{cit}$ that takes the value one if the establishment i is using at least one abatement activity on the chemical c at time t.

Finally, production is measured by the variable $\text{Production ratio}_{cit}$, which is the ratio of the output at time t over the output at time $t-1$ from which the chemical is used. The TRI production ratio is self-reported and often based on proxies or inconsistent scopes, making it a noisy measure of true output changes.

[6] The reporting thresholds are complex and depend on the chemical and industry.

[7] Indeed, Cohn, Liu, and Wardlaw (2022) shows that the semi-elastitiy λ of $1+Y$ is related to the semi-elasticity γ of Y by the following equation: $\lambda = \frac{E[Y|x]}{E[Y|x]+1} \times \gamma$, where $E[Y|x]$ is the conditional expectation of Y.

1.3 Descriptive statistics

There are 3,400 establishments from 1,200 firms between 1992 and 1999. The treated group consists of 250 establishments. Table 2 reports the descriptive statistics of the entire sample. The average facility in our sample is large: it employs 590.9 workers who are paid, on average, $37,510 per year; it invests $44,770,000 in its building and other structures, and it generates $243,700,000 of real output per year. The average firm in the sample accrues a significant amount of debt but is not close to bankruptcy, with a leverage ratio equal to 0.7439 and an average Z-score of 3.54. Moreover, the average firm has a Tobin's Q of 1.719.

Most firms use secured debt. The variable LM in Compustat reports the amount of debt-mortgages and other secured debts. This variable plausibly underestimates the true use of secured debt by firms (Rampini and Viswanathan, 2020). With this caveat, 72% of firms used secured debt, and 92% of the firms in the sample used secured debt at least once during the period 1992 through 1999.

The average pollution generated by a firm is significant. A firm generates 29,000 pounds of on-site waste annually for a given chemical. Slightly more than one-third (10,500) is regulated under CERCLA, and the rest is released through the air. Almost half of these chemicals are highly toxic. In the sample, 5.7% of facilities report investing in process-related abatement technology and 11.9% of observations in the firm-year sample have at least one environmental violation.

Table H.1 of the Internet Appendix investigates how establishments within the 11th Circuit and those outside differ according to observable characteristics before 1994 (included). Two stylized facts emerge. First, consistent with the notion that firms in the 11th Circuit face more scrutiny by lenders, they display lower levels of pollution and better environmental compliance outcomes on average. Specifically, on-site CERCLA pollution$_{cit}$ is two times greater for firms in the control group than for firms in the treated group, but the difference is not statistically significant. Firms in the treated group invest more in abatement activities (1.59 percentage points more), and the num-

ber of other liabilities is almost halved. The differences are statistically significant. Interestingly, this cross-sectional pattern holds for regulated pollution exclusively. In particular, firms display almost the same amount of air pollution, which is not regulated under CERCLA: 22,290 pounds for the treated group and 21,020 for the control group.

Second, firms in the treated and control groups show similar non-environmental outcomes before the treatment for the main variables. In particular, they have the same Tobin's Q, ROE, ROA, capital intensity, cash flow, cash holding, cost of capital, tangibility ratio, leverage, and production ratio. This finding is consistent with the notion that the cross-sectional variation in lender liability standards as adjudicated by Federal Circuit Courts is not driven by firm characteristics.

2 Institutional background

2.1 Overview of CERCLA liability

In 1980, Congress passed the Comprehensive Environmental Response, Compensation, and Liability Act (CERCLA, also known as "Superfund"). The main goals of CERCLA are to finance the cleanup of toxic waste and make the responsible party pay for these cleanup costs.[8] CERCLA provides several legal tools to achieve these goals. First, it gives the EPA the legal right to clean up contaminated sites and to compel entities responsible for that contamination to pay cleanup costs. Second, the EPA can seek voluntary settlement agreements under which the responsible party resolves the contamination. Third, CERCLA lets private parties recoup their cleanup costs from responsible entities.

Operators found responsible under CERCLA are exposed to significant cleanup costs, which usually cannot be paid in full. Remedial actions undertaken under CERCLA can total more than a

[8]The online appendix section B describes some salient facts on contaminated sites in the United States.

billion dollars.[9] Despite the high cost of remedial actions, CERCLA places no limit on the potential liability it can impose. Moreover, failure to comply with a CERCLA order exposes the responsible party to punitive damages.

CERCLA defines a potentially responsible party based on the central tenet that "polluters should pay." Consequently, the current owner of the contaminated site is a potentially responsible party. Regulators also included the facility's new owner as a potentially responsible party; otherwise, a current owner could escape CERCLA liability by transferring all its assets to a newly formed company. Lenders that write a debt contract using the facility as a secured asset, therefore, possess a form of ownership that could expose them to CERCLA liabilities.

2.2 Lender exposures to CERCLA liability

The 1980 statute of CERCLA is vague and imprecise on whether secured lenders are excluded from environmental liabilities. As stated by Madden (1990): "The statutory definition of owner or operator, however, provides courts with little guidance in determining who may be liable as an owner or operator." The statute excludes "a person, who, without participating in the management of a vessel or facility, holds indicia of ownership primarily to protect his security interest in the vessel or facility." Although secured lenders holding indicia of ownership cannot be considered as owners or operators, they are nonetheless subject to environmental liabilities if they do not participate in management activities, known as the "secured lender exemption." Courts and Congress have made several interpretations of the meaning of "participating in the management" over time.

[9]The contamination can trigger firms' liquidation for two reasons. First, remedial actions are costly. Second, a contaminated facility operates once the remedial action is completed, which reduces the firm's cash flows, thus increasing bankruptcy risks.

2.2.1 Mirabile

The first standard is consistent with an actual-control standard, as defined in *United States v. Mirabile (E.D. Pa. September 4, 1985)*.[10] Given the importance of this District Court judgment, I expose the facts, the procedural history, and the court's rationale for its decision in the online appendix Section C. The court found that secured lenders were not responsible for environmental cleanup costs as long as they did not interact with the day-to-day production aspects of a business. The day-to-day production aspects of a business include, "the participation in operational, production or waste disposal activities," and differ from the financial aspects of a business, such as providing financial advice to a company. The ability to influence and participate indirectly in the company's financial management was insufficient to find a secured lender liable.

Several courts adopted a standard close to the Mirabile case before 1996. For instance, in *Hill v. East Asiatic Co. (In re Bergsoe Metal)*, the Court of Appeals did not develop an interpretation of what constitutes "actual participation in management." However, it highlighted that, "whatever the precise parameters of 'participation,' there must be some actual management of the facility before a secured creditor will fall outside the exception," thus reiterating the interpretation of Mirabile.

2.2.2 Fleet Factors

The second interpretation defined in *United States v. Fleet Factors Corp (11th Cir. 1990)* is consistent with an ability-to-control test.[11] Given the importance of this judgment, Section D of the online appendix contains a case study of the court ruling, where I explore the facts, the procedural history, and the court's rationale for its decision. The 11th Circuit Court stipulated that any lender that has the ability to influence the management team is liable for the cleanup cost in case of environmental contamination by hazardous waste. Actual participation in the debtor's

[10]The lawsuit in *United States v. Mirabile* predates the sample period analyzed in this study and therefore lies outside the temporal scope of our empirical analysis.

[11]The lawsuit in *United States v. Fleet Factors Corp.* likewise predates the sample period analyzed in this study and therefore lies outside the temporal scope of our empirical analysis.

activity was not required for a lender to be found liable, contrary to Mirabile. Secured creditors were responsible if they "could affect hazardous waste disposal decisions if it [they] so chose." Lenders usually influence how corporations operate their facilities through covenants and on-site inspections to ensure that the value of their collateral is protected. The judgment implies that such monitoring activities would expose lenders to environmental liability, "by participating in the financial management of a facility to a degree indicating a capacity to influence the corporation's treatment of hazardous wastes." This new understanding represented a significant shift in the way lenders could be held responsible: "the court enunciated a radical new standard for determining lender liability under CERCLA" (Madden, 1990).[12] In particular, this "radical new standard" increased the likelihood for a lender to be required under CERCLA to pay for the cleanup costs of its debtors. This means that lenders with collateral within the 11th Circuit faced heightened risks of being held liable for environmental costs, primarily due to lawsuits from other potentially responsible parties or private litigants.[13]

2.2.3 Lender Liability Act

Congress uniformized and reduced the scope of CERCLA liabilities for secured lenders in the Asset Conservation Lender Liability and Deposit Insurance Protection Act (henceforth the Lender Liability Act) of 1996. The statute was passed on September 30, 1996, but was first introduced in February 1995.

The statute adopted some dimensions of the actual-control tests of Mirabile and listed some activities for which lenders were protected from liabilities. Under the Lender Liability Act, a lender becomes responsible as soon as it takes part in any day-to-day activities of a facility that are unre-

[12]Following Fleet Factors, the EPA issued a lender liability rule to limit the extent of CERCLA liability *(57 Fed.Reg. 18,344 (April 29, 1992))*. As soon as the rule was issued, states and chemical manufacturers' associations filed a petition to review it. The rule was vacated in *Kelley v. EPA (D.C. Cir.)* on the grounds that the EPA has no authority to adjudicate the extent of CERCLA liability and that this right is given only to courts.

[13]Throughout the paper, the terms "changes in environmental lender liability" and "changes in environmental lender liability risks" are used interchangeably.

lated to financial or administrative functions but that include environmental compliance activities. The Lender Liability Act also listed some activities that do not lead to CERCLA liability for lenders and that therefore constitute a "safe harbor." The presence of "safe harbors" makes the Lender Liability Act more protective of lenders than ever, even under the Mirabile standards.

In particular, the statute explicitly removed the interpretation given in the Fleet Factors decision that a secured lender can be held liable if it can influence its debtors, as "participate in management [...] (ii) does not include merely having the capacity to influence, or the unexercised right to control, vessel or facility operations."

3 Empirical design

The main empirical design exploits the succession of legal events described above and the rich panel dimension of the dataset.

According to the stare decisis principle, U.S. courts follow the precedents set by other courts. Circuit Courts of Appeals have binding authority over their respective circuits, which implies that lower courts must follow the interpretation they provide. As shown in the previous section, circuit courts adjudicated different lender liability standards before 1996, which I use to construct the treated and control groups.

In Figure 1, facilities in states not circled in red are in the control group. Before 1996, these states had either no specific liability standard or a standard close to the Mirabile interpretation and, as a result, limited lenders' exposure to the risk of being held liable for environmental cleanup costs tied to their collateral. Facilities in states circled in red are in the treated group. Before 1996, these states were subject to the ability-to-control test, which heightened the risk for lenders of incurring environmental liabilities. The treatment is the Lender Liability Act of 1996, which broadly enhanced lender protections against such risks. Although facilities in control group states were affected by the Lender Liability Act of 1996, they were affected affected to a lesser degree than

those in the treated group.[14]

The Lender Liability Act was introduced, discussed, and debated in Congress at the beginning of 1995. Firms and lenders make forward-looking decisions. This view is put forth by Ivanov, Kruttli, and Watugala (2023), who show strong anticipatory effects among lenders and debtors' debt contracts during discussion of the American Clean Energy and Security Act of 2009 (also known as the Waxman-Markey cap-and-trade bill). One might therefore expect some anticipatory effects in 1995. To be conservative, I define the treatment year as 1996. If some effects occur in 1995 instead of 1996, then defining the treatment year as 1996 will attenuate the true treatment effect. Ultimately, the extent of the anticipatory effects is an empirical question. I plot the difference-in-differences coefficients in dynamic event-study windows to show the immediacy of the effects around 1995 and 1996.

This empirical design translates into the following econometric equation, which I estimate by ordinary least square (OLS):[15]

$$\begin{aligned} Y_{cit} = {} & \textit{Chemical FE}_c \times \textit{Facility FE}_i + \textit{Chemical FE}_c \times \textit{Year FE}_t + \textit{Legal Status FE}_i \times \textit{Year FE}_t \\ & + \textit{NAICS FE}_i \times \textit{Year FE}_t + \textit{Firm-level Controls}_{it} + \textit{Post}_t \times \textit{11th Circuit}_i + \epsilon_{cit} \end{aligned} \tag{1}$$

The specifications include several fixed effects. *Chemical fixed effects* (Chemical FE$_c$) group identical chemicals using their CAS number. *Year fixed effects* (Year FE$_t$) control for shocks common

[14]The common trend assumption could be violated if a major regional macroeconomic shock affected plants located in the 11th Circuit after 1996 but not plants outside the 11th Circuit. Table H.2 in the online appendix shows that 12 variables defined at the state level do not show a statistically significant and economically meaningful difference between states within and outside the 11th Circuit after 1996. These state-level variables are the following: corporate income tax, general revenues (state), gross domestic product, personal income tax, property taxes, sales tax, state budget balance, total revenues (state), unemployment insurance, unemployment insurance base wage, unemployment insurance rate, and unemployment rate. Similarly, Table H.3 and Table H.4 in the online appendix show that EPA enforcement measures do not predict the treatment status for the 11th Circuit.

[15]Recent literature has shown econometric biases when using a difference-in-differences specification with two-way fixed effects estimators undergoing multiple treatments (De Chaisemartin and d'Haultfoeuille (2020), Borusyak and Jaravel (2017)). This problem does not arise here, because the treatment takes place for one specific group and time. Indeed, the empirical design used in this paper does not have a staggered structure, wherein different groups receive different treatments at different times.

to all observations in a given year. *Facility fixed effects* (Facility FE$_i$) and *industry fixed effects* (NAICS FE$_i$) control for time-invariant characteristics of each facility and industry. *Legal status fixed effects* (Legal status FE$_i$) group firms by legal status to account for differences in how they use chemicals over time. *NAICS-by-year fixed effects* (NAICS FE$_i$ $\times$ Year FE$_t$) control for industry-specific trends, while *chemical-by-facility fixed effects* (Chemical FE$_c$ $\times$ Facility FE$_i$) control for stable patterns in how each facility uses a specific chemical. *Legal status-by-year fixed effects* (Legal status FE$_i$ $\times$ Year FE$_t$) absorb trends that vary by legal status.

The dependent variable Y_{cit} is at the chemical c, facility i, and time t level. When the dependent variable does not vary by chemical, chemical-level controls and fixed effects are excluded.

Firm-level controls$_{it}$ include 15 time-varying controls defined at the firm level, which are commonly used in empirical corporate finance. These controls are the capital intensity, capx, cash flow, cash holding, cost of capital, firm sales, the log of firm size, R&D intensity, the return on asset (ROA), the return on equity (ROE), the tangibility ratio, the net income, Tobin's Q, total asset, and total firms' liability. The precise definition of the controls and fixed effects is explained in Table 1.[16]

The main coefficient of interest is the interaction Post$_t$ $\times$ 11th Circuit$_i$. Post$_t$ is a dummy variable that takes the value one after 1996 and zero otherwise. 11th Circuit$_i$ is a dummy variable that takes the value one for plants located in the 11th Circuit and zero otherwise. The direct inclusion of Post$_t$ and 11th circuit$_i$ are omitted because they are absorbed respectively by Year FE$_t$ and Facility FE$_i$. If the treatment is conditionally exogenous, then the interaction term Post$_t$ $\times$ 11th Circuit$_i$ measures the causal impact of the Lender Liability Act among plants in the treated group. The treatment is at the circuit level. Therefore, the standard errors are clustered at this level as in Akey and Appel (2021). As shown in the robustness section, the results remain similar with different levels of

[16]For a small fraction of observations, some variables are missing. To obtain the same number of observations when the controls are included, I input the missing observation by the firm average. If this quantity is missing, then I set the value equal to zero. I verify that the results are robust without the inclusion of these controls.

clustering and when the standard errors are computed using a bootstrapping approach or different clusterings at the firm or chemical level.

4 Baseline impact on pollution

The goal of this section is to document that lower environmental lender liability leads to a plausible causal increase in pollution among their borrowers.

4.1 Impact on pollution and Economic magnitudes

Table 3 shows how a decrease in environmental lender liability affects borrowers' pollution practices. Column (1) normalizes pollution by employment, while column (2) normalizes by the level of employment in the year before the shock. The magnitudes are statistically and economically significant. For instance, for column (2), the estimate implies an increase in pollution that is equal to an increase of 38% of the sample mean or 5.9% of the standard deviation.

However, interpreting these results solely in terms of percentage changes can be misleading, particularly when baseline pollution levels are low, as shown by Hartzmark and Shue (2022), which is the case for the treated group before the shock. Indeed, as shown in Table H.1 of the Online Appendix, prior to and including 1994, establishments exposed to high environmental lender liability risk report an average of 6,495 pounds of on-site CERCLA pollution, compared with 12,650 pounds for non-treated establishments. Small absolute changes may translate into large percentage swings that distort the perceived economic relevance.[17] To provide a more grounded interpretation of the magnitudes, I perform two complementary exercises.

First, I show that the magnitudes align with the interpretation that the pollution per employment levels in the treated group reverted to the sample average following the shock. Prior to 1994, the

[17] For example, an increase in pollution from 6 to 12 tonnes constitutes a 100% rise, while a decrease from 12 to 6 tonnes is only a 50% reduction.

pollution-to-employment ratio (using 1994 employment) for the treated group was 1.231. The same ratio for the full sample after 1996 is 1.621. Thus, the observed 0.564 increase can be interpreted as the treated group nearly reverting to the post-1996 sample mean, with a percentage error of approximately 10%.

Second, I examine whether the coefficient recovered aligns with pre-shock differences in level between treated and control groups. Table H.5 in the Online Appendix replicates the baseline regression using only pre-shock data, with the 11th Circuit dummy only and without the establishment fixed effects that are, by construction, colinear with the 11th Circuit dummy variable. The estimated difference suggests that, prior to 1994 (included), pollution in the control group was 0.259 units lower than in the treated group in level. A t-test cannot reject that this difference is statistically different from 0.564. This finding further supports the interpretation that, following the reduction in environmental lender liability, pollution in the treated group reverts to a level similar to that of the control group.

Column (3) of Panel A, Table 3 shows that firms are more likely to incur accidental releases when their lenders face less environmental liability. The increase is significant, equal to an increase of 25.8% of the baseline rate or 4.06% of the baseline standard deviation.

Columns (1) and (2) of Table 3, Panel B, decompose the overall effect into extensive and intensive margins, respectively. The results indicate that the increase in pollution reflects adjustments along both margins, reinforcing the baseline findings based on the pollution-to-employment ratio.

Finally, Column (3) examines the impact of lender liability on log(Pollution+1), yielding a coefficient of 0.132. With the caveat that we are comparing different samples with different treated firms, the economic magnitudes align with previous works that rely on the TRI. Specifically, Ohlrogge (2020) documents a coefficient of 0.13 to 0.30 for the same dependent variable when the priority of cleanup obligations increases. Among public corporations, Akey and Appel (2021) find that subsidiaries increase pollution by 22% when their parent company becomes less liable for their cleanup costs. Observing a lower point estimate than Akey and Appel (2021) makes sense, as

lenders have less control over their debtors than parent companies have over their subsidiaries.

I next estimate the dynamic event window version of Equation 1, replacing the interaction term $\text{Post}_t \times \text{11th Circuit}_i$ with a set of year-specific interactions: $\sum_{\substack{j=1992 \\ j\neq 1994}}^{1999} \gamma_j \cdot \mathbb{1}_{t=j} \times \text{11th Circuit}_i$. Here, $\mathbb{1}_{t=j}$ is an indicator variable equal to one if year $t = j$, and zero otherwise. The coefficient γ_j captures the conditional average difference in the outcome variable Y_{cit} between treated and control groups in year j, relative to the omitted base year, 1994.

Figure 3.A shows the raw average for the two groups, while Figure 3.B contains the estimated γ_j. The results suggest that some firms began adjusting their pollution behavior in anticipation of the 1996 law, with observable increases as early as 1995. Such anticipatory effects are plausible, as the Lender Liability Act was introduced and debated in Congress in early 1995.[18] Nevertheless, the bulk of the increase in pollution occurred after 1996, peaking around 1997 or 1998.

4.2 Heterogeneity according to the shock's exposure

In this subsection, I show that the effect is significantly stronger among firms with greater exposure to the shock. Exposure intensity varies across firms. To capture this heterogeneity, I estimate the following triple difference-in-differences specification, building on the baseline specification of equation (1) introduced earlier:

$$\begin{aligned} Y_{cit} = \; & \textit{Chemical FE}_c \times \textit{Facility FE}_i + \textit{Chemical FE}_c \times \textit{Year FE}_t + \textit{Legal Status FE}_i \times \textit{Year FE}_t \\ & + \textit{NAICS FE}_i \times \textit{Year FE}_t + \textit{Firm-level Controls}_{it} + \textit{Post}_t \times \textit{11th Circuit}_i \\ & + \textit{Exposure Intensity}_i \times \textit{11th Circuit}_i + \textit{Post}_t \times \textit{Exposure Intensity}_i \\ & + \textit{Post}_t \times \textit{Exposure Intensity}_i \times \textit{11th Circuit}_i + \epsilon_{cit} \end{aligned} \tag{2}$$

[18] The Lender Liability Act was introduced, discussed, and debated in Congress at the beginning of 1995. Firms and lenders are forward-looking, and may adjust behavior in response to expected policy changes. This forward-looking behavior is also documented by Ivanov, Kruttli, and Watugala (2023), who find strong anticipatory effects in lending and contract structures during the legislative process surrounding the 2009 Waxman-Markey cap-and-trade bill.

where the variable *Exposure Intensity*$_i$ equals one for firms more likely to be affected by the shock, and zero otherwise. *Exposure intensity*$_i \times$ *11th Circuit*$_i$ is collinear with the fixed effects and is thus omitted in the estimation. I also show the results when the fixed effect *Year*$_t \times$ *11th Circuit*$_i$ is included. The interaction term, *Post*$_t \times$ *Exposure Intensity*$_i \times$ *11th Circuit*$_i$ allows for a formal test of whether the effect is statistically stronger among firms with greater exposure to the shock. I use several different measures of exposure intensity. Notice that such a specification allows us to include *Circuit FE*$_i \times$ *Year FE*$_t$, which absorbs potential regional shocks taking place at the Circuit level, but then absorbs the coefficient on *Post*$_t \times$ *11th Circuit*$_i$.

First, Panel A of Table 4 shows that the impact on pollution is statistically significantly stronger for borrowers who are more financially distressed. I use two proxies for financial distress: a firm's Z-score and its leverage. Firms with high leverage, all else being equal, are more likely to file for bankruptcy if a random productivity shock hits them (Leland (1994)). Firms that are more financially distressed are more affected by changes in environmental lender liability. The reason is that lenders that have distressed borrowers are implicitly more exposed to potential environmental lender liability than lenders with deep-pocket borrowers. Indeed, litigation is costly. As a result, private parties, other potential responsible parties, or affected landowners may pursue deep-pocketed PRPs as they are more likely to obtain significant recoveries or payments.[19]

Second, Panel B of Table 4 shows that the effect is entirely driven by firms for which it is more difficult to use collateral outside the 11th Circuit. In particular, columns (1) and (3) of this table show that the effect of environmental lender liability is economically small and statistically insignificant for firms that can easily pledge assets located outside the 11th Circuit as collateral, whereas the effects for firms whose establishments are mostly located within the 11th Circuit are economically large and statistically significant.

I use two proxies to capture a firm's ability to rely on collateral outside the 11th Circuit: (i) whether the majority of its establishments (weighted by employment as a proxy for size) are located within

[19] See for instance Hinerman and Taft (n.d.), CHAPTER 26

the 11th Circuit, and (ii) the fraction of its debt that is secured. When most of a firm's assets are located within the 11th Circuit, it becomes more difficult to rely exclusively on assets as collateral that are outside the 11th Circuit. Likewise, if a large share of the firm's debt is secured, it is harder to avoid pledging assets located within the 11th Circuit.

Third, Panel C of Table 4 shows that the baseline effects are statistically stronger for firms potentially exposed to higher contamination risks or more expensive cleanup costs. Specifically, columns (1) and (2) condition on firms' chemical releases that have experienced a recent, unexpected release of toxic chemicals. Such an event is a plausible statistical signal for higher environmental risks among firms. Next, columns (3) and (4) look at the effects of chemicals that are more toxic. Sites that exploit more toxic components are thus more likely to create more critical contamination, with more severe damage to the local population and the environment. EPA responses are more robust for sites with higher environmental and public health threats. I detail the process that the EPA uses to respond to a contamination in the online appendix section G.

4.3 Are the effects driven by a change in production?

I run two tests to show that the baseline effects are not mechanically driven by an increase in production scale.

Table 5 replicates the baseline estimation, but controls for production. Specifically, column (1) replicates the baseline specification, but control for the variable production ratio$_{cit}$. Column (2) and (3) control for input variables and output variables from the CFM/ASM survey, which explain why the sample size is lower. The difference between column (2) and column (3) is that column (3) omits the employment variable, as this variable in 1994 is used to normalize pollution. Overall, controlling for production does not significantly change the magnitudes of main effect.

I next consider the impact of the shock on process-related abatement activities, which consist of modifying how the product is made to reduce pollution, similar to Akey and Appel (2021). For

instance, firms can reuse chemicals or reduce the packaging or the chemicals contained in their product. According to the EPA, as shown in the Waste Management Hierarchy (see Figure H.1 in the online appendix), this approach to reducing pollution is the most preferred one as it reduces pollution at the source.

Panel A of Figure 4 contains the normalized raw averages per year of both the treated and control groups with respect to 1994 for $\mathbb{1}(\text{Process-related abatement})_{cit}$. Panel B of Figure 4 plots the dynamic event window of Equation 1 for the same variable. Both figures show that the effect is not driven by the existence of a pre-trend and support the view that the effect took place when the law was introduced in Congress.

Panel A of Table 6 contains the regression results of Equation 1 where the dependent variable is equal to $\mathbb{1}(\text{Process-related abatement})_{cit}$. When all the controls and fixed effects are included, treated establishments reduced investment in process-related activities on average by 2.917 percentage points. This decrease consists of a reduction of 36.64%, which is economically meaningful. The coefficient is also statistically significant at the 1% level.

Panel B of Table 6 looks at a different way of constructing the abatement measure. Specifically, the variable $\mathbb{1}(\text{All abatement})_{cit}$ takes the value one if the chemical c has at least one type of abatement activity for establishment i at time t. When all the controls and fixed effects are included, treated establishments reduced abatement investment on average by 1.857 percentage points. The coefficient is also statistically significant at the 1% level. This decrease consists of a reduction of 14.69%, which is economically meaningful. Overall, the results are consistent with the view that firms reduced their environmental efforts when their secured lenders were less responsible for the environmental cleanup costs of their collateral.

4.4 Robustness tests

I test the baseline Equation 1 under two alternative control-group definitions. First, Table H.8 displays the baseline estimations that restrict controls to plants that are located in the 4th, 5th, and 6th Circuits, which are all adjacent to the 11th Circuit. Doing so ensures that both treated and control plants are more likely to share the same regional economic shocks. Second, Table H.9 replicates the baseline specifications, but instead excludes these adjacent circuits from the control groups. Doing so mitigate the role of output or labor shifting across nearby facilities. Together, these exercises show that the main findings are robust to a control group that maximizes geographic similarity or minimizes potential spillovers.

Choosing the time window of the analysis faces a trade-off: longer panels provide more observations and thus greater statistical power, but they also make it more likely that unrelated, state-level shocks differentially hit treated and control plants. As a result, Table H.10 replicates the analysis over three nested horizons: 1993-1998 (Panel A), 1994-1997 (Panel B), and 1995-1996 (Panel C). The time horizon of 1995 through 1996 is highly restrictive, as it compares pollution decisions for firms within the time the legislation was discussed in Congress (1995) and passed (1996). The results are always statistically significant.

The list of chemicals that eligible establishments must report changes with time because of evolving public needs and EPA priorities, as well as advances in innovation and information technology. Enforcement and the ability to report also change with time. For instance, the chemical component hydrogen sulfide was supposed to be added to the reporting list in 1995, but some issues were raised in an administrative stay, so the chemical was not added to the list that year. To ensure that the results are not potentially driven by this time-varying coverage in the number of chemicals, Table H.11 shows the results when all the chemicals that were never reported before 1995 are dropped. Overall, the results remain statistically and economically significant.

The environmental compliance variables used are potentially highly serially correlated. Firms'

pollution decisions depend on many factors, some of which could be fixed for several years. Even if these factors are strictly exogenous, they could create a bias in the way standard errors are computed, as shown by Bertrand, Duflo, and Mullainathan (2004). As suggested by Bertrand, Duflo, and Mullainathan (2004), one credible way to address this problem is to aggregate the data into a pre-treatment (before 1996) period and a post-treatment (after 1996) period and estimate the baseline model using this transformed dataset. Aggregating the data in such a way removes the time-series dimension, which limits the time-series correlation problem. Column (1) of Table H.12 reports the results when such aggregation is made. The economic magnitudes of this before and after comparison are slightly higher than in the baseline specification. Notably, the results remain statistically significant.

I perform different ways of clustering and computing the standard errors, beginning with a bootstrapped approach.[20] Column (2) of Table H.12 shows the standard errors and point estimate without the bootstrapped approach, while column (3) of Table H.12 reports the p-value of the significance of the coefficient of interests using the bootstrapped standard error. Finally, I cluster the standard errors at the firm and chemical levels (columns 4 and 5 of Table H.12, respectively). The significance of the results remains identical.

One potential alternative explanation is that the observed effect reflects changes in EPA enforcement activity in areas under the 11th Circuit. This alternative explanation is less plausible given that the geographic boundaries of federal appellate courts do not align with those of EPA administrative regions. That said, I perform two tests to rule out the possibility that the effect is entirely driven by changes in EPA enforcement.

First, I exploit the fact that air pollution is not regulated under CERCLA, which implies that we should expect no effect on air pollution. Panel H.6 of the online appendix shows that air pollution remains unchanged for the treated group after 1996. This result provides a natural falsification

[20] Because the many fixed effects in the specifications make this approach time-consuming, I take the first difference of the before and after sample. These transformations of the sample allow me to compute almost instantaneously an unbiased and asymptotically consistent estimator, which makes the bootstrapped approach implementable.

test, since air emissions should remain unaffected if the mechanism operates specifically through changes in CERCLA.

Second, I show that the shock is not predicted by measures of enforcement activities from the EPA. Following Trebbi and Zhang (2022), I use data from the U.S. Office of Personnel Management's (OPM) Enterprise Human Resources Integration System to construct proxies for EPA enforcement activities.[21] Specifically, I compute the log level, growth rate, three-year log differences, and log differences relative to 1995 of the total number of full-time employees and total pay at the state level. Table H.3 and H.4 of the online appendix show that these enforcement proxies do not predict treatment status, supporting the view that large shifts in EPA enforcement are not driving our results.

Finally, I investigate the sensitivity of the main estimate to different sets of controls and fixed effects. Figure H.3 and Figure H.4 report coefficients from 432 different regressions that explore different combinations of controls and fixed effects to transparently show how a specific set of controls affects the results, in a way similar to the specification curves of Cookson (2018) and Simonsohn, Simmons, and Nelson (2019).

5 Mechanism

In this section, I show a potential mechanism explaining how lenders can influence the environmental practices of their borrowers. Specifically, this mechanism is described in subsection 5.1, and its empirical predictions are then tested in subsection 5.2.

5.1 Conceptual framework and predictions

In this section, I describe a mechanism for how environmental lender liability affects borrowers' pollution decisions. The stylized model in the online appendix section A formalizes these intu-

[21] I thank Miao Ben Zhang for sharing this data with me.

itions.

An entrepreneur with monopolistic competition produces a good with two production inputs: pollution and labor. He combines the two inputs using a cobb-douglas production function, as it is standard in the economic literature (Copeland and Taylor, 2017; Shapiro and Walker, 2018).[22] Consistent with Pitchford (1995), pollution increases the risk of having an environmental contamination, which creates a cleanup liability. The key friction is that the entrepreneur is only exposed to part of this cleanup liability. Many reasons could explain such avoidance, such as the entrepreneur's lack of deep pockets, bankruptcy filings, or the location of the headquarters outside of the plant's jurisdiction. Consequently, the entrepreneur's pollution-reduction effort is below the socially optimal level.

I next introduce short-term and long-term debt into the conceptual framework. Each debt contract generates a fixed contractual fee. Long-term debt is repaid in the last period, and minimizes transaction costs. An entrepreneur who initially borrows short-term debt must roll over that debt in the interim stage. In the interim stage, the lender fully observes pollution. Lenders cannot write perfect contracts specifying to debtors which environmental practices to adopt, as in Hart and Moore (1994). As a result, pollution decision affects refinancing costs when short-term debt is rolled over. I analyze the set of optimal contracts that are incentive-compatible between the lender and the entrepreneur, where the entrepreneur has full bargaining power.

Lenders offer two types of optimal contracts when exposed to high environmental lender liability. First, a short-term debt contract that exposes the entrepreneur to higher borrowing costs during the refinancing if lenders receive information that their debtors have poor environmental practices.[23]

[22] There are three different ways of thinking about this production function. First, we can interpret the pollution choice as a byproduct of production, which has a cost imposed by regulators. Second, we could think of pollution as an input to production, which has an implicit price. Third, we could interpret pollution as an optimization under an emission constraint. A strength of this Cobb-Douglas framework is that it accommodates all these perspectives, treating them as equivalent, without requiring a commitment to any single interpretation.

[23] This implicit monitoring mechanism is consistent with a class of models in corporate finance in which lenders use short-term debt to make it more possible to punish debtors' behaviors. The punishment takes the form of higher borrowing costs, as in Haubrich (1989) and Hu, Varas, and Ying (2021).

Second, an expensive long-term contract. The long-term debt contract is expensive, because the lender knows that after the long-term contract is granted, the firm will have no obligation to reduce pollution. In the high environmental liability regime, the contract that is chosen and maximizes the borrower's profit is the short-term contract, as the abatement costs are cheaper than the equilibrium increase in financing costs.

Comparative statics when lender liability is reduced give five key predictions: (i) Pollution increases; (ii) Firms borrow more long-term; (iii) The ex ante impact on financing costs is underdetermined; (iv) Conditional on high pollution, firms face lower financing costs; (v) Conditional on high pollution, firms face longer debt maturity. These empirical predictions are discussed in great detail in Section A of the Online Appendix.

One alternative mechanism could be that lenders offer unsecured debt instead of secured debt. The model extension of Section A of the Online Appendix shows that the main prediction holds if two conditions are met. First, unsecured debt needs to be more expensive to firms than secured debt. Second, lenders can observe some signal of firms' pollution. Under these two conditions, when lenders face high environmental liability, the optimal contract is to offer secured debt if the firm pollutes less, and unsecured debt if the firm pollutes. Firms trade off the increase in financing costs coming from the use of unsecured debt with the abatement costs of reducing pollution.

In equilibrium, some firms, especially those that have low abatement costs, reduce pollution and use secured debt. Other firms, especially those with high abatement costs, still pollute but use unsecured debt.

If unsecured debt has a shorter debt maturity than secured debt, then all the previous predictions hold. The key difference in the previous mechanism is that firms will switch to secured debt when their lenders face lower environmental liability risks.

5.2 Empirical Results

The conceptual framework implies that treated loans should be defined as those secured by buildings located within the jurisdiction of the 11th Circuit Court of Appeals, or by buildings that could plausibly have been used as collateral. Unfortunately, we do not observe this latter piece of information, which poses a challenge. While there is no perfect solution, we adopt an econometric approach based on the assumption that firms whose assets are predominantly located within the 11th Circuit are more likely to use, or be able to use, collateral situated in that jurisdiction.

More specifically, for each debt facility j at time t, let $Z_{ijt} \in \{0,1\}$ denotes an indicator variable equal to one if the true collateral, or the establishment that would have been used as collateral had the loan been secured, is establishment i, and zero otherwise. The model of interest is given by:

$$\begin{aligned} Z_{ijt} \times Y_{ijt} = Z_{ijt} \times \; & \textit{Post}_t \times\times \textit{11th Circuit}_i \times\times \textit{Secured}_j + \textit{Controls}_{ijt} \\ & + \textit{Legal Status FE}_i \times \textit{Year FE}_t + \textit{NAICS FE}_i \times \textit{Year FE}_t \\ & + \textit{Circuit FE}_i \times \textit{Year FE}_t + \epsilon_{ijt} \end{aligned} \tag{3}$$

Where the variables' definition follows the same definition as equation (1). The variable $\textit{Controls}_{ijt}$ includes the same firm-level controls but also adds the other features of the debt contract and the bank's loan share. $\textit{Secured}_j$ is a dummy variable that takes the value one if the loan facility j is secured, and zero otherwise. The notation $\times\times$ denotes that all lower-order and higher-order interaction terms between $\textit{Post}_t$, $\textit{11th Circuit}_i$, and $\textit{Secured}_j$ are included in the specification, except for those collinear with the fixed effects.

The specification is thus a triple difference-in-differences. We compare the change in the outcome variable between secured and unsecured loans, both within and outside the 11th Circuit Court of Appeals, and both before and after the Lender Liability Act. This triple-difference framework allows us to isolate the causal effect of the Act on secured lending, netting out time effects, regional differences, and any baseline disparities between secured and unsecured credit. Notice that such a

specification allows us to include *Circuit FE*$_i$ $\times$ *Year FE*$_t$, which absorbs potential regional shocks taking place at the Circuit level, but then absorbs the coefficient on *Post*$_t$ $\times$ *11th Circuit*$_i$.

The variable Z_{ijt} is unobserved in our dataset. To address this, we assume that Z_{ijt} satisfies, for each firm $f$with a set of establishments Ω_f, $\mathbb{E}[Z_{ijt}] = p_{ijt} = \frac{emp_{ift}}{\underset{i \in \Omega_f}{emp_{ift}}}$. This assumption implies, in economic terms, that firm f is more likely to pledge establishment i as collateral when that establishment is larger, as proxied by its employment share within the firm. As a robustness test, we estimate the model by assuming instead that firms use an establishment as collateral with the same probability.[24]

Panels A and B of Table 7 present results that support the model's core predictions. In Panel A, we observe that the average maturity of secured debt increases by 17% relative to unsecured debt. In Panel B, the variable *Secured*$_j$ is replaced with a dummy variable, $High, Pollution_i$, which equals one if the potential collateral's pollution level in 1995 is above the sample median, and zero otherwise. The results indicate a stronger effect for long-maturity debt and a decline in firms' financing cost, measured by the All-In-Drawn and All-In-Undrawn Spreads, among firms with high initial pollution.

One potential concern is that firms may have shifted toward greater use of secured lending following the shock. Theoretically, as shown in Section A of the Online Appendix, such a mechanism could also discipline borrowers if lenders offer more expensive unsecured credit to more polluting firms. However, Columns (5) of both Panel A and Panel B show no evidence of a substitution effect toward secured debt after the shock. Specifically, affected firms are not more likely to switch to secured debt, and the estimated coefficient (0.00667) is both economically small and statistically insignificant. Similarly, when we decompose the baseline effect between high-polluting and low-polluting collateral in Panel B, the results in Column (5) reveal no statistically or economically meaningful effect.

[24]That is, $p_{ijt} = \frac{1}{card(\Omega_f)}$. The results, contained in Table H.13 of the online appendix, confirm the baseline results of Table 7.

6 Distortions on economic production and labor

Theoretically, a change in environmental lender liability translates into a lower implicit cost of using polluting inputs, as shown in Section A of the Online Appendix. When the price of an input changes for a firm with a Cobb-Douglas production technology, there is a substitution and wealth effect. The substitution effect implies that the firm uses more pollution and fewer labor, because the relative price of pollution becomes lower. The wealth effect implies that because the firm will spend a smaller amount of money on pollution (now that pollution is cheaper), the firm can use more of both labor and pollution.

The strength of the impact of lower environmental lender liability on input and production choices depends on the output elasticity of pollution and the firm's market power. In the context of US industrial firms in the 1990s, there is strong evidence that market power is high and the output elasticity of pollution is low. First, previous papers document low magnitudes of output elasticity of pollution and the presence of market power (Shapiro and Walker, 2018). Second, the firms in the sample are large, and large firms are more likely to have more market power. Moreover, the low ratio of pollution to labor suggests that pollution is either relatively expensive compared to labor and/or that its output elasticity is low.

If the output elasticity of pollution is low, then this substitution effect is limited, because the firm is more reliant on labor, and cannot pivot much toward pollution.

If the output elasticity of pollution is low and the firm's market power is high, then the wealth effect is limited. Indeed, if the output elasticity of pollution is low $(1-\alpha)$, pollution is not important in total costs. As a result, even if the unit cost goes down, the impact on total costs is limited. Moreover, firms restrict output below competitive levels to preserve markups, which limit the increase of production if firms have more monopolistic competition.

Table 8 supports these predictions, showing limited responses along the production and employment margins. The estimated effect on labor corresponds to a 2.01% increase in employment,

while the production ratio rises by 0.0178. Relative to the sample mean of 0.669, this reflects a 2.66% increase. Both estimates point to modest economic impacts on real outcomes. Although the employment effect is statistically significant and the production ratio is not, the production ratio is likely measured with greater error, which would inflate its standard errors.

While a structural estimation is outside the scope of the paper, there is a set of parameters for the model in section A of the online appendix that match the observed response on pollution, employment, and production documented in this paper. In particular, suppose that firms have an output elasticity of pollution equal to $1-\alpha = 0.05$. Then, suppose we have an elasticity of substitution equal to $\sigma = 4.76$, implying some market power. These parameters are reasonable. Shapiro and Walker (2018), using a sample from the same type of firms for the same decade, estimate a range that goes from 0.0005 to 0.0557 for $1-\alpha$, and a range for σ that goes from 2.89 to 8.18. Then, log-linearizing the model, we obtain a 2% increase in labor, a 2.6% increase in production, and a 13% increase in pollution if the implicit cost of pollution is reduced by 11%. All three magnitudes on labor, pollution, and employment are within the range of what the paper finds.

A final prediction is that profit margin should go up, which we document in panel H.14 of the online appendix. We only observe profitability at the firm level. This is problematic because with firm-level data, we cannot add the establishment-level fixed effects that improve the comparability between the treated and control groups. That said, Panel H.14 of the online appendix documents a statistically significant increase in profitability for firms that have the majority of their establishments located in the 11th Circuit Court of Appeal.

7 Conclusion

This paper examines the impact of reducing environmental lender liability risks on firms' pollution, production, labor decisions, and debt contracts. To that end, I match U.S. Census Bureau data with datasets from the EPA and debt information from Dealscan. The identification strategy

exploits the Lender Liability Act of 1996. The act reduced environmental lender liability risks at the federal level. It overruled the pre-existing legal interpretations of Circuit Courts that had shaped the intensity of ex-ante lender liability risks. This setting is well-suited for the estimation of difference-in-differences specifications.

The first part of the paper shows that good environmental practices deteriorate for facilities that experience a reduction in the environmental liability risks they pose to their lenders. Specifically, these facilities increased on-site pollution (15% on the intensive margin and 2% on the extensive margin), reduced process-related abatement by 36.6%, and were 25% more likely to experience accidental environmental releases.

The second part of the paper investigates how lender behavior influences borrowers' environmental practices. A stylized model shows that under high environmental liability risks, lenders use short-term debt to discipline firms by adjusting borrowing costs based on observed pollution, creating incentives for pollution reduction. When lender liability risks decline, the incentive to monitor decreases, firms shift toward long-term debt, and pollution levels rise. The empirical evidence supports these predictions: treated firms increase debt maturity by 16.6%, and more-polluting firms see substantial drops in borrowing costs (28% and 40% in all-in-drawn and undrawn spreads, respectively).

The third part examines how reduced environmental lender liability affects firms' real decisions. Since pollution is a relatively minor input and firms have market power, lower liability leads to limited changes in output or employment but boosts profitability. These predictions follow from the conceptual framework, where pollution is an input in a Cobb-Douglas framework with monopolistic competition. Empirical results confirm these predictions: treated firms show only a small, statistically insignificant increase in production (2.38%) and a modest 2% rise in employment, with no significant change in total lending. However, return on assets increases significantly, suggesting that lower pollution costs primarily translate into higher profit margins rather than expanded operations.

References

Akey, P., and I. Appel. 2021. The limits of limited liability: Evidence from industrial pollution. *The Journal of Finance* 76:5–55.

Balkenborg, D. 2001. How liable should a lender be? The case of judgment-proof firms and environmental risk: Comment. *American Economic Review* 91:731–8.

Bartram, S. M., K. Hou, and S. Kim. 2022. Real effects of climate policy: Financial constraints and spillovers. *Journal of Financial Economics* 143:668–96.

Bellon, A. 2020. Does private equity ownership make firms cleaner? the role of environmental liability risks. *The Role Of Environmental Liability Risks (May 18, 2020)* .

Bellon, A., and Y. Boualam. 2024. Pollution-shifting vs. downscaling: How financial distress affects the green transition. *Downscaling: How Financial Distress Affects the Green Transition (March 15, 2024)* .

Bénabou, R., and J. Tirole. 2010. Individual and corporate social responsibility. *Economica* 77:1–19.

Bertrand, M., E. Duflo, and S. Mullainathan. 2004. How much should we trust differences-in-differences estimates? *The Quarterly journal of economics* 119:249–75.

Boomhower, J. 2019. Drilling like there's no tomorrow: Bankruptcy, insurance, and environmental risk. *American Economic Review* 109:391–426.

Borusyak, K., and X. Jaravel. 2017. Revisiting event study designs. *Available at SSRN 2826228* .

Boyer, M., and J.-J. Laffont. 1997. Environmental risks and bank liability. *European Economic Review* 41:1427–59.

Brandon, R. G., S. Glossner, P. Krueger, P. Matos, and T. Steffen. 2020. Responsible institutional investing around the world .

Brehm, J., and J. T. Hamilton. 1996. Noncompliance in environmental reporting: Are violators ignorant, or evasive, of the law? *American Journal of Political Science* 444–77.

Bui, L. T., and C. J. Mayer. 2003. Regulation and capitalization of environmental amenities: evidence from the toxic release inventory in massachusetts. *Review of Economics and statistics* 85:693–708.

Chakraborty, I., I. Goldstein, and A. MacKinlay. 2018. Housing price booms and crowding-out effects in bank lending. *The Review of Financial Studies* 31:2806–53.

Chen, J. 2022. Redeploying dirty assets: The impact of environmental liability. In *Redeploying Dirty Assets: The Impact of Environmental Liability: Chen, Jason (Pang-Li)*. [Sl]: SSRN.

Chen, J., and J. Roth. 2024. Logs with zeros? some problems and solutions. *The Quarterly Journal of Economics* 139:891–936.

Chow, M. C., T. C. Fort, C. Goetz, N. Goldschlag, J. Lawrence, E. R. Perlman, M. Stinson, and T. K. White. 2021. Redesigning the longitudinal business database. Working Paper, National Bureau of Economic Research.

Chu, Y., and D. Zhao. 2019. Green hedge fund activists. *Available at SSRN 3499373* .

Cohn, J., and T. Deryugina. 2018. Firm-level financial resources and environmental spills. Working Paper, National Bureau of Economic Research.

Cohn, J. B., Z. Liu, and M. I. Wardlaw. 2022. Count (and count-like) data in finance. *Journal of Financial Economics* 146:529–51.

Cookson, J. A. 2018. When saving is gambling. *Journal of Financial Economics* 129:24–45.

Copeland, B. R., and M. S. Taylor. 2017. North-south trade and the environment. In *International trade and the environment*, 205–38. Routledge.

De Chaisemartin, C., and X. d'Haultfoeuille. 2020. Two-way fixed effects estimators with heterogeneous treatment effects. *American Economic Review* 110:2964–96.

De Haas, R., and A. A. Popov. 2019. Finance and carbon emissions .

Di Giuli, A., and L. Kostovetsky. 2014. Are red or blue companies more likely to go green? politics and corporate social responsibility. *Journal of Financial Economics* 111:158–80.

Environment America Research and Policy Center. 2021. Superfund underfunded how taxpayers have been left with a toxic financial burden .

Friedman, M. 1970. of business is to increase its profits. *New York Times Magazine, September* 13:122–6.

Giannetti, M., M. Jasova, M. Loumioti, and C. Mendicino. 2023. "glossy green" banks: The disconnect between environmental disclosures and lending activities .

Green, D., and B. Vallee. 2023. Can finance save the world? measurement and effects of bank coal exit policies. Working Paper, Working Paper, Harvard Business School.

Hart, O., and J. Moore. 1994. A theory of debt based on the inalienability of human capital. *The Quarterly Journal of Economics* 109:841–79.

Hart, O., and L. Zingales. 2017. Companies should maximize shareholder welfare not market value. *ECGI-Finance Working Paper* .

Hartzmark, S. M., and K. Shue. 2022. Counterproductive sustainable investing: The impact elasticity of brown and green firms. *Available at SSRN 4359282* .

Haubrich, J. G. 1989. Financial intermediation: Delegated monitoring and long-term relationships. *Journal of Banking & Finance* 13:9–20.

Heyes, A. G. 1996. Lender penalty for environmental damage and the equilibrium cost of capital. *Economica* 311–23.

Hinerman, P. L., and J. L. Taft. n.d. Settling cercla actions. https://www.thomasneches.com/settling-cercla-actions/. Accessed: 2025-08-08.

Houston, J. F., and H. Shan. 2022. Corporate esg profiles and banking relationships. *The Review of Financial Studies* 35:3373–417.

Hu, Y., F. Varas, and C. Ying. 2021. Debt maturity management. Working Paper, Working Paper.

Ivanov, I., M. S. Kruttli, and S. W. Watugala. 2023. Banking on carbon: Corporate lending and cap-and-trade policy. *Available at SSRN 3650447* .

Kacperczyk, M. T., and J.-L. Peydró. 2021. Carbon emissions and the bank-lending channel .

Khanna, N. 2019. *A narrative on two weaknesses of the tri for research purposes*. Department of Applied Economics and Management, Cornell University.

Kiaghadi, A., H. S. Rifai, and C. N. Dawson. 2021. The presence of superfund sites as a determinant of life expectancy in the united states. *Nature communications* 12:1–12.

Klemick, H., H. Mason, and K. Sullivan. 2020. Superfund cleanups and children's lead exposure. *Journal of environmental economics and management* 100:102289–.

Krueger, P., Z. Sautner, and L. T. Starks. 2020. The importance of climate risks for institutional investors. *The Review of Financial Studies* 33:1067–111.

Leland, H. E. 1994. Corporate debt value, bond covenants, and optimal capital structure. *The journal of finance* 49:1213–52.

Levine, R., C. Lin, Z. Wang, and W. Xie. 2019. Finance and pollution: Do credit conditions affect toxic emissions? .

Li, W., Q. Xu, and Q. Zhu. 2021. Ceo hometown favoritism in corporate environmental policies. *Available at SSRN 3859116* .

Liang, H., and L. Renneboog. 2017. On the foundations of corporate social responsibility. *The Journal of Finance* 72:853–910.

Madden, S. P. 1990. Will the cercla be unbroken–repairing the damage after fleet factors. *Fordham L. Rev.* 59:135–.

Naaraayanan, S. L., and K. M. Nielsen. 2021. Does personal liability deter individuals from serving as independent directors? *Journal of Financial Economics* 140:621–43.

Naaraayanan, S. L., K. Sachdeva, and V. Sharma. 2019. The real effects of environmental activist investing. *Available at SSRN 3483692* .

Office of Brownfields and Land Revitalization. 2015. Air and water quality impacts of brownfields redevelopment .

Ohlrogge, M. 2020. Bankruptcy claim dischargeability and public externalities: Evidence from a natural experiment. *Available at SSRN 3273486* .

———. 2023. Down the tubes: Financial distress, bankruptcy, and industrial water pollution. *Bankruptcy, and Industrial Water Pollution (August 4, 2023)* .

Paull, E. 2008. The environmental and economic impacts of brownfields redevelopment. *Northeast Midwest* .

Persico, C., D. Figlio, and J. Roth. 2020. The developmental consequences of superfund sites. *Journal of Labor Economics* 38:1055–97.

Persico, C. L., and J. Venator. 2021. The effects of local industrial pollution on students and schools. *Journal of Human Resources* 56:406–45.

Pitchford, R. 1995. How liable should a lender be? the case of judgment-proof firms and environmental risk. *The American Economic Review* 1171–86.

Rampini, A. A., and S. Viswanathan. 2020. Collateral and secured debt. *Unpublished working paper, Duke University* .

Schiller, C. 2018. Global supply-chain networks and corporate social responsibility. In *13th Annual Mid-Atlantic Research Conference in Finance (MARC) Paper*.

Shapiro, J. S. 2022. Institutions, comparative advantage, and the environment: Why are developing countries polluted? .

Shapiro, J. S., and R. Walker. 2018. Why is pollution from us manufacturing declining? the roles of environmental regulation, productivity, and trade. *American Economic Review* 108:3814–54.

Shavell, S. 1997. The optimal level of corporate liability given the limited ability of corporations to penalize their employees. *International Review of Law and Economics* 17:203–13.

Simonsohn, U., J. P. Simmons, and L. D. Nelson. 2019. Specification curve: Descriptive and inferential statistics on all reasonable specifications. *Available at SSRN 2694998* .

Steinzor, R. I., and M. Clune. 2006. *The toll of superfund neglect: Toxic waste dumps & communities at risk*. Center for Progresive Reform.

Trebbi, F., and M. B. Zhang. 2022. The cost of regulatory compliance in the united states. Working Paper, National Bureau of Economic Research.

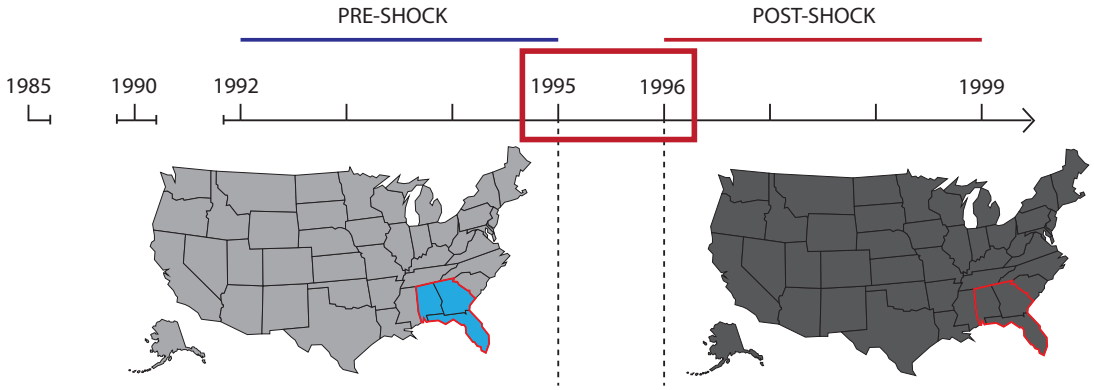
US EPA. 2016. Superfund's 25th anniversary: Capturing the past, charting the future .

Wittry, M. D. 2021. (debt) overhang: Evidence from resource extraction. *The Review of Financial Studies* 34:1699–746.

Xu, Q., and T. Kim. 2022. Financial constraints and corporate environmental policies. *The Review of Financial Studies* 35:576–635.

Figures

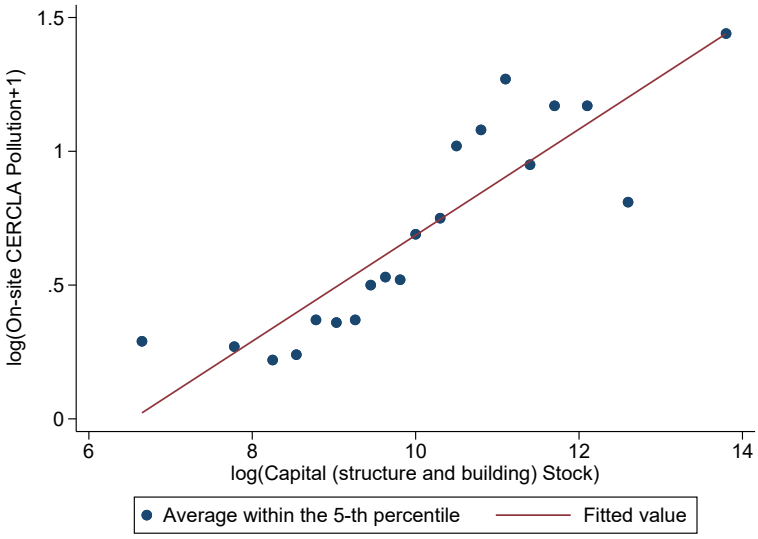
Figure 1: Treated and Control Groups



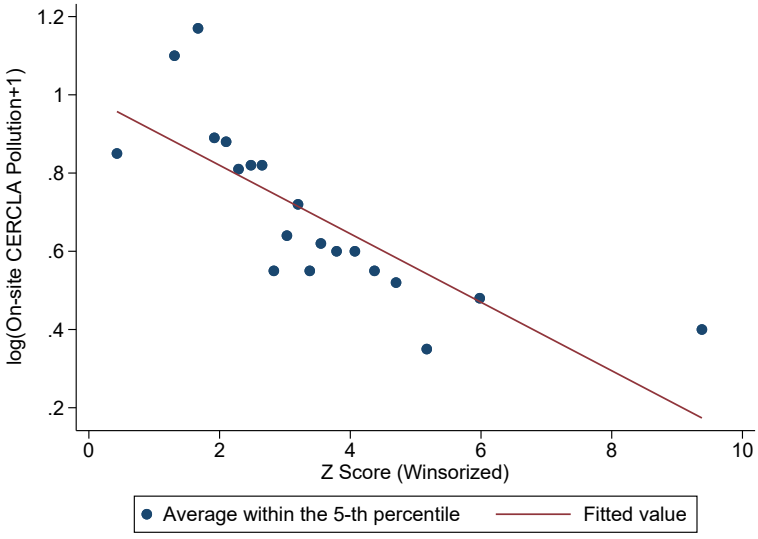
This figure plots the treated and control groups before and after the Lender Liability Act. The treated states, namely those of the 11th Circuit Court of Appeal, are circled in red. The U.S. map on the left represents different environmental lender liability standards with different colors, before the year when the Lender Liability Act was passed. Specifically, places with a high lender liability standard are colored in light blue, and areas in light gray have low or no specific lender liability standard. The U.S. map on the right represents the environmental lender liability standards after the Lender Liability Act, which became uniform across all circuit courts. The identification strategy compares plants in the 11th Circuit Court of Appeal (treated group) with those not located in the 11th Circuit Court of Appeal (control group) before and after the Lender Liability Act in a difference-in-differences framework.

Figure 2: Validation of the Dataset

Panel A: Capital Stock and Pollution



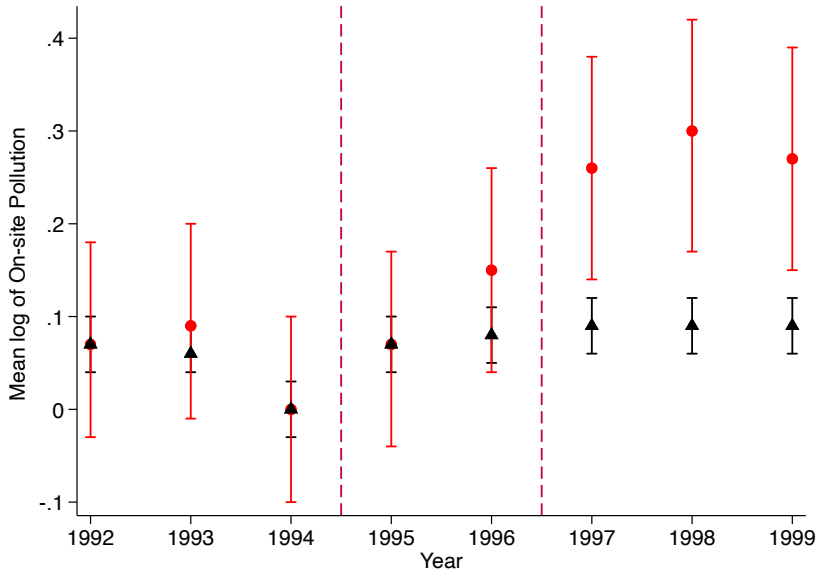
Panel B: Altman Z-score and Pollution



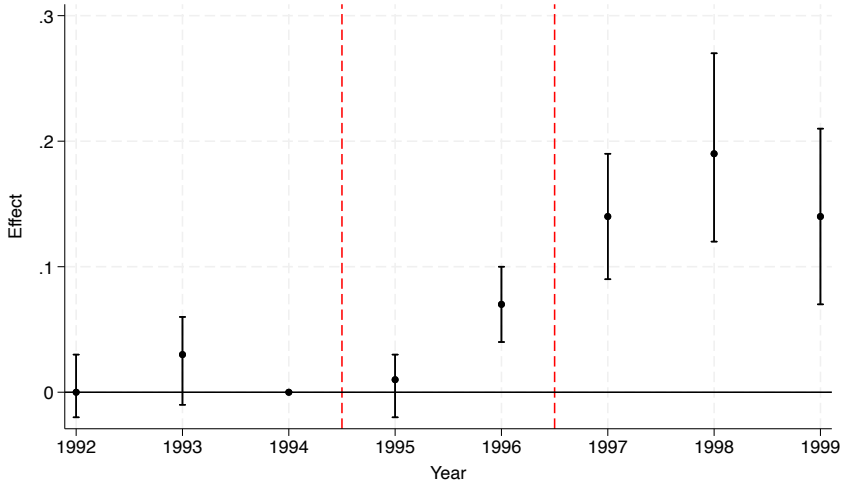
Note: The goal of these figures is to investigate the quality of the matching, by replicating the well-known relationships between firms' production size and pollution as well as the probability of bankruptcy and pollution. Specifically, Panel A uses a measure of the facility's scale, namely the capital invested in the building and other structures of the facility. For each 5th percentile of this variable, it plots the average of log(on-site CERCLA pollution+1)$_{cit}$. Similarly, there is a positive relationship between pollution and facility size. Finally, Panel B reports the relationship between the probability a firm will file for bankruptcy, as proxied by the Altman Z-score, and the variable log(on-site CERCLA pollution+1)$_{cit}$. For each 5th percentile of the Z-score, the graph plots the average of log(on-site CERCLA pollution+1)$_{cit}$. Consistent with economic theory, firms that are more likely to file for bankruptcy and thus have a lower Z-score are more likely to pollute. Estimates have been rounded to four significant digits according to the disclosure avoidance practices in place at the Census Bureau.

Figure 3: Effect on environmental practices

Panel A



Panel B

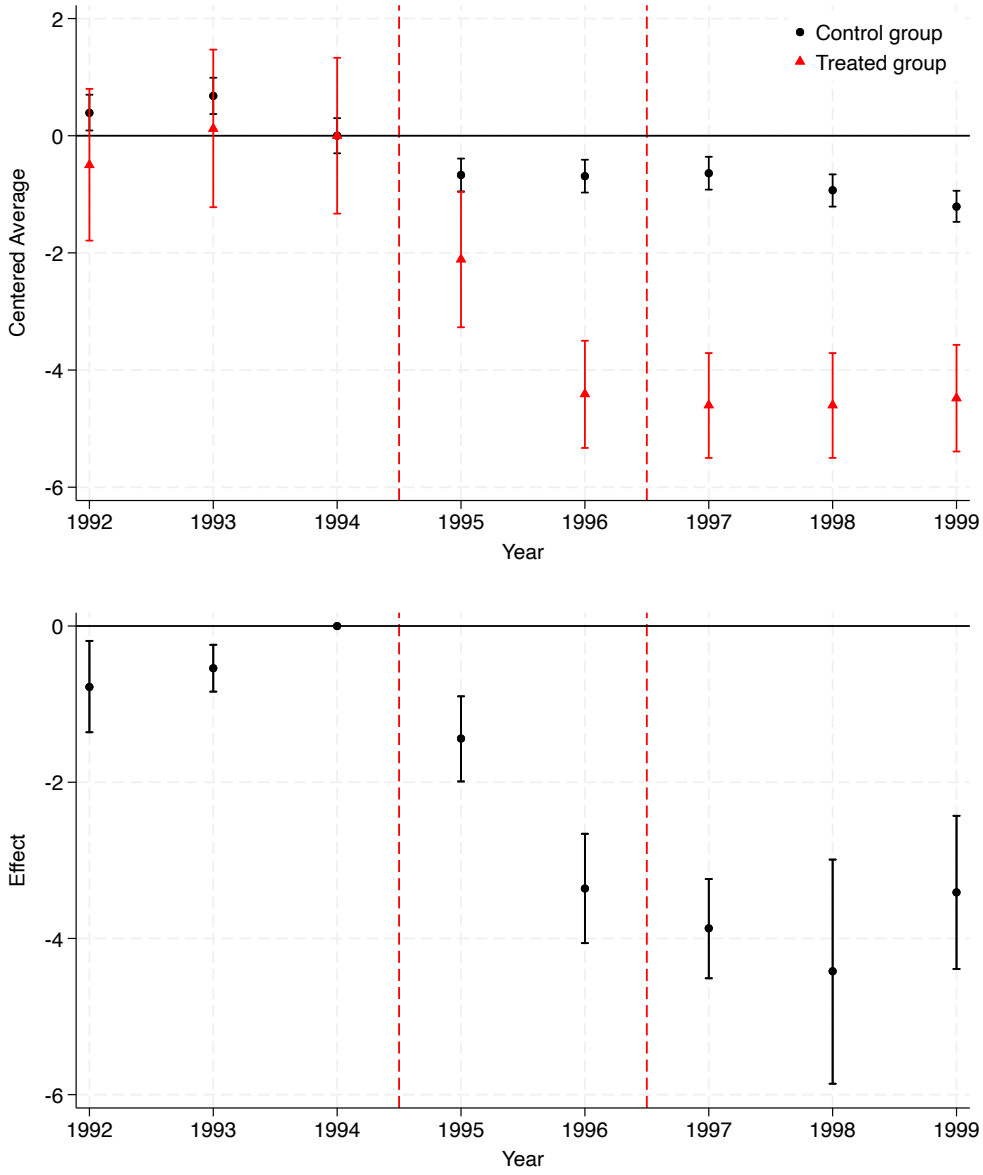


Note: All figures represent the dynamics of outcomes around the Lender Liability Act, which was introduced to Congress in 1995 and enacted in 1996. The figure of Panel A reports the sample average of log(on-site CERCLA pollution+1)$_{cit}$ per year for both the treatment and control groups. The averages are taken with respect to their level in 1994 for each year for both the treated (in red) and control groups (in black). Confidence intervals are constructed at the 5% level.
Panels B reports the coefficients of event study difference-in-differences, where the dependent variable Y_{cit} is log(on-site CERCLA pollution+1)$_{cit}$. Specifically, the estimated coefficients (γ_k) of the following equation are reported:

$$\begin{aligned} Y_{cit} = \quad & \text{Chemical FE}_c \times \text{Facility FE}_i + \text{Chemical FE}_c \times \text{Year FE}_t + \text{Legal status FE}_i \times \text{Year FE}_t \\ & + \text{NAICS FE}_i \times \text{Year FE}_t + \text{Firm-level controls}_{it} + \sum_{\substack{k=1992 \\ k \neq 1994}}^{1999} \gamma_k . \text{Year}_{tk} \times \text{11th Circuit}_i + \epsilon_{cit} \end{aligned}$$

where Year$_{tk}$ is a dummy variable that takes the value one if t is equal to k and zero otherwise. 11th Circuit$_i$ takes the value one for plants located within the 11th Circuit and zero otherwise. Chemical FE$_c$ is a chemical fixed effect defined at the CAS registry number level. Year FE$_t$ is a year fixed effect, Facility FE$_i$ is a facility fixed effect, and NAICS FE$_i$ is an industry fixed effect. The two-digit NAICS code is defined in the LBD dataset. Legal status FE$_i$ is a fixed effect that groups together firms with the same legal status as defined by the LBD. Firm-level controls$_{it}$ includes 15 time-varying controls defined at the firm level that are commonly used in empirical corporate finance. These controls are the capital intensity, capx, cash flow, cash holding, cost of capital, firm sales, the log of firm size, R&D intensity, the return on asset (ROA), the return on equity (ROE), the tangibility ratio, the net income, Tobin's Q, total asset, total firms' liability. Table 1 provides additional descriptions of how the variables are defined.

Figure 4: Effect on Abatement Investment



Both figures represent the dynamics of outcomes around the Lender Liability Act, which was introduced to Congress in 1995 and enacted in 1996. The figure of Panel A reports the sample average of $\mathbb{1}(\text{Process-related abatement})_{cit}$ per year for both the treatment and control groups. The averages are taken with respect to their level in 1994 for each year, for both the treated (in red) and control groups (in black). Confidence intervals are constructed at the 5% level. Panel B reports the coefficients of an event study difference-in-differences, where the dependent variable Y_{cit} is $\mathbb{1}(\text{Process-related abatement})_{cit}$. Specifically, the estimated coefficients (γ_k) of the following equation are reported:

$$\begin{aligned} Y_{cit} = \quad & \text{Chemical FE}_c \times \text{Facility FE}_i + \text{Chemical FE}_c \times \text{Year FE}_t + \text{Legal status FE}_i \times \text{Year FE}_t \\ & + \text{NAICS FE}_i \times \text{Year FE}_t + \text{Firm-level controls}_{it} + \sum_{\substack{k=1992 \\ k \neq 1994}}^{1999} \gamma_k . \text{Year}_{tk} \times \text{11th Circuit}_i + \epsilon_{cit} \end{aligned}$$

where Year_{tk} is a dummy variable that takes the value one if t is equal to k and zero otherwise. 11th Circuit_i takes the value one for plants located within the 11th Circuit and zero otherwise. Chemical FE_c is a chemical fixed effect defined at the CAS registry number level. Year FE_t is a year fixed effect, Facility FE_i is a facility fixed effect, and NAICS FE_i is an industry fixed effect. The two-digit NAICS code is defined in the LBD dataset. Legal status FE_i is a fixed effect that groups together firms with the same legal status as defined by the LBD. Firm-level controls_{it} includes 15 time-varying controls defined at the firm level that are commonly used in empirical corporate finance.These controls are the capital intensity, capx, cash flow, cash holding, cost of capital, firm sales, the log of firm size, R&D intensity, the return on asset (ROA), the return on equity (ROE), the tangibility ratio, the net income, Tobin's Q, total asset, total firms' liability. Table 1 provides additional descriptions of how the variables are defined. The dataset is at the chemical-year level from 1992 through 1999.

Tables

Table 1: Variables

Variable Description
pollution$_{cit}$ The on-site pollution minus air pollution that does not expose the facility to CERCLA liabilities, for facility i, in year t and for the toxic component c, in pounds. (Source: Business Register and TRI)
$\mathbb{1}$**(Pollution)**$_{cit}$ A dummy variable taking the value 100 if on-site CERCLA pollution is strictly positive and zero otherwise. (Source: Business Register and TRI)
Post$_t$ A dummy variable that takes the value 1 after 1996 and 0 otherwise. (Source: Business Register)
11th Circuit$_i$ A dummy variable that takes the value 1 for facilities in the 11th Circuit Court of Appeals and zero otherwise. (Source: Business Register)
$\mathbb{1}$**(Process-related abatement)**$_{cit}$ A dummy variable that takes the value 100 if the establishment invested in an abatement technology that changes the production process and zero otherwise. (Source: Business Register and EPA)
Air pollution$_{cit}$ Total fugitive and stack air emissions for pollutant in pounds. (Source: Business Register and EPA)
toxic$_{ict}$ A dummy variable taking the value one if the chemical poses a threat to human health and zero otherwise. A component is toxic if people exposed to it consequently develop alterations in their biological system, such as the cardiovascular or dermal system, or if it causes cancer. (Source: EPA Integrated Risk Information System)
Total releases$_{cit}$ Total releases of the chemical *c* at time *t*, for the plant *i* in pounds. (Source: Business Register and EPA)
On-site releases$_{cit}$ Total on-site releases of the chemical *c* at time *t*, for the plant *i* in pounds. (Source: Business Register and EPA)
Production ratio$_{cit}$ The ratio of the output at time t over the output at time $t-1$ from which the chemical c is used. (Source: Business Register and EPA)
log(Q)$_{it}$ The logarithm of the real output of the facility in thousand of US dollars. (Source: Business Register and CMF/ASM)
log(emp)$_{it}$ The logarithm of the number of employees. (Source: Business Register and LBD)
log(wage)$_{it}$ The total payroll amount in thousands of U.S. dollars divided by the number of employees working at the facility i at time t. (Source: Business Register and LBD)
**Other liabilities (LO, thousand, $)**$_{it}$ Variable lo in Compustat for the firm with plant i at time t. This variable captures non-financial liability, including accrual for expected future environmental costs.
Capital intensity$_{it}$ CAPEX (variable CAPX in Compustat) over Total assets (variable AT in Compustat) for the firm with plant i at time t.
Cash Flow$_{it}$ Income Before Extraordinary Items (variable IBC in Compustat) + Depreciation and Amortization over Total assets (variable AT in Compustat) for the firm with plant i at time t.
(Secured debt>0)$_{it}$ A dummy variable that takes the value one if debt-Mortgages and Other Secured (DM; Item241) is above zero and zero otherwise.
Cash Holding$_{it}$ Cash and Short-Term Investments (variable CHE in Compustat) over Total assets (variable AT in Compustat) for the firm with plant i at time t.
Cost of Capital$_{it}$ Interest and Related Expense - Total (variable XINT in Compustat) over Debt in Current Liabilities - Total (variable DLC in Compustat) for the firm with plant i at time t.
ROA$_{it}$ Net Income (variable NI in Compustat) over Total assets (variable AT in Compustat) for the firm with plant i at time t.
ROE$_{it}$ Net Income (variable NI in Compustat) over Book value of equity (Variables CSHO times PRCC_F in Compustat) for the firm with plant i at time t.
Tangibility$_{it}$ Net Total of Property, Plant, and Equipment (variable PPENT in Compustat) over Total assets (variable AT in Compustat) for the firm with plant i at time t.
Tobin's Q$_{it}$ Total assets (variable AT in Compustat) + Book value of equity (Variables CSHO times PRCC_F in Compustat) minus Total Common/Ordinary Equity (variable CEQ in Compustat) over Total assets (variable AT in Compustat) for the firm with plant i at time t.
Leverage$_{it}$ (Total Long-Term Debt (variable DLTT in Compustat) + Total Debt in Current Liabilities (variable DLC in Compustat)) over Stockholders Equity - Parent (variable SEQ in Compustat) for the firm with plant i at time t.
Emp$_{it}$ Total Employment at the establishment level for the firm with plant i at time t.
Z Score$_{it}$ Calculated using the formula in WRDS and Compustat variable as follows: (EBIT/AT)*3.3+(SALE/AT)*0.99+((CSHO*PRCC_F)/LT)*0.6+(WCAP/AT)*1.2+(RE/AT)*1.4 for the firm with plant i at time t.
Maturity (log) Log of the maturity of the loan in months. (Source: Dealscan)
Amount (log) Log of the loan amount in million $. (Source: Dealscan)
All-In-Drawn (log) Log of Loan Spread + Annual Fee. (Source: Dealscan)
All-In-Undrawn (log) Log of Commitment Fee + Annual Fee. (Source: Dealscan)

Table 2: Descriptive statistics

Variables	Mean	Standard Deviation
Total releases$_{cit}$	33,150	562,000
On-site releases$_{cit}$	29,000	547,000
$\mathbb{1}$(Pollution)$_{cit}$	12	32.5
$\frac{\text{(Pollution)}_{cit}}{\text{Emp}_{it}}$	1.338	9.147
$\mathbb{1}$(Accidental releases)$_{cit}$	2.428	15.39
$\frac{\text{Pollution}_{cit}}{\text{Emp}_{it=1994}}$	1.464	9.5
pollution$_{cit}$	10,500	389,100
Toxic$_{cit}$	0.467	0.498
$\mathbb{1}$(Process-related abatement)$_{cit}$	5.62	23
Air pollution$_{cit}$	18,500	381,100
IHS(on-site CERCLA pollution)$_{cit}$	0.7476	2.344
Production ratio$_{cit}$	0.669	0.793
Log(pollution)$_{cit}$	5.585	3.432
Log(pollution+1)$_{cit}$	0.6719	2.155
Log(air pollution+1)$_{cit}$	3.497	4.228
On-site CERCLA pollution per facility capital$_{cit}$	0.6628	63.7
$\mathbb{1}$(All abatement)$_{cit}$	12.64	33.23
Capital intensity$_{it}$	0.06388	0.03929
Cash Flow$_{it}$	0.09876	0.06952
Cash Holding$_{it}$	0.05419	0.07527
Cost of Capital$_{it}$	6.055	106.2
ROA$_{it}$	0.04629	0.09525
ROE$_{it}$	-0.5101	54.59
(Secured debt>0)$_{it}$	0.7229	0.4475
Tangibility$_{it}$	0.371	0.1585
Tobin's Q$_{it}$	1.719	0.8519
Leverage$_{it}$	0.7439	1.825
Employment$_{it}$	590.9	1374
Wage$_{it}$ (thousand, \$)	37.51	14,15
Z Score$_{it}$	3.54	2.139
Capital (structure, thousand, \$)$_{it}$	44,770	168,000
Other liabilities (LO, thousand, \$)$_{it}$	606.9	1,250
Amount (million \$)	500	877
Maturity (months)	57.78	39.7
All-In-Drawn Spread	94.12	76.8
All-In-Undrawn Spread	18.12	9.92

This table reports the descriptive statistics of the main variables used in the empirical analysis. Estimates have been rounded to 4 significant digits according to the disclosure avoidance practices in place at the Census Bureau. Table 1 provides additional descriptions of how the variables are defined.

Table 3: Net Effect on Pollution Measures

Panel A

	(1) $\frac{\text{Pollution}_{cit}}{\text{Emp}_{it}}$	(2) $\frac{\text{Pollution}_{cit}}{\text{Emp}_{it=1994}}$	(3) $\mathbb{1}(\text{Accidental releases})_{cit}$
$\text{Post}_t \times$ 11th Circuit_i	0.610***	0.564***	0.626***
	(0.0872)	(0.107)	(0.193)
Observations	210,000	210,000	210,000
R-squared	67	68	40.2
Chemical $\text{FE}_c \times$ Year FE_t	x	x	x
Facility $\text{FE}_i \times$ Chemical FE_c	x	x	x
Legal $\text{status}_i \times$ Year FE_t	x	x	x
NAICS $\text{FE}_i \times$ Year FE_t	x	x	x
Firm-level controls_{it}	x	x	x

Panel B

	(1) $\mathbb{1}(\text{Pollution})_{cit}$	(2) $\text{Log}(\text{Pollution}_{cit})$	(3) $\text{Log}(\text{Pollution+1}_{cit})$
$\text{Post}_t \times$ 11th Circuit_i	2.105***	0.154**	0.132***
	(0.279)	(0.0536)	(0.0158)
Observations	210,000	23,500	210,000
R-squared	69.9	90.6	72.5
Chemical $\text{FE}_c \times$ Year FE_t	x	x	x
Facility $\text{FE}_i \times$ Chemical FE_c	x	x	x
Legal $\text{status}_i \times$ Year FE_t	x	x	x
NAICS $\text{FE}_i \times$ Year FE_t	x	x	x
Firm-level controls_{it}	x	x	x

These tables report the difference-in-differences between the 11th Circuit (treated group) and the others (control group) after 1996, the year that the Lender Liability Act was passed. The dependent variable of column (1) Panel A is $\frac{\text{Pollution}_{cit}}{\text{Emp}_{it}}$, the total on-site CERCLA pollution for chemical c normalized by establishment i employment at time t. The dependent variable of column (2) Panel A is $\frac{\text{Pollution}_{cit}}{\text{Emp}_{it=1994}}$, the total on-site CERCLA pollution for chemical c normalized by establishment i employment in the year of 1994. The dependent variable of column (3) is $\mathbb{1}(\text{Accidental releases})_{cit}$, a dummy variable that takes the value one if the establishment experiences non-routine, unplanned, or accidental releases for chemical c. The dependent variable of column (1), Panel B is $\mathbb{1}(\text{pollution})_{cit}$, a dummy variable that takes the value 100 if the on-site CERCLA releases of chemical c, for facility i at time t, are strictly positive and zero otherwise. The dependent variable of column (2), Panel B, is $\log(\text{pollution})_{cit}$, the log of the CERCLA on-site releases for chemical c, facility i, and time t. The dependent variable of column (3), Panel B, is $\log(\text{pollution+1})_{cit}$, the log of the CERCLA on-site releases plus one for chemical c, facility i, and time t. The estimations are at the chemical-year level from 1992 through 1999. Post_t is a variable that takes the value one after 1996 (included) and zero otherwise. 11th Circuit_i takes the value one for plants located within the 11th Circuit and zero otherwise. Chemical FE_c is a chemical fixed effect defined at the CAS registry number level. Year FE_t is a year fixed effect, Facility FE_i is a facility fixed effect, and NAICS FE_i is an industry fixed effect based on the two-digit NAICS code as defined in the LBD dataset. Legal status FE_i is a fixed effect that groups together firms with the same legal status as defined by the LBD. Firm-level controls_{it} includes 15 time-varying controls defined at the firm level that are commonly used in empirical corporate finance. These controls are the capital intensity, capx, cash flow, cash holding, cost of capital, firm sales, the log of firm size, R&D intensity, the return on asset (ROA), the return on equity (ROE), the tangibility ratio, the net income, Tobin's Q, total asset, total firms' liability. Table 1 provides additional descriptions of how the variables are defined. Estimates and sample size have been rounded to four significant digits according to Census Bureau disclosure guidelines. The symbols*, **, and *** denote statistical significance at the 10%, 5%, and 1% levels, respectively.

Table 4: Cross-Sectional Tests (1/2)

Panel A

	$\frac{\text{Pollution}_{cit}}{\text{Emp}_{it=1994}}$			
	(1)	(2)	(3)	(4)
$\text{Post}_t \times$ 11th Circuit_i	0.0132	absorbed	-0.0269	absorbed
	(0.0835)		(0.0591)	
$\text{Post}_t \times$ 11th $\text{Circuit}_i \times$ High Leverage in 1995_i	0.941***	0.941***		
	(0.0310)	(0.0314)		
$\text{Post}_t \times$ 11th $\text{Circuit}_i \times$ Distress in 1995_i			1.048***	1.052***
			(0.0952)	(0.0957)
Observations	170,000	170,000	135,000	135,000
R-squared	84.1	84.1	89.5	89.5
Circuit $\text{FE}_i \times$ Year FE_t		x		x
Facility $\text{FE}_i \times$ Chemical FE_c	x	x	x	x
Legal status $\text{FE}_i \times$ Year FE_t	x	x	x	x
Chemical $\text{FE}_c \times$ Year FE_t	x	x	x	x
NAICS $\text{FE}_i \times$ Year FE_t	x	x	x	x
Firm-level controls_{it}	x	x	x	x

Panel B

	$\frac{\text{Pollution}_{cit}}{\text{Emp}_{it=1994}}$			
	(1)	(2)	(3)	(4)
$\text{Post}_t \times$ 11th Circuit_i	0.0867	absorbed	0.00887	absorbed
	(0.0866)		(0.108)	
$\text{Post}_t \times$ 11th $\text{Circuit}_i \times$ High Exposure_i	0.910***	0.914***		
	(0.0631)	(0.0597)		
$\text{Post}_t \times$ 11th $\text{Circuit}_i \times$ High fraction of secured loans_i			0.713***	0.728***
			(0.0439)	(0.0432)
Observations	210,000	210,000	175,000	175,000
R-squared	68	68.1	81.7	81.7
Circuit $\text{FE}_i \times$ Year FE_t		x		x
Facility $\text{FE}_i \times$ Chemical FE_c	x	x	x	x
Legal status $\text{FE}_i \times$ Year FE_t	x	x	x	x
Chemical $\text{FE}_c \times$ Year FE_t	x	x	x	x
NAICS $\text{FE}_i \times$ Year FE_t	x	x	x	x
Firm-level controls_{it}	x	x	x	x

Table 4: Cross-Sectional Tests (2/2)

Panel C

	$\frac{\text{Pollution}_{cit}}{\text{Emp}_{it=1994}}$			
	(1)	(2)	(3)	(4)
$\text{Post}_t \times$ 11th Circuit_i	0.560***	absorbed	0.253**	absorbed
	(0.108)		(0.0990)	
$\text{Post}_t \times$ 11th $\text{Circuit}_i \times \mathbb{1}(\text{Accidental releases in 1995})_{ci}$	0.223**	0.223**		
	(0.0843)	(0.0824)		
$\text{Post}_t \times$ 11th $\text{Circuit}_i \times \text{Toxic}_{cit}$			0.659**	0.653**
Observations	210,000	210,000	210,000	210,000
R-squared	68	68.1	68	68.1
Circuit $\text{FE}_i \times$ Year FE_t		x		x
Facility $\text{FE}_i \times$ Chemical FE_c	x	x	x	x
Legal status $\text{FE}_i \times$ Year FE_t	x	x	x	x
Chemical $\text{FE}_c \times$ Year FE_t	x	x	x	x
NAICS $\text{FE}_i \times$ Year FE_t	x	x	x	x
Firm-level controls_{it}	x	x	x	x

These tables report triple difference-in-differences between the 11th Circuit (treated group) and the others (control group) after 1996, the year that the Lender Liability Act was passed. Panel A groups firms based on their financial health, Panel B groups firms based on the exposure of their collateral to the 11th circuit, and Panel C groups firms based on the use of toxic chemicals and accidental releases. Specifically, columns (1) and (2) of Panel A, the variable High Leverage in 1995_i, is a dummy variable that takes the value one if the firm has a leverage in 1995 that is above the median sample value of the state where the establishment is located, and zero otherwise. In columns (3) and (4) of panel A, the variable Distress in 1995_i, is a dummy variable that takes the value one if the firm has a Z-score that is below the median sample value of the state where the establishment is located, and zero otherwise. In columns (1) and (2) of Panel B, the variable high Exposure_i takes the value one if the firm has a number of establishments (weighted by employment) in 1995 above the sample median, and zero otherwise. In columns (3) and (4) of Panel B, Secured Lender_i, is a dummy variable that takes the value one if the firm has an age that is above the median sample value of the state where the establishment is located, and zero otherwise. In columns (1) and (2) of Panel C, the variable toxic_c, takes the value one if the chemical is toxic according to the IRIS database and zero otherwise. Finally, in columns (3) and (4) of Panel C, the variable $\mathbb{1}(\text{Accidental releases in 1995})_{ci}$ takes the value one if the establishment experienced an accidental release of chemical c. The dataset is at the chemical-year level from 1992 through 1999. Post_t is a variable that takes the value one after 1996 (included) and zero otherwise. 11th Circuit_i takes the value one for plants located within the 11th Circuit and zero otherwise. Chemical FE_c is a chemical fixed effect defined at the CAS registry number level. Year FE_t is a year fixed effect, Facility FE_i is a facility fixed effect, and NAICS FE_i is an industry fixed effect based on the two-digit NAICS code as defined in the LBD dataset. Legal status FE_i is a fixed effect that groups together firms with the same legal status as defined by the LBD. Circuit $\text{FE}_i \times$ Year FE_t is a circuit fixed effect times a year fixed effect. Firm-level controls_{it} includes 15 time-varying controls defined at the firm level that are commonly used in empirical corporate finance. These controls are the capital intensity, capx, cash flow, cash holding, cost of capital, firm sales, the log of firm size, R&D intensity, the return on asset (ROA), the return on equity (ROE), the tangibility ratio, the net income, Tobin's Q, total asset, total firms' liability. Table 1 provides additional descriptions of how the variables are defined. Estimates and sample size have been rounded to four significant digits according to Census Bureau disclosure guidelines. The symbols*, **, and *** denote statistical significance at the 10%, 5%, and 1% levels, respectively.

Table 5: Controls for production

Panel A

	(1)	$\frac{\text{Pollution}_{cit}}{\text{Emp}_{it=1994}}$ (2)	(3)
$\text{Post}_t \times$ 11th Circuit_i	0.554***	0.598***	0.599***
	(0.105)	(0.0923)	(0.0922)
	0.682	0.661	0.661
Observations	210,000	183,000	183,000
R-squared	68.2	66.1	66.1
Production ratio_{cit}	x		
CMF/ASM Controls		x	x
Facility $\text{FE}_i \times$ Chemical FE_c	x	x	x
Legal $\text{status}_i \times$ Year FE_t	x	x	x
Chemical $\text{FE}_c \times$ Year FE_t	x	x	x
NAICS $\text{FE}_i \times$ Year FE_t	x	x	x
Firm-level controls_{it}	x	x	x

These tables report the difference-in-differences between the 11th Circuit (treated group) and the others (control group) after 1996, the year that the Lender Liability Act was passed. The dependent variable is $\frac{\text{Pollution}_{cit}}{\text{Emp}_{it=1994}}$, the total on-site CERCLA pollution for chemical c normalized by establishment i employment in the year of 1994. Column (1) reports the baseline estimates when the Production ratio_{cit} is included as a control. Columns (1) and (2) add controls from the CMF/ASM. Specifically, column (1) adds Production ratio, and column (2) includes controls from CMF/ASM. The estimations are at the chemical-year level from 1992 through 1999. Post_t is a variable that takes the value one after 1996 (included) and zero otherwise. 11th Circuit_i takes the value one for plants located within the 11th Circuit and zero otherwise. Chemical FE_c is a chemical fixed effect defined at the CAS registry number level. Year FE_t is a year fixed effect, Facility FE_i is a facility fixed effect, and NAICS FE_i is an industry fixed effect based on the two-digit NAICS code as defined in the LBD dataset. Legal status FE_i is a fixed effect that groups together firms with the same legal status as defined by the LBD. Firm-level controls_{it} includes 15 time-varying controls defined at the firm level that are commonly used in empirical corporate finance. These controls are the capital intensity, capx, cash flow, cash holding, cost of capital, firm sales, the log of firm size, R&D intensity, the return on asset (ROA), the return on equity (ROE), the tangibility ratio, the net income, Tobin's Q, total asset, total firms' liability. Table 1 provides additional descriptions of how the variables are defined. Estimates and sample size have been rounded to four significant digits according to Census Bureau disclosure guidelines. The symbols*, **, and *** denote statistical significance at the 10%, 5%, and 1% levels, respectively.

Table 6: Impact on Investment

Panel A

	$\mathbb{1}$(Process-related abatement)$_{cit}$			
	(1)	(2)	(3)	(4)
Post$_t$ × 11th Circuit$_i$	-2.935***	-3.001***	-2.999***	-2.917***
	(0.231)	(0.226)	(0.222)	(0.242)
Observations	210,000	210,000	210,000	210,000
R-squared	50.4	51.4	51.4	51.5
Facility FE$_i$× Chemical FE$_c$	x	x	x	x
Year FE$_t$	x			
Legal status$_i$× Year FE$_t$		x	x	x
Chemical FE$_c$× Year FE$_t$		x	x	x
NAICS FE$_i$× Year FE$_t$			x	x
Firm-level controls$_{it}$				x

Panel B

	$\mathbb{1}$(All abatement)$_{cit}$			
	(1)	(2)	(3)	(4)
Post$_t$ × 11th Circuit$_i$	-1.875***	-1.842***	-1.901***	-1.857***
	(0.339)	(0.326)	(0.298)	(0.317)
	0.51	0.529	0.529	0.529
Observations	210,000	210,000	210,000	210,000
R-squared	51	52.9	52.9	52.9
Facility FE$_i$× Chemical FE$_c$	x	x	x	x
Year FE$_t$	x			
Legal status$_i$× Year FE$_t$		x	x	x
Chemical FE$_c$× Year FE$_t$		x	x	x
NAICS FE$_i$× Year FE$_t$			x	x
Firm-level controls$_{it}$				x

These tables report the difference-in-differences between the 11th Circuit (treated group) and the others (control group) after 1996, the year that the Lender Liability Act was passed. The dependent variable of Panel A is $\mathbb{1}$(Process-related abatement)$_{cit}$, a dummy that takes the value 100 if the establishment invested in an abatement technology that changes the production process and zero otherwise. The dependent variable of Panel B is $\mathbb{1}$(All abatement)$_{cit}$, a dummy variable that takes the value one if the chemical c has at least one type of abatement activity at time t and for establishment i. The dataset is at the chemical-year level from 1992 through 1999. Post$_t$ is a variable that takes the value one after 1996 (included) and zero otherwise. 11th Circuit$_i$ takes the value one for plants located within the 11th Circuit and zero otherwise. Chemical FE$_c$ is a chemical fixed effect defined at the CAS registry number level. Year FE$_t$ is a year fixed effect, Facility FE$_i$ is a facility fixed effect, and NAICS FE$_i$ is an industry fixed effect based on the two-digit NAICS code as defined in the LBD dataset. Legal status FE$_i$ is a fixed effect that groups together firms with the same legal status as defined by the LBD. Firm-level controls$_{it}$ includes 15 time-varying controls defined at the firm level that are commonly used in empirical corporate finance. These controls are the capital intensity, capx, cash flow, cash holding, cost of capital, firm sales, the log of firm size, R&D intensity, the return on asset (ROA), the return on equity (ROE), the tangibility ratio, the net income, Tobin's Q, total asset, total firms' liability. Table 1 provides additional descriptions of how the variables are defined. Estimates and sample size have been rounded to four significant digits according to Census Bureau disclosure guidelines. The symbols*, **, and *** denote statistical significance at the 10%, 5%, and 1% levels, respectively.

Table 7: Impact on Loans

Panel A

	Maturity$_{itj}$ (log) (1)	All-In-Drawn Spread$_{itj}$ (log) (2)	All-In-Undrawn Spread$_{itj}$ (log) (3)	Amount$_{itj}$ (log) (4)	Secured$_{itj}$ (5)
Post$_t$ × 11th Circuit$_i$	absorbed	absorbed	absorbed	absorbed	0.00667
					(0.0117)
Post$_t$ × 11th Circuit$_i$ × Secured$_{itj}$	0.170***	-0.0357	-0.0624	0.0165	
	(0.0410)	(0.0210)	(0.0373)	(0.0573)	
Observations	10,000	10,000	10,000	10,000	10,000
R-squared	45.8	73.6	66.3	76.5	63.4
Controls$_{itj}$	x	x	x	x	x
Establishment FE$_i$	x	x	x	x	x
Legal status$_i$ × YearFE$_t$	x	x	x	x	x
NAICS FE$_i$ × Year FE$_t$	x	x	x	x	x
Circuit $_i$ × Year FE$_t$	x	x	x	x	

Panel B

	Maturity$_{itj}$ (log) (1)	All-In-Drawn Spread$_{itj}$ (log) (2)	All-In-Undrawn Spread$_{itj}$ (log) (3)	Amount$_{itj}$ (log) (4)	Secured$_{itj}$ (5)
Post$_t$ × 11th Circuit$_i$ ×High Pollution$_i$	0.155**	-0.245***	-0.329***	0.147	0.00486
	(0.0507)	(0.0668)	(0.0574)	(0.0892)	(0.0255)
Observations	10,000	10,000	10,000	10,000	10,000
R-squared	45.6	71.1	64.7	76.5	64.9
Controls$_{itj}$	x	x	x	x	x
Establishment FE$_i$	x	x	x	x	x
Legal status$_i$ × Year FE$_t$	x	x	x	x	x
NAICS FE$_i$ × Year FE$_t$	x	x	x	x	x
Circuit$_i$ × Year FE$_t$	x	x	x	x	x

This table reports the difference-in-differences between the 11th Circuit (treated group) and the others (control group) after 1996, the year that the Lender Liability Act was passed. The dependent variable of Column (1) is Maturity$_{itj}$ log, which is the log of the maturity of the loan's facility. The dependent variable of Column (2) is AllInDrawn$_{itj}$ (log), which is the log of Loan Spread + Annual Fee. The dependent variable of Column (3) is AllInUndrawn$_{itj}$ (log), which is the log of Commitment Fee + Annual Fee. The dependent variable of Column (4) is Amount$_{itj}$ (log), which is the log of the loan's amount. The dependent variable of Column (5) is Secured$_{itj}$, which is a dummy variable that takes the value one if the loan facility is secured, and zero otherwise. The dataset is at the loan facility-bank-year level from 1992 through 1999. The definition of lead arrangers follows closely Chakraborty, Goldstein, and MacKinlay (2018). $Post_t$ is a variable that takes the value one after 1996 (included) and zero otherwise. 11th Circuit$_i$ takes the value one for plants located within the 11th Circuit and zero otherwise. Secured$_{itj}$ is a dummy variable that takes the value one if the facility is secured, and 0 otherwise. We include all interaction terms among *Post$_t$*, *11th Circuit$_i$*, and *Secured$_j$* (or *High Pollution$_i$*), unless already absorbed by the fixed effects. Year FE$_t$ is a year fixed effect, Establishment FE$_i$ is an establishment fixed effect, and NAICS FE$_i$ is an industry fixed effect based on the two-digit NAICS code as defined in the LBD dataset. Legal status FE$_i$ is a fixed effect that groups together firms with the same legal status as defined by the LBD. Firm-level controls$_{it}$ includes time-varying controls defined at the firm level that are commonly used in empirical corporate finance. These controls are the capital intensity, capx, cash flow, cash holding, cost of capital, firm sales, the log of firm size, R&D intensity, the return on asset (ROA), the return on equity (ROE), the tangibility ratio, the net income, Tobin's Q, total asset, total firms' liability, other debt contracts' features, and the loan share of the bank. Table 1 provides additional descriptions of how the variables are defined. Estimates and sample size have been rounded to four significant digits according to Census Bureau disclosure guidelines. The symbols *, **, and *** denote statistical significance at the 10%, 5%, and 1% levels, respectively.

Table 8: Impact on Production and Employment

Panel A: Employment

	log(emp)$_{it}$			
	(1)	(2)	(3)	(4)
Post$_t$ × 11th Circuit$_i$	0.0187*	0.0222**	0.0226**	0.0208**
	(0.00859)	(0.00831)	(0.00898)	(0.00873)
Observations	27,000	27,000	27,000	27,000
R-squared	96	96.1	96.1	96.1
Facility FE$_i$	x	x	x	x
Year FE$_t$	x			
NAICS FE$_c$× Year FE$_t$		x	x	x
Legal status$_i$× Year FE$_t$			x	x
Firm-level controls$_{it}$				x

Panel A: Production

	Production ratio$_{cit}$			
	(1)	(2)	(3)	(4)
Post$_t$ × 11th Circuit$_i$	0.00756	0.0151	0.0152	0.0178
	(0.0107)	(0.0116)	(0.0115)	(0.0113)
Observations	210,000	210,000	210,000	210,000
R-squared	35.6	41.3	41.4	41.4
Facility FE$_i$× Chemical FE$_c$	x	x	x	x
Year FE$_t$	x			
Legal status$_i$× Year FE$_t$		x	x	x
Chemical FE$_c$× Year FE$_t$		x	x	x
NAICS FE$_i$× Year FE$_t$			x	x
Firm-level controls$_{it}$				x

These tables report the difference-in-differences between the 11th Circuit (treated group) and the others (control group) after 1996, the year that the Lender Liability Act was passed. The dataset is at the establishment-year level for Panel A and at the chemical-establishment-year level for Panel B. They both range from 1992 through 1999. The dependent variable of Panel A is log(emp)$_{it}$, which is the log of the employment for facility i at time t. The dependent variable of Panel B is the Production ratio$_{cit}$ of the component c for plant i at time t. Post$_t$ is a variable that takes the value one after 1996 (included) and zero otherwise. 11th Circuit$_i$ takes the value one for plants located within the 11th Circuit and zero otherwise. Chemical FE$_c$ is a chemical fixed effect defined at the CAS registry number level. Year FE$_t$ is a year fixed effect, Facility FE$_i$ is a facility fixed effect, and NAICS FE$_i$ is an industry fixed effect based on the two-digit NAICS code as defined in the LBD dataset. Legal status FE$_i$ is a fixed effect that groups together firms with the same legal status as defined by the LBD. Firm-level controls$_{it}$ includes 15 time-varying controls defined at the firm level that are commonly used in empirical corporate finance. These controls are the capital intensity, capx, cash flow, cash holding, cost of capital, firm sales, the log of firm size, R&D intensity, the return on asset (ROA), the return on equity (ROE), the tangibility ratio, the net income, Tobin's Q, total asset, total firms' liability. Table 1 provides additional descriptions of how the variables are defined. Estimates and sample size have been rounded to four significant digits according to Census Bureau disclosure guidelines. The symbols*, **, and *** denote statistical significance at the 10%, 5%, and 1% levels, respectively.

ONLINE APPENDIX

This Internet Appendix presents the following additional analyses, additional institutional background, and robustness tests:

- **Section A:** Describes a stylized model generating the predictions
- **Section B:** Contains some salient facts on contaminated sites in the United States
- **Section C:** Case study of *United States v. Mirabile (E.D. Pa. Sept 4, 1985)*
- **Section D:** Case study of *United States v. Fleet Factors Corp. (11th Cir. 1990)*
- **Section E:** Describes the matching approach between the Business Register and the TRI dataset
- **Section F:** Describes the data cleaning steps
- **Section G**: Describe how the EPA response to a contamination threat
- **Section H:** Provides additional tables and figures
  - **Table H.1:** Balance tests before the shock
  - **Table H.2:** Prediction of the Shock with State-Level Macroeconomic Variables
  - **Table H.3:** Prediction of the Shock with enforcement variables (1/2)
  - **Table H.4:** Prediction of the Shock with enforcement variables (2/2)
  - **Table H.5:** Simple difference in level before the shock
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  - **Table H.8:** Include adjacent Circuits as control group
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  - **Table H.10:** Different Time Ranges for the Sample: 1993-1998, 1994-1997, and 1995-1996
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  - **Figure H.2:** Types of On-site Pollution
  - **Figure H.3:** Specification Curves: Pollution Outcomes
  - **Figure H.4:** Specification Curves: Pollution Outcomes with Real Controls
  - **Figure H.5:** Active Contaminated Sites in the USA

A Conceptual framework

A.1 Assumptions

Production

Following standard approaches in the macroeconomics and trade literature, we consider a continuum of firms (or entrepreneurs) indexed by $i \in [0, 1]$, each producing an intermediate good y_i that is supplied to a competitive final goods sector producing Y. We model firm i pollution and production choices in a way that is consistent with the economic literature on this topic (Copeland and Taylor, 2017; Shapiro and Walker, 2018). Specifically, firm i produces a quantity of good equal to y_i. The firm allocates a fraction $1 - a_i$ of its labor force l_i to produce the final good, and a_i to reduce pollution. As a result, the total quantity of output is $y_i = (1 - a_i).l_i$. Pollution is exposed to implicit costs equal to $\tilde{I}_z$, such as tax, regulation, and permitting costs, and workers are paid at a wage $\tilde{w}_l$. As in Copeland and Taylor (2017) and Shapiro and Walker (2018), firm i pollution z_i is created according to the following technology:

$$z_i = (1 - a_i)^{\frac{1}{1-\alpha}}.l_i$$

Solving for $1 - a_i$ gives: $z_i^{1-\alpha} l_i^{\alpha-1} = (1 - a_i)$, so

$$y_i = (1 - a_i).l_i = z_i^{1-\alpha} l_i^{\alpha}$$

As a result, total output is given by a Cobb-Douglas production function where pollution is used as an input. As described by Copeland and Taylor (2017) and Shapiro and Walker (2018), this specification allows for a wide range of interpretation. First, we can interpret the pollution choice as a byproduct of production, which has a cost imposed by regulators. Second, we could think of pollution as an input to production, which has an implicit price. Third, we could interpret pollution as an optimization under an emission constraint. A strength of this Cobb-Douglas framework is that it accommodates all these perspectives, treating them as equivalent, without requiring a commitment to any single interpretation.

Imperfect competition

The final goods sector operates under perfect competition and aggregates intermediate goods y_i using a constant elasticity of substitution (CES) production function, each intermediate goods producer possesses some degree of market power:

$$Y = \left(\int_0^1 {q_i}^{\rho} \, di \right)^{\frac{1}{\rho}}, \quad 0 < \rho < 1$$

Define the elasticity of substitution as:

$$\sigma = \frac{1}{1-\rho} > 1$$

Each intermediate good i is sold at price p_i. The price of the final good is normalized to 1. Solving the cost minimization problem of the final goods sector yields the following demand for intermediate goods:

$$y_i = (\tfrac{p_i}{P})^{-\sigma}.Y \text{ with } P = \left[\int_0^1 {p_i}^{1-\sigma}\, di\right]^{\frac{1}{1-\sigma}}$$

Since the final good sells at price 1 in equilibrium, $P = 1$, so we have $y_i = (\frac{p_i}{P})^{-\sigma}.Y = (p_i)^{-\sigma}.Y$. As a result, the price at which the intermediate input i is sold becomes: $p_i = (\frac{y_i}{Y})^{-\frac{1}{\sigma}}$.

Liability and lending

As in Pitchford (1995), the benefit of polluting z_i is that it increases the probability of having a major toxic release defined by $z_i\beta$, where β ensures that the quantity $z_i\beta$ is between 0 and 1. Investing in z_i projects means that the firms face an expected liability cost from a contamination that is equal to $(z_i\beta).C_E$.

Without loss of generality, and for simplicity and tractability, we assume that C_E is independent of the entrepreneur's wealth and profit. Let C_{ES} denote the socially optimal level of liability such that $C_E < C_{ES}$. Many reasons could explain that $C_E < C_{ES}$. For example, the entrepreneur may lack deep pockets, discharge cleanup costs through bankruptcy, or reside outside the plant's jurisdiction; courts themselves may also be imperfect or subject to bias.

Finally, the lender faces a potential expected liability of C_L, which reflects the product of the probability of being held liable and the magnitude of the resulting damages. This expectation arises because the lender's liability depends on uncertain legal outcomes, specifically, whether a court holds the lender responsible and whether a private or third party brings a claim. The empirical analysis in this paper exploits variation in these probabilities. Without loss of generality, we assume that there is no one-to-one relationship between C_E and C_L. Implicitly, the idea is that there are other stakeholders, such as taxpayers and local governments, who can partly pay for the damages if what the firm or the lender ends up paying is too low. Finally, the entrepreneur's outside option is normalized to 0.

Without loss of generality, the entrepreneur has an initial wealth of 0. The entrepreneur borrows $D = I$, where I is the cost of all inputs. The entrepreneur receives an interim cash flow of f_b at time $t = 1$. There are two types of debt contracts: short-term and long-term debt contracts.

Debt can have a long-run maturity, and in this case, the interest payment will take place at time $t = 2$. Debt can also have a short maturity, and an interest payment will be paid for the next period. Debt can be rolled over without constraints. Issuing debt costs an underwriting fee of $f < f_b$.

We assume that the cost of debt from period $t = 0$ to period $t = 1$ is zero, then the cost of debt from period $t = 1$ to period $t = 2$ faces interest payments that can be decomposed into two parts. The first part, r_e, depends on the firm's fundamentals and is paid as a fraction of the amount borrowed. Because the firm borrows a quantity equal to all its inputs, we can factor this cost of debt into the input costs as follows: $w_l = r_e + \tilde{w}_l$ and $I_z = r_e + \tilde{I}_z$. The second part depends only on the firm's pollution practices and is independent of the amount borrowed. This reflects the joint and several nature of environmental liability, whereby the lender can be held responsible for the full cost of cleaning up a contaminated site, regardless of the size of their loan or their specific contribution to the contamination.

Lenders have no bargaining power, and their outside option is normalized to $r_e.D = (l_i + z_i).r_e$, which is the cost of lending to a comparable firm that does not expose the lender to environmental lender liability risk.

The timing is as follows. At time $t = 0$, the entrepreneur invests in the project by maximizing the expected profit and by choosing the optimal level of pollution and labor inputs. Inputs are paid at this stage. At time $t = 1$, the fraction of low-pollution projects is perfectly observed by the bank and is irreversible. At time $t = 1$, the entrepreneur rolls over the new debt if short-term debt is used. The entrepreneur receives an interim cash flow of $f < f_b$ at time $t = 1$. At time $t = 2$, projects that are invested generate their cash flow, the debt and interest rates are paid back, and if a major environmental contamination happens, then the firm and bank face a cleanup liability. Both discount rates are normalized to 1.

A.2 Optimal abatement effort when there is no debt

PROPOSITION 1: *The optimal pollution z_i^* decreases in the amount of liability the entrepreneur faces. As a result, if the entrepreneur faces an amount of environmental liabilities that are below the social level ($C_E < C_{ES}$), the entrepreneur will overpollute compared to the social optimum.*

Proof and intuition: The expected profit of the entrepreneur at time $t = 0$ is as follows:

$$p_i.y_i - z_i\beta.C_E - z_i\tilde{I}_z - l_i\tilde{w}_l$$

As we detail in Subsection A.8, we can solve for the optimal input demand of firm i as follows:

$$\frac{z_i}{l_i} = \frac{\tilde{w}_l}{\tilde{I}_z + \beta_z C_E} . \frac{(1-\alpha)}{\alpha}$$

The pollution intensity is decreasing with the firm's exposure to environmental liability. The same happens with total pollution, which decreases in the amount of environmental liability:

$$z_i^* = \xi_z \cdot \tilde{w}_l^{\alpha(1-\sigma)} \cdot (\tilde{I}_z + \beta C_E)^{-\sigma+\alpha.\sigma-\alpha} \text{ where } \xi_z = \frac{(1-\alpha)}{\alpha} . \quad \frac{(\frac{(1-\alpha)}{\alpha})^{\frac{1-\alpha}{\sigma}+\alpha}}{Y^{\frac{1}{\sigma}}(1-\frac{1}{\sigma})(1-\alpha)} \quad ^{-\sigma}$$

In particular, if the firm faces an amount of liability that is below the social optimum, then the firm pollutes more than in the social optimum.

A.3 Introducing debt contracts and lender liability

Long-term debt

PROPOSITION 2:*The optimal entrepreneur's abatement effort if the entrepreneur uses long-term debt is* $z_{i,LT}^* = \xi_z \cdot w_l^{\alpha(1-\sigma)} \cdot (I_z + \beta C_E)^{-\sigma+\alpha.\sigma-\alpha}$*, and its borrowing costs are* $r_{LT} = f + \beta . z_{i,LT}^* . C_L$

Proof and intuition: We solve the problem when the debtor uses long-term debt. The expected profit $\mathbb{E}_0(\Pi_E)$ at time 0 of the entrepreneur becomes, as $w_l = (1+r_e).\tilde{w}_l$ and $I_z = (1+r_e).\tilde{I}_z$:[25]

$$\mathbb{E}_0(\Pi_E) = p_i.y_i - z_i\beta.C_E - z_i\tilde{I}_z - l_i\tilde{w}_l - (l_i\tilde{w}_l + z_i\tilde{I}_z).r_e - r_{LT} + f_b,$$

$$\mathbb{E}_0(\Pi_E) = p_i.y_i - z_i\beta.C_E - z_iI_z - l_iw_l - r_{LT},$$

$$\text{s.t. } \mathbb{E}_0(\Pi_E) = p_i.y_i - z_i\beta.C_E - z_iI_z - l_iw_l - r_{LT} \geq 0.$$

Therefore, given a debt contract (r, D) the entrepreneur optimal pollution becomes:

$$z_i^* = \xi_z \cdot w_l^{\alpha(1-\sigma)} \cdot (I_z + \beta C_E)^{-\sigma+\alpha.\sigma-\alpha} \text{ where } \xi_z = \frac{(1-\alpha)}{\alpha} . \quad \frac{(\frac{(1-\alpha)}{\alpha})^{\frac{1-\alpha}{\sigma}+\alpha}}{Y^{\frac{1}{\sigma}}(1-\frac{1}{\sigma})(1-\alpha)} \quad ^{-\sigma}$$

The expected profit of the lender is:

$$\mathbb{E}_0\Pi_B(z_{i,LT}^*; C_L) = (l_i\tilde{w}_l + z_i\tilde{I}_z).r_e + r_{LT} - f - z_{i,LT}^*\beta.C_L,$$

Lenders face the outside opportunity normalized to $(l_i\tilde{w}_l + z_i\tilde{I}_z).r_e$ when providing long-term debt contracts and f denotes a fixed transaction cost of writing a debt contract. Lending markets are competitive, which implies that their participation constraint is saturated. The interest rate r_{LT} is set so that: $\mathbb{E}_0\Pi_B(z_i; C_L) = (l_i + z_{i,LT}^*).r_e$. These conditions gives: $r_{LT} = f + (z_{i,LT}^*\beta.).C_L$. Borrowing cost for long-term debt increases in the amount of the lender's liabilities C_L, but decreases if polluting is relatively expensive (I_z high).

[25] $(l_i\tilde{w}_l + z_i\tilde{I}_z)$ is paid at time 0 and $(l_i\tilde{w}_l + z_i\tilde{I}_z).r_e$ is paid at time $t = 2$. We can simplify because discount rates are normalized to 1.

Short-term debt

PROPOSITION 3: *When the entrepreneur uses riskless interim short-term debt, the optimal pollution becomes* $z^*_{i,ST}(C_L) = \xi_z \cdot w_l^{\alpha(1-\sigma)} \cdot (I_z + \beta C_L + \beta C_E)^{-\sigma+\alpha.\sigma-\alpha}$ with $\xi_z > 0$, *and the borrowing costs between the interim stage* $t = 1$ *and the last stage* $t = 2$ *are equal to* $r_{2,ST} = f + z^*_{i,ST}.C_L$. *The borrowing cost of the initial riskless debt contract is equal to the transaction fee* $r_{1,ST} = f$.

Proof and intuition: We solve for the optimal contract by backward induction, where the entrepreneur chooses for each stage a debt contract $(D_1, r_{1,ST}, D_2, r_{2,ST})$.

Specifically, at time $t = 1$,the borrower needs to repay $r_{1,ST}$ and D_1, and decides how much to borrow D_2. As the cash flow $p_i.y_i$ has not been generated yet, the firm will roll over the debt: $D_2 = D_1$. Without loss of generality, we assume that the firm has enough cash flow to repay the short-term interest rate $r_{1,ST}$ [26] Now that z_i is observed at time $t = 1$, lenders' profit function write: $\mathbb{E}_1\Pi_B(z_i; C_L) = r_{2,ST} - f - z_i.\beta.C_L + (l_i\tilde{w}_l + z_i\tilde{I}_z).r_e$

As lending markets are competitive, their participation constraint is saturated. The outside option of lenders is $D_2.r_e = (l_i\tilde{w}_l + z_i\tilde{I}_z).r_e$. As a result, the cost of debt for the next period becomes: $r_{2,ST} = f + z_i\beta.C_L$ This cost of debt increases in the amount of high-pollution projects z_i, and decreases in the amount of lender liability. Importantly, it implies that the marginal effect of lender liability on borrowing costs is stronger for high-polluting firms, namely: $\frac{\partial r_{2,ST}}{\partial C_L} = z_i\beta_z > 0$ Borrowing costs become more sensitive to firms' polluting efforts when lenders are more liable.

The default will not happen at time $t = 1$. As a result, the cost of debt between time 0 and 1 is equal to the fixed transaction cost f: $r_{1,ST} = f$, and $D_1 = (l_i\tilde{w}_l + z_i\tilde{I}_z)$.

Given the new refinancing cost at time $t = 1$, the entrepreneur solves at time $t = 0$:

$$p_i.y_i - z_i\beta.C_E - z_iI_z - l_iw_l - r_{2,ST}(z_i) - r_{1,ST} + f_b$$

Notice here that as the short-term contract is made after the pollution choice z_i is made and therefore observed, the choice of pollution z_i will directly affect the expected discounted borrowing cost at time $t = 1$. The threat to borrow at a higher cost disciplines firms' pollution choices. Higher lender liability makes this threat credible, as it commits lenders to increase borrowing costs if their borrowers pollute.

Given the anticipated rollover cost of debt, the entrepreneur's optimal pollution choice becomes:

[26] We could have assumed, without loss of generality, that the firm borrows $r_{1,ST}$ at time $t = 1$ using the rate r_e and repays it back at period $t = 2$

$$z^*_{i,ST} = \xi_z \cdot w_l^{\alpha(1-\sigma)} \cdot (I_z + \beta C_L + \beta C_E)^{-\sigma+\alpha.\sigma-\alpha} \text{ where } \xi_z = \frac{(1-\alpha)}{\alpha} . \quad \frac{(\frac{(1-\alpha)}{\alpha})^{\frac{1-\alpha}{\sigma}+\alpha}}{Y^{\frac{1}{\sigma}}(1-\frac{1}{\sigma})(1-\alpha)}^{-\sigma}$$

Notice here that borrowers pollute less with short-term debt than with long-term debt. The reason is that the pollution effort gets rewarded through lower borrowing costs, incentivizing the borrowers to reduce pollution. Plugging the optimal abatement effort on the cost function gives: $r_{2,ST} = f + z^*_{i,ST}.\beta.C_L$.

Choice between long-term and short-term debt

PROPOSITION 4: *The incentives of the entrepreneur to use short-term debt increase monotonically with the amount of environmental lender liability. Moreover, when the amount of environmental lender liability decreases, the incentives of high-polluting firms to choose long-term debt are higher than those of low-polluting firms. firms.*

Proof. Let's define $g(C_L) = \Pi_{LT}(z_{i,LT}) - \Pi_{ST}(z_{i,ST})$. We have $g(0) > 0$, as in a world without lender liability, long-term debt maximizes the entrepreneur's profit.

Let's show that $g'(C_L) < 0$. The envelop theorem gives us $\frac{\partial \Pi_{ST}(z^*_{i,ST})}{\partial C_L} = -\beta z^*_{i,ST}$ and $\frac{\partial \Pi_{LT}(z^*_{i,LT})}{\partial C_L} = -\beta z^*_{i,LT}$, so $g'(C_L) = -z^*_{i,LT}\beta + \beta z^*_{i,ST} < 0$, as $z_{i,LT} > z_{i,ST}$.

We next want to show the effect of higher lender liability for firms that face a higher natural cost of reducing pollution. To that end, we take the derivative of $g'(C_L)$ with respect to $\tilde{I}_z$. This gives: $\frac{\partial g(C_L)}{\partial C_L \tilde{I}_z} = -\beta \frac{dz^*_{i,LT}}{d\tilde{I}_z} + \beta \frac{dz^*_{i,ST}}{d\tilde{I}_z}$. One can show that $\frac{dz^*_{i,ST}}{d\tilde{I}_z} > \frac{dz^*_{i,LT}}{d\tilde{I}_z}$, so $\frac{\partial g(C_L)}{\partial C_L \tilde{I}_z} = -\frac{dz^*_{i,LT}}{dI_z}\beta + \beta \frac{dz^*_{i,ST}}{dI_z} > 0$.

The sign of this cross-derivative means that if lender liability is increased, and if the natural cost of polluting is increasing, then firms have a higher profit in using long-term debt. When lender liability is high, cleaner firms (those with a high $\tilde{I}_z$) prefer long-term debt compared to high-polluting firms (those with a low $\tilde{I}_z$). Conversely, if lender liability drops, cleaner firms prefer short-term debt more than high-polluting firms. Another way of saying this last sentence is that if lender liability drops, high-polluting firms prefer long-term debt more than cleaner firms.

Overall, these results are intuitive. When lenders face the environmental liabilities of their debtors, long-term debt becomes costly for the entrepreneur. With long-term debt, the entrepreneur cannot credibly commit to lower pollution: once the contract is signed, the entrepreneur increases his utility by polluting more without any punishment from the lender. As a result, the lender anticipates that the entrepreneur will pollute more in a long-term contract, and increases the borrowing cost. A short-term debt contract allows the entrepreneur who

diminishes pollution to be rewarded at the interim stage. As a result, more lender liability makes short-term debt more appealing to the entrepreneur.

A.4 Impact on firms' output, profitability, and labor

Prediction on real outcomes: If firms' market power is high and the output elasticity of pollution is low, then a decrease in environmental lender liability leads to a high reduction in pollution, low changes in employment and production, and a high increase in profitability.

Intuition: Suppose the implicit cost of the pollution input goes down. There is a substitution and a wealth effect.

The substitution effect implies that the firm uses more pollution and fewer labor. However, if the output elasticity of pollution is low $(1-\alpha)$, then this substitution effect is limited, because the firm is more reliant on labor, and cannot pivot much toward pollution.

The wealth effect implies that the firm can use more of both labor and pollution because the firm will spend a smaller amount of money on pollution (now that pollution is cheaper). However, this wealth effect will be limited for two reasons. First, because the output elasticity of pollution is low $(1-\alpha)$, pollution was not important in total costs. As a result, even if the unit cost goes down, the impact on total costs is limited. Second, firms restrict output below competitive levels to preserve markups, which limit the increase of production if firms have more monopolistic competition.

Illustration with a numerical exercise: While a structural estimation or a calibration is outside the scope of the paper, as the model is too stylized and does not include realistic productivity heterogeneity, we can still provide a numerical simulation to show the mechanism in greater detail. Suppose that firms have a low share of pollution so that $1-\alpha = 0.05$. Then, suppose we have $\sigma = 4.76$. These parameters are reasonable. Shapiro and Walker (2018) using a sample from the same type of firms for the same decade, estimate a range that goes from $0.0005 - 0.0557$ for $1-\alpha$, and a range for σ that goes from to $2.89 - 8.18$. Then, log-linearizing the optimal costs, profits, and revenues functions, we obtain that a 11% drop in the implicit cost of pollution generates a 2% increase in labor, a 2.38% increase in production, and a 13% increase in pollution. All three magnitudes are within the range of what the paper finds.

A.5 Empirical predictions

The model, while highly stylized, generates the following predictions, which are tested in the paper. These predictions are obtained by comparing the equilibrium debt contracts and firms' input choices when C_L lender liability varies exogenously. These comparative statics naturally parallel the difference-in-differences specifications, which also compare equilibrium debt contracts and firms' input choices after a reduction in lender liability risks.

Prediction I: Less environmental lender liability causes an increase in pollution and a reduction in abatement activity efforts

If lenders face less environmental lender liability risks, then their expected profit becomes less dependent on the pollution of their borrowers. Because lenders have zero expected profit in equilibrium, all the increase in profit coming from lower liability is passed through to their borrowers as lower financing costs. This implies that borrowers who pollute are less penalized for polluting than in a world with higher lender liability. Debtors who face fewer negative impacts on their borrowing costs from increased pollution are incentivized to increase pollution. This incentive to increase pollution comes from the fact that the entrepreneur fully benefits from the abatement cost savings, but is exposed to only part of the environmental costs.

It is important to note that the effects take place conditional on production happening. One alternative consequence and extreme impact of less environmental liability could be that the entrepreneur participation constraint starts to be met. As a result, less environmental liability reduces borrowing costs, which could make the participation constraint of the entrepreneur more likely to be held.

***Prediction II*:** Less environmental lender Liability leads firms to use more long-term debt in equilibrium.

This is a direct implication of Proposition 4. In the model, short-term debt lets debtors benefit from reduced borrowing costs if pollution is reduced. As they have full bargaining power, they choose short-term debt. However, if borrowing costs become less sensitive to pollution practices, then no incentive to use short-term debt exists. As a result, firms rely more on long-term debt, exposing them to lower transaction costs.

***Prediction III*:** Less environmental lender liability generates an indeterminate effect on the cost of debt.

Two forces influence borrowing costs. First, lenders who face less lender liability will punish their debtors less by lower increases in borrowing costs if they pollute more. This effect brings borrowing costs close to 0. Second, lenders react to borrowers who pollute more by raising borrowing costs, because they are more likely to be exposed to these lower environmental lender liabilities. This effect raises borrowing costs away from 0. The net effect is thus indeterminate in general.

***Prediction IV*:** Less environmental lender liability reduces the cost of debt more for high-polluting firms.

When borrowing long-term debt, pollution levels will not matter because they are not yet observed, and the borrower cannot credibly commit to the lender a lower future pollution path. However, pollution can be observed in the renegotiation phase. Specifically, at time $t = 1$, we have $r_{2,ST} = f + z_i \beta . C_L$, with the pollution z_i observed. The sensitivity of environmental lender liability to borrowing costs is increasing with the level of pollution. Intuitively, more pollution exposes the lender to more environmental liability. As a result, because lenders' participation constraint is saturated, lending to high-polluting firms is costlier, and even more so when environmental lender liabilities are high.

***Prediction V*:** Less environmental lender liability generates an increase in debt maturity that is stronger for high-polluting firms.

This is a direct implication of Proposition 4. The benefit of using short-term debt is high for high-polluting firms, especially when environmental lender liability is higher. The reason is that they benefit from a stronger reduction in their financing costs by using short-term debt. The reverse is also true, so if lender liability drops, high-polluting firms prefer long-term debt more than cleaner firms. With long-term debt, the entrepreneur cannot credibly commit to lower pollution: once the contract is signed, the entrepreneur increases his utility by polluting more without any punishment from the lender. The commitment problem is stronger for firms with high incentives to pollute (firms with low natural cost of polluting), because lenders anticipate a much stronger level of pollution after the long-term debt is signed.

Prediction VI: If firms' market power is high and the output elasticity of pollution is low, then a decrease in environmental lender liability leads to low changes in employment and production

A decline in the implicit cost of the pollution input generates both a substitution effect (encouraging firms to use more pollution and less labor) and a wealth effect (freeing up resources to increase use of both inputs). The substitution effect is muted when the output elasticity of pollution $(1 - a)$ is low, since firms remain heavily dependent on labor and cannot substantially pivot toward pollution. The wealth effect is also limited because pollution constitutes a small share of total costs, so even a lower unit cost has only a modest impact on overall expenditures. Moreover, in markets with monopolistic competition, firms restrict output to preserve markups. As a result, any potential production gains from cheaper pollution inputs are further constrained.

Prediction VII: If firms have significant market power and the output elasticity of pollution is low, then a decrease in environmental lender liability results in a substantial increase in profitability.

Because the implicit cost of the pollution input falls when environmental lender liability drops, a firm's marginal and average costs decline for any given output level. So with output prices almost unchanged, the gap between price and cost (i.e. the per-unit profit) widens. Even though the substitution and wealth effects on input mix and output volume are limited, each unit sold now costs less to produce while still almost fetching the same market price, boosting the profit margin. Finally, any modest expansion in output enabled by lower costs generates additional incremental profits at that higher margin.

A.6 Discussion: using secured debt and unsecured debt

Baseline framework

The shock that the paper exploits only applies to secured debt. Another margin of adjustment could be that banks offer unsecured debt when they are subject to higher environmental lender liability risks. Suppose the only choice variable that is available to banks is to decide whether to offer secured debt or unsecured debt. Under the assumption that banks can observe a signal on the environmental effort of their debtor, we show that the main prediction holds: less environmental lender liability risks will lead to higher pollution.

Specifically, let's assume that in a world without environmental lender liability, banks always prefer secured debt over unsecured debt. Such an assumption is credible for several reasons. First, secured loans carry lower risk: the pledged collateral can be seized upon default, raising the lender's recovery rate and reducing its loss-given-default. Second, secured creditors enjoy payment priority in liquidations or restructurings, further shielding banks from insolvency losses. Third, having a lien simplifies the valuation and monitoring of the underlying

assets. Finally, regulatory frameworks generally assign lower risk-weights to secured exposures, which in turn lowers the capital banks must hold against potential losses.

Let's denote r_s the interest rate on secured debt, r_u the interest rate on unsecured debt, f_s the bank's underwriting cost of secured debt, and f_u the bank's underwriting cost of unsecured debt. We assume that the bank does not keep the whole benefits of these lower costs, but transmits part of them to their borrowers through a lower interest rate for secured debt and a higher maturity than what the bank could offer with an unsecured loan. As a result, $r_s < r_u$ and $f_s < f_u$.

We simplify the firm's problem to make the discussion easier. The firm borrows a fixed investment amount of $D = I$. They decide whether to pollute $z > 0$ or not (and pollution equals 0). Polluting increases profit by ΔR. Polluting creates an expected liability to the bank equal to $p.C$. As a result, for a debt contract d, we have the following profit for the entrepreneur if he pollutes: $\Pi_{d,t} = R + \Delta R - D - r_d$. In this highly simple framework, when there is no constraint, it is always optimal for the firm to have a high level of pollution.

Banks' profit is thus: $r_s - f_s - p.C1_{z>0}$ if it offers secured debt, and $r_u - f_u$ if it offers unsecured debt. Suppose that it is optimal for the bank to offer secured debt if the debtor does not pollute and optimal to offer unsecured debt if the debtor pollutes, i.e., we have: $\Pi_b - p.C < r_u - f_u < \Pi_b$, given $\Pi_b = r_s - f_s$.

Then, the entrepreneur chooses to take secured debt if: $R - D - r_s > R + \Delta R - D - r_u$. This simplifies to: $r_s < r_u - \Delta R$.

Thus, as long as the unsecured rate r_u is sufficiently high, the risk of losing collateralized financing deters pollution. If r_u is high enough, then banks can discipline firms to not pollute, by threatening to use unsecured debt if firms deviate.

A.7 Different types

Let's assume that there are two types of firms. The framework will still have the core insight of the baseline model, where the threat of using unsecured debt is used to discipline debtors, but only for a subset of firms, yielding a set of predictions that are close to what the paper finds.

Specifically, let γ be the proportion of firms that benefit the most from polluting (high type), which have ΔR_H benefits from polluting. Let $1 - \gamma$ be the proportion of firms that have a gain from polluting equal to ΔR_L. By construction, we have: $\Delta R_H > \Delta R_L$.

Suppose that r_s satisfies the following conditions: $r_s < r_u - \Delta R_L$ and $r_s > r_u - \Delta R_H$. This implies that the high type is better off polluting and using unsecured debt, while the low type prefers not to pollute and use secured debt, if the bank punishes pollution with unsecured debt. Indeed:

For the high type, the profit of polluting becomes: $R - D - r_s < R + \Delta R_H - D - r_u$ which simplifies to $r_s > -\Delta R_H + r_u$. For the low type, the firm is better off not polluting: $R - D - r_s > R + \Delta R_L - D - r_u$, which simplifies to: $r_s < -\Delta R_L + r_u$.

Now, suppose that lender liability is reduced so that it becomes a dominant strategy for the lender to always offer secured debt, even if the firm pollutes, i.e., $C_L < C$, such that: $\Pi_b - p.C_L > r_u - f_u$. As a result, the bank offers secured debt only, and both firms pollute. With this smaller liability, the bank never finds unsecured debt more attractive. Consequently, it offers secured contracts to all firms, eliminating the threat of higher r_u. Both high- and low-type firms now optimally pollute, raising aggregate pollution.

Notice that in this example with two different types, the use of secured debt in the economy increased following the reduction in lenders' liability risk. If we assume that secured debt is cheaper and has longer maturity, we would also observe an overall increase in debt maturity and a reduction in financing costs following the reduction in lender liability risks.

A.8 Derivation

We detail the calculations here.

Optimal input demands under monopolistic competition

Let's $p_i.y_i - q_i w_q - z_i w_z$ be the profit of the entrepreneur. Depending on the debt contracts and the amount of liability that the firm is using, we will have different value for w_z.

We can rewrite the expected profit as follows:

$$Y^{\frac{1}{\sigma}} \cdot (q_i^{\alpha} z_i^{1-\alpha})^{1-\frac{1}{\sigma}} - C_E - q_i w_q - z_i w_z$$

As $p_i.y_i = Y^{\frac{1}{\sigma}} \cdot (q_i^{\alpha} z_i^{1-\alpha})^{1-\frac{1}{\sigma}}$.

The First-Order Condition with respect to q_i is:

$$Y^{\frac{1}{\sigma}} \cdot \quad 1 - \frac{1}{\sigma} \quad \cdot (q_i^{\alpha} z_i^{1-\alpha})^{-\frac{1}{\sigma}} \cdot \alpha q_i^{\alpha-1} z_i^{1-\alpha} = w_q$$

And with respect to z_i:

$$Y^{\frac{1}{\sigma}} \cdot \left(1-\frac{1}{\sigma}\right) \cdot (q_i^{\alpha} z_i^{1-\alpha})^{-\frac{1}{\sigma}} \cdot (1-\alpha) q_i^{\alpha} z_i^{-\alpha} = w_z$$

Taking the ratio of the two FOCs and simplifying gives:

$$\frac{Y^{\frac{1}{\sigma}} \cdot \left(1-\frac{1}{\sigma}\right) \cdot (q_i^{\alpha} z_i^{1-\alpha})^{-\frac{1}{\sigma}} \cdot \alpha q_i^{\alpha-1} z_i^{1-\alpha}}{Y^{\frac{1}{\sigma}} \cdot \left(1-\frac{1}{\sigma}\right) \cdot (q_i^{\alpha} z_i^{1-\alpha})^{-\frac{1}{\sigma}} \cdot (1-\alpha) q_i^{\alpha} z_i^{-\alpha}} = \frac{\alpha . z_i}{(1-\alpha) . q_i} = \frac{w_q}{w_z}$$

This gives the following ratio: $\frac{z_i}{q_i} = \frac{w_q}{w_z} . \frac{(1-\alpha)}{\alpha}$. Substitute $z_i = q_i . \frac{w_q}{w_z} . \frac{(1-\alpha)}{\alpha}$ into the First-Order Condition with respect to z_i gives:

$$Y^{\frac{1}{\sigma}} (1-\frac{1}{\sigma})(1-\alpha) (\frac{w_q}{w_z} . \frac{(1-\alpha)}{\alpha})^{-\frac{1-\alpha}{\sigma}} q_i^{-\frac{1}{\sigma}} \cdot q_i^{\alpha} (\frac{w_q}{w_z} . \frac{(1-\alpha)}{\alpha})^{-\alpha} q_i^{-\alpha} = w_z$$

Simplifying and solving for q_i gives:

$$q_i^* = \left(\frac{I_z (\frac{w_q}{w_z} . \frac{(1-\alpha)}{\alpha})^{\frac{1-\alpha}{\sigma}+\alpha}}{Y^{\frac{1}{\sigma}} (1-\frac{1}{\sigma})(1-\alpha)} \right)^{-\sigma}$$

As $z_i = q_i . \frac{w_q}{w_z} . \frac{(1-\alpha)}{\alpha}$, we have:

$$z_i^* = \frac{w_q}{w_z} . \frac{(1-\alpha)}{\alpha} . \left(\frac{I_z (\frac{w_q}{w_z} . \frac{(1-\alpha)}{\alpha})^{\frac{1-\alpha}{\sigma}+\alpha}}{Y^{\frac{1}{\sigma}} (1-\frac{1}{\sigma})(1-\alpha)} \right)^{-\sigma}$$

For clarity, we can factor each input demand based on the input prices:

$$q_i^* = \xi_q \cdot w_q^{-(1-\alpha+\sigma\alpha)} \cdot w_z^{-(\sigma-1)(1-\alpha)} \text{ where } \xi_q = \left(\frac{(\frac{(1-\alpha)}{\alpha})^{\frac{1-\alpha}{\sigma}+\alpha}}{Y^{\frac{1}{\sigma}} (1-\frac{1}{\sigma})(1-\alpha)} \right)^{-\sigma}$$

and

$$z_i^* = q_i . \frac{w_q}{w_z} . \frac{(1-\alpha)}{\alpha} = \xi_z \cdot w_q^{\alpha(1-\sigma)} \cdot w_z^{-\sigma+\alpha.\sigma-\alpha} \text{ where } \xi_z = \frac{(1-\alpha)}{\alpha} . \left(\frac{(\frac{(1-\alpha)}{\alpha})^{\frac{1-\alpha}{\sigma}+\alpha}}{Y^{\frac{1}{\sigma}} (1-\frac{1}{\sigma})(1-\alpha)} \right)^{-\sigma}$$

B Contaminated areas in the United States

To put the results of this paper into perspective, this section describes some salient facts on contaminated sites in the United States.

B.1 Clean up cost of all currently contaminated sites in the United States

Estimates about the total cost of cleaning up all contaminated areas do not exist and the problem "is plagued by a lack of quantitative data" (Paull (2008)).

One way to provide a quantification is to collect information using public balance sheet and recent policy proposals. The U.S. government's environmental liability amounted to $577 billion in fiscal year 2018. However,

this number does not take into account the cleanup of sites contaminated by per- and polyfluoroalkyl substances (PFAS). During a House Defense Appropriations Subcommittee meeting in 2021, Richard Kidd, deputy assistant secretary of defense for environment and energy resilience, estimated that cleaning military sites to remove this substance would cost $29 billion. In President Biden's proposal of March 2021, $16 billion were allocated for cleaning abandoned mines and orphaned oil and gas wells (E&E News PM, March 31). These numbers omit the cleanup of abandoned sites owned by private entities that are not in the oil and gas or mining industries, but provide a conservative lower-bound estimate of $622 billion.

Precise estimates about the magnitudes of the problem for other private sites are not readily available. There is an agreed estimated number of 450,000 brownfields in the United States. Moreover, there were 1,374 sites registered in the National Priorities List (NPL) and awaiting remedial action. On average, a Superfund site costs between $25 and $50 million and the average per-site cost for brownfield remediation is estimated at $602,000 according to the Northeast Midwest Institute, which is based on cleanup data from EPA (Paull (2008)). The extrapolation ignores the large variability in cleanup costs. With these caveats in mind, simple extrapolation and back-of-the-envelope calculations give a total cost of $339.6 billion.

B.2 Social gain of environmental cleanup

While the cleaning of a contaminated area necessitates considerable upfront costs, the benefits are diffuse, scattered and even more challenging to identify and quantify precisely. The literature has identified several ways in which contaminated areas reduce welfare and in which remediation of those areas improves welfare.

First, contaminated areas pose a public health problem to a significant fraction of the U.S. population. One out of three Americans lives within three miles of a federal Superfund site (US EPA (2016); Persico, Figlio, and Roth (2020)), and 11 million Americans, including 3 to 4 million children, are located within one mile of a Superfund site (Steinzor and Clune (2006); Persico, Figlio, and Roth (2020)). People are exposed to the contaminants from Superfund sites by drinking or swimming in contaminated water or eating food grown on toxic land. As a result, people exposed to contaminated areas are more likely to suffer from health problems, including cancer (Environment America Research and Policy Center (2021)), which reduces life expectancy (Kiaghadi, Rifai, and Dawson (2021)). Young children and pregnant women are particularly susceptible to these effects. In particular, children living close to Superfund sites have higher lead levels in their blood (Klemick, Mason, and Sullivan (2020)), which causes anemia, weakness, kidney failure, and brain damage. Children living close to

a contaminated site or who experienced prenatal exposure also have lower cognitive and behavioral outcomes (Persico and Venator (2021); Persico, Figlio, and Roth (2020)).

Second, the release of hazardous substances by contaminated sites endangers the survival of ecosystems (Environment America Research and Policy Center (2021)). It is difficult to evaluate how prevalent this damage is because of data scarcity. Moreover, it is challenging to quantify how the loss of ecosystems affects human welfare, as it depends on unknown parameters, such as the cash flows of ecosystems, their discount rates, and the irreversibility of their losses.

Third, cleaning up contaminated areas allows for urban redevelopment with better energy efficiency uses at the city level, because they are often previous industrial sites located in densely developed urban areas with high location efficiency. For instance, the Office of Brownfields and Land Revitalization (2015) shows that brownfield redevelopment in five areas[27] would decrease the volume of carbon dioxide and air pollutant emissions per capita by 32-57%. It would also reduce stormwater runoff more than other conventional developments by reducing daily vehicle miles and trips per capita.

[27] Baltimore (MD), Dallas-Forth Worth (TX), Emeryville (CA), Minneapolis-Saint Paul (MN), and Seattle (WA)

C Case study 1: *United States v. Mirabile (E.D. Pa. Sept 4, 1985)*

I describe the case in three steps. Subsection C.1 presents the facts of the case. Subsection C.2 gives its procedural history.[28] Subsection C.3 explains the bankruptcy court's rationale for its decision.

C.1 Case facts

In 1976, Turco Coatings, Inc. (henceforth Turco) purchased a facility in Phoenixville, Pennsylvania (henceforth Turco site) and opened a paint manufacturing factory on the site. The purchase was made from Arthur C. Mangels Industries Inc. (henceforth Mangels). The previous owner of the facility had taken a loan from the American Bank and Trust Company (henceforth ABT), which was partially secured by a mortgage on the Turco site. Turco filed for bankruptcy under Chapter 11 in January 1980, but the petition was rejected in 1981 by a bankruptcy court. Turco ceased operating in December 1980.

In 1976, the predecessor lender of Mellon Bank, provided a credit line to advance working capital at Turco, secured by the inventory and assets of the company. One board member in charge of supervising Turco's operations was the loan officer initially responsible for the loan. The monitoring effort increased after Turco filed the petition.

Finally, Turco took out a loan from the Small Business Administration (henceforth, SBA) in July 1979, secured by a second lien on equipment, inventory, account receivables, and real estate. An SBA representative monitored the site three times to inspect how the assets were sold in 1981. SBA contracts contained several limitations placed on Turco's actions. Specifically, Turco was not allowed to enter into management consulting services withtout SBA approval, and the bank set a cap on the total remuneration of operating officers. Moreover, the purchase of any life insurance or dividend required SBA approval.

The bank ABT repossessed the facility and sold it to Thomas A. Mirabile and Anna Mirabile on December 15, 1981. Between the time of the foreclosure and the sale to the Mirabiles, the bank performed several tasks on the property: "It secured the building against vandalism by boarding up windows and changing locks, made inquiries as to the approximate cost of disposal of various drums located on the property, and, through its loan officer Donald Hans, visited the property on various occasions for the purpose of showing it to prospective purchasers."

[28] The lawsuit in *Mirabile v. United States* predates the sample period analyzed in this study and therefore lies outside the temporal scope of our empirical analysis.

The predecessor of Mellon Bank took possession of the inventory from the Turco site with the approval of the bankruptcy court.

In December 1981, the Pennsylvania Department of Environmental Resources (D.E.R.)[29] informed Mr. Mirabile that toxic leaking drums were on the Turco site, contaminating the surroundings. Mirabile undertook some actions to clean up the site, regrouping the drums into a warehouse, but no further action was undertaken. However, there is an absence of evidence that the Mirabiles increased the overall pollution when they purchased the site at auction.

In February of 1983, a representative from the EPA visited the site and noticed that many drums were in poor condition and that access to the site was not sufficiently protected from trespassers. Evidence that some trespassers could access the site was noted. Immediate removal of the drums was ordered, and the EPA started the cleanup on February 11 of that year using Superfund money.

C.2 Procedural history and Final disposition

The court entered a summary judgment in favor of the SBA and ABT. However, the court rejected the motion for a summary judgment from Mellon Bank.

C.3 Court rationale

The court's reasoning was based on both statutory arguments and the legislative history of CERCLA. First, the statutory argument derived from CERCLA implied that a secured lender must participate in the management of a facility to be held liable. The court enunciated a narrow standard of what it means to "participate in the management," stating, "before a secured creditor such as ABT may be held liable, it must, at a minimum, participate in the day-to-day operational aspects of the site." The court accepted participation in management affairs as equating to participation at the site; that is, "the participation in operational, production or waste disposal activities," to incur environmental liabilities for secured lenders.

Second, the court justified this standard as being closer to the principle of CERCLA, which is to make polluters pay for their pollution, as they bore the fruit of negligent environmental practices. The court referred to a decision from the District Court of Missouri (*United States v. Northeastern Pharmaceutical & Chemical Co. (NEPACCO), 579 F. Supp. 823, 849 (W.D. Mos. 1984)*), where this principle was first stated.

[29] On July 1, 1995, the DER was replaced by the Pennsylvania Department of Environmental Protection and the Pennsylvania Department of Conservation and Natural Resources.

Given this standard, the court rejected the motion for a preliminary judgment from Mellon Bank, as one of its loan officers was part of the board attached to the management of Turco's site. However, SBA and ABT did not participate in the day-to-day operational aspects of the business and were therefore exempt according to this test.

D Case study 2: United States v. Fleet Factors Corp. (11th Cir. 1990)

This case is presented similarly to the discussion above, outlining the details of the matter and reviewing the court's final judgment.[30]

D.1 Case details

In 1976, Fleet Factors Corporation (henceforth Fleet), a factoring company, made a collateralized factoring agreement with Swainsboro Print Works (henceforth SPW), a cloth printing facility. In the agreement, Fleet advanced funds against the assignment of SPW's accounts receivable. Fleet took a security interest in SPW's equipment, inventory, and fixtures. In August 1979, SPW filed for Chapter 11 bankruptcy, but the loan contract between SPW and Fleet continued. At the beginning of 1981, Fleet stopped providing funds to SPW because the company estimated that the ongoing debt of SPW exceeded the value of its accounts receivable. On February 27, 1981, SPW ceased operations, and in December 1981, the company filed for Chapter 7 bankruptcy.

As soon as SPW stopped operating, Fleet started to participate directly in the management of the facility in a several of dimensions to maximize the amount collected on the accounts receivable.

(1) Fleet tried to sell "the twenty to twenty-five million yards of grey goods and finished cloth remaining" and collected from the accounts receivable from those goods. Specifically, Fleet took care of resolving disputes and ensured both consumer reliability and the lack of delinquent accounts.

(2) Fleet wired money to SPW's account to pay the remaining workers to maintain production at facility. No workers directly employed by Fleet were on-site. Fleet also directly paid some suppliers that were not accepting SPW checks.

(3) Finally, Fleet participated in the tax management of SPW. Fleet provided advice to the company and used the EIN of SPW. It was involved with the "tax deposit reports" of the company.

One disputed fact concerns whether Fleet blocked the sale of SPW chemicals and, by doing so, maintained the leaking drums on-site. The government used the argument as evidence that Fleet participated in the management of the company. However, the evidence at trial suggested otherwise. Managers instead testified at trial that they were constrained by the Fleet lien in relation to SPW's chemicals. A communication problem ocurred among

[30]The lawsuit in *Fleet Factors Corp. v. United States* likewise predates the sample period analyzed in this study and therefore lies outside the temporal scope of our empirical analysis.

SPW's managers that did not transmit correctly their intentions to sell the drums. Subsequently, the non-response by Fleet was interpreted by SPW's managers as a refusal to sell the drums.

However, Fleet never foreclosed on its whole collateral but did so for the equipment and machines.

After 1981, Fleet hired two contractors. The first contractor, Baldwin Industrial Liquidators, was responsible for auctioning off the remaining equipment. The task of the second contractor, Nix Riggers (henceforth Nix) was to clean the facility and make it "broom clean". Nix testified that he was allowed to do anything possible so that no equipment or machinery remained.

The EPA inspected the facility on January 20, 1984, and found evidence of environmental contamination. The agency proceed to clean up and then sued the two principal owners of the facility as well as Fleet to cover the cleanup costs.

D.2 Procedural history and final judgment

Fleet filed for a summary judgment, which was rejected in *United States v. Fleet Factors Corp., 724 F.Supp. 955, 960 (S.D.Ga. 1988)*, on the basis that its participation in the management of the facility made the applicability of the secured creditor exemption questionable. Fleet filed an interlocutory appeal following the denial of its summary judgment. The 11th Circuit Court of Appeals stated in *United States v. Fleet Factors Corp., 901 F.2d 1550 (11th Cir. 1990) (Fleet II)* that, as a matter of law, the secured creditor exemption was not applicable. As a result, the court concluded that rejecting Fleet's motion for summary judgment was correct and remanded the case.

D.3 Court rationale

The 11th Circuit Court explicitly rejected the interpretation given during *Mirabile*. The court put forward two main arguments in *United States v. Fleet Factors Corp., 901 F.2d 1550 (11th Cir. 1990) (Fleet II)* to justify the final disposition.

The first argument is that the legislative history of CERCLA supports a broader interpretation of lender liability. Indeed, senators made a clear distinction between an operator and a secured lender that participates in managing a facility. The interpretation of environmental lender liability that *Mirabile* gives is similar to the one related to operators. However, when the senators wrote the initial CERCLA law, they made a clear distinction between a secured lender that "participates in the management" and an "operator." The definition of the secured

lender exemption is similar to the definition of an owner or operator under *Mirabile*, thus making it redundant. The 11th Circuit Court interpreted the statement made by Representative Harsha when the amendment was introduced as consistent with the narrow interpretation of CERCLA liability. Indeed, the word "affiliated" was used to describe which lenders would be exposed to CERCLA liabilities under the new statement. Affiliation implies less involvement in day-to-day activities made by lenders as compared with that of an owner.

The second argument is that more robust environmental lender liability helps lenders influence their debtors to adopt better environmental practices, consistent with one of CERCLA's goals. "Our ruling today should encourage potential creditors to investigate thoroughly the waste treatment systems and policies of potential debtors." Lenders that are more exposed to the cleanup costs of their secured assets have a stronger incentive to require better environmental compliance when negotiating their loan terms with debtors.

E Fuzzy matching

To match the Business Register to the EPA database and obtain a cross-link between the two administrative sources, I perform a fuzzy matching algorithm that contains multiple steps.

Step 1: Normalization of fields

Names are normalized so that each word has the exact same spelling and is capitalized. For instance, INC. and Incorporated are transformed into INCORPORATED, as their meanings are the same. I exploit the four different names available in the Business Register (BR) from the Census Bureau and the two names available in the TRI data from the EPA.

The street number from the address field in both databases is extracted. Next, the street name is normalized. For instance, rd is the abbreviation for road, so I replace road with rd. For the BR, I use the physical address when available. If this field is missing, then the mailing address is used instead.

Step 2: Drop duplicate observations

From the BR and TRI, I use a time span from 1992 through 1999. I drop the duplicates according to the field that will be used for the matching, namely the establishment names, addresses, ZIP, state and county codes, NAICS, and other identifiers.

Step 3: Run matching without pre-processing the data

I run a *perfect* matching of variables using names and addresses within establishments that share the same state number, street number and two-digit NAICS code. As we have two names for the TRI and four for the BR, the score for the name field is the highest score of all name pairs. I first keep all perfect matching sets before pre-processing so as to avoid dropping relevant information.

Step 4: Pre-processing of variables

I then pre-process the variables to keep the most relevant information in a firm's name. Intuitively, if one term is used by many firms, then its usefulness in terms of matching is more limited than a unique term. I compute the frequency of each term for each database separately. I create a list of terms to be dropped from the database if the term is among the top 1% of the most frequent terms for each database. This step results in dropping common terms, such as "America," "group," or "LLC."

Step 5: Coarse fuzzy matching

I then perform a first fuzzy matching on the address and company names, using a bigram approach with no weight, where the similarity scoring is based on the Jaccard index. The final score is the unweighted average of the best score for the name matching, between the combination of the four names from the BR and the two names from the TRI, with the score for the street name. The matching is done at the establishment level with the same ZIP code and industry code. If the physical address is used, then I review whether the street number matches perfectly. All matches with a score above 90% accuracy are kept. I manually check the accuracy of matches when the score is 70% to 90%.

Finally, all establishments for which I find a linkage are dropped from the BR and TRI files before I move to the next step.

Step 6: Broader fuzzy matching

I then perform a second fuzzy matching on the address and company names. Similar to the previous approach, I use a bigram approach with no weight, where the similarity scoring is based on the Jaccard index. The final score is the unweighted average of the best score for the name matching, between the combination of the four names from the BR and the two names from the TRI, with the score for the street name.

If the physical address is used, then I review whether the street number matches perfectly. I then verify that the two establishments are located in the same county. All matches with a score above 90% accuracy are retained. I manually check the accuracy of matches when the score is 70% to 90%. In particular, I use the industry code when this information can be located.

Finally, all establishments for which I find a linkage are dropped from the BR and TRI files before I move to the next step.

Step 7: Visual inspection of perfect matches for addresses

I then perform matching within counties but use only the address name. I keep all matches with a score above 90%. I manually check the names for these observations. Sometimes, a database will report a name through abbreviations or only the initials, resulting in a low matching rate for the name score. Alternatively, one name will be a lengthy description of the business, with the name inside the description. In these cases, the matching score for the name field will be below 70%, although a visual inspection makes it evident that the company is the same.

Step 8: Visual inspection of perfect matches for addresses

I manually check all the matched links. As I keep all links with a score above 90%, I find multiple links for the same establishment in some rare cases. When that happens, I visually inspect the name and detailed industry code to keep the most relevant linkage.

F Data cleaning steps

I create a balanced panel at the establishment level for the sample period 1992 through 1999, which retains 374,158 observations representing 71.53% of the unbalanced sample. Several reasons motivate that action. First, I want to make my results comparable with those of other papers that use the TRI, such as Ohlrogge (2020), which also adopts this assumption.

Second, unbalanced panels introduce noise into the regressions. If this noise is exogenous, then it attenuates the coefficients. However, if the selection is due to the reporting framework of the TRI, then the unbalanced panel will put more weight on some specific industries or establishments with more employees.

Third, focusing on firms that report consistently every year lets me focus on the highest quality part of the dataset. Previous works, such as Brehm and Hamilton (1996), have shown that misreporting in the TRI results from ignorance rather than from strategic misconduct, as the errors are concentrated among firms that report small amounts of chemicals.[31] However, firms that consistently report a higher quantity of chemicals are more likely to have a specific team dedicated to TRI reporting. This also means that they are not "new firms" that had to learn how to report to the survey, e.g., the utilities included in 1998. Note that creating a balanced panel does not mean that bankrupt firms are automatically dropped from the sample. If an establishment is liquidated and sold to another firm, then the establishment will still report to the TRI and will be observed.

I impute the missing chemicals by a zero. This imputation is motivated by the fact that when facilities report to the TRI, they can either mention a chemical that they do not report as using zero pounds of the component, or simply omit it. Assigning zero to the missing components standardizes the data.

Finally, I focus on publicly listed firms. This brings the sample to 210,000 observations.[32] The first reason for this choice is that the cross-sectional tests and the firm-level controls are based on Compustat, which is available only for publicly listed firms. Another advantage of focusing on publicly listed firms is that it concentrates the analysis on the highest quality segment of the TRI. As Brehm and Hamilton (1996) have shown, there are fewer inconsistencies in reporting among firms that report higher pollution measures. Publicly listed firms have a higher scale of production, and, as a result, report more pollution.

[31] Bui and Mayer (2003) found that there is little systematic over- or underreporting in the TRI.
[32] All sample sizes are rounded to four significant digits following Census Bureau disclosure guidelines

G EPA response to a contamination threat

During the Preliminary Assessment phase, an EPA team performs initial and limited inspections to assess the danger of a site. The assessment is made using the Hazard Ranking System (HRS).This system assigns a grade between 0 and 100 that denotes the risk level a site represents to human health and the environment. The risk is multidimensional because it takes into account different pathways through which toxic releases could affect human health or environmental systems. The EPA evaluates four main pathways: groundwater migration, soil exposure, surface water migration, and air migration. For instance, groundwater migration relates to the likelihood of toxic components traveling to aquifers and drinking water wells. Sites ranked highly are registered in the National Priorities List (NPL). In April 2021, there were 1,374 sites on this list, with an average score of 43.5. Figure H.5 shows their locations.

The EPA team then decides the type of response actions it requires. A release may necessitate an emergency response to eliminate immediate risks to human health, such as in the case of a road accident in which toxic chemicals may enter directly into contact with the population. A site could necessitate an early action to block a near-future threat of contamination, or a long-term action if the risk of contamination may take several years or decades to materialize. For instance, if drums storing chemical components leak from an industrial site, then an early action would consist of removing the leaking drums, and a long-term action would be a cleanup of the contaminated soil and underground water formation from the chemical component.

The EPA's plans for long-term responses are subject to public comments at least 30 days before the remedial action begins. The plan describes the options possible to perform the cleanup as well as the remediation preferred by the EPA. Public concerns are taken into account, and then the EPA issues a Record of Decision (ROD) that describes how the cleanup will be performed. Once the cleanup is completed, the site is removed from the NPL.

H Additional tables and figures

Table H.1: Balance tests

Variables	Treated group	Control group	Diff	P-value
Production ratio$_{cit}$	0.727	0.74	0.0133	0.192
Total releases$_{cit}$	35,110	36,890	1,777	0.8235
On-site CERCLA pollution$_{cit}$	6,495	12,650	6,157	0.2613
$\mathbb{1}$(on-site CERCLA pollution)$_{cit}$	11.5	11.6	0.0511	0.975
Toxic$_{cit}$	0.475	0.466	-0.0094	0.238
$\mathbb{1}$(Process-related abatement)$_{cit}$	7.96	6.37	-1.59**	0.0162
Air pollution$_{cit}$	22,290	21,020	-1,272	0.7458
Capital intensity$_{it}$	0.0455	0.0467	0.0012	0.6541
Cash Flow$_{it}$	0.0702	0.0698	-0.0004	0.8554
Cash Holding$_{it}$	0.0456	0.0435	-0.0021	0.3336
Cost of Capital$_{it}$	4.974	4.363	-0.611	0.2785
ROA$_{it}$	0.0258	0.0249	-0.0009	0.4583
ROE$_{it}$	0.0078	-0.0011	-0.0089	0.1667
Tangibility$_{it}$	0.3122	0.3035	-0.0087	0.6782
Tobin's Q$_{it}$	1.086	1.043	-0.0434	0.1357
Leverage$_{it}$	0.7477	0.7243	-0.0234	0.8959
Other liabilities (LO)$_{it}$	855.8	1,469	613.2***	0.0003

This table reports the descriptive statistics of the main variables used in the empirical analysis, for both the treated group (facilities in the 11th Circuit) and control group (facilities not located in the 11th Circuit), before 1994 (included), that is the year before the Lender Liability Act is first introduced. Estimates have been rounded to four significant digits according to Census Bureau disclosure guidelines. Table 1 provides additional descriptions of how the variables are defined. The symbols*, **, and *** denote statistical significance at the 10%, 5%, and 1% levels, respectively.

Table H.2: Prediction of The Shock with State-Level Variables (1/3)

	Dependent variable: Treated group × Post 1996 (included)						
	(1)	(2)	(3)	(4)	(5)	(6)	(7)
Corporate income tax	-0.001			-0.001	-0.003		-0.001
	(0.003)			(0.003)	(0.004)		(0.003)
Sales tax	-0.000			-0.002	-0.002		-0.003
	(0.003)			(0.004)	(0.003)		(0.005)
Personal income tax	-0.006			-0.007	-0.006		-0.006
	(0.006)			(0.006)	(0.006)		(0.005)
Property taxes (state)	0.000			-0.000	-0.000		0.000
	(0.000)			(0.000)	(0.000)		(0.000)
Unemployment insurance		0.000		0.000		0.000	0.000
		(0.000)		(0.000)		(0.000)	(0.000)
Unemployment insurance rate		-0.024		-0.033		-0.026	-0.032
		(0.024)		(0.029)		(0.024)	(0.029)
Unemployment insurance base wage		-0.000		-0.000		-0.000	-0.000
		(0.000)		(0.000)		(0.000)	(0.000)
Unemployment rate		-0.012		-0.010		-0.009	-0.010
		(0.013)		(0.012)		(0.011)	(0.012)
Gross domestic product		0.000		0.000		0.000	0.000
		(0.000)		(0.000)		(0.000)	(0.000)
Total revenues (state)			-0.000		-0.000	-0.000	0.000
			(0.000)		(0.000)	(0.000)	(0.000)
General revenues (state)			0.000		0.000	-0.000	-0.000
			(0.000)		(0.000)	(0.000)	(0.000)
State budget balance			0.043		-0.025	0.058	-0.009
			(0.055)		(0.043)	(0.061)	(0.054)
Observations	459	459	459	459	459	459	459
R-squared	0.020	0.055	0.0070	0.074	0.026	0.065	0.079

Note: This table reports state-year level regressions to investigate whether the treated group after 1996 can be predicted by changes in state-level characteristics. It reports regressions whereby the dependent variable takes the value one if the state is in the 11th Circuit after the year 1996 (included) and zero otherwise. The symbols*, **, and *** denote statistical significance at the 10%, 5%, and 1% levels, respectively.

Table H.3: Prediction of The Shock with State-Level Variables: enforcement (2/3)

A. EPA employees

	Dependent variable: Treated group × Post 1996 (included)				
$\log(\frac{\text{EPA employees}_{it}}{\text{EPA employees}_{it-3}})$	0.003				
	(0.008)				
$\log(\frac{\text{EPA employees}_{it}}{\text{EPA employees}_{it=1995}})$		0.001			
		(0.016)			
$\text{EPA employees}_{it}$			-0.000		
			(0.000)		
$\log(\text{EPA employees}_{t})$				0.005	
				(0.005)	
$\frac{\text{EPA employees}_{t}-\text{EPA employees}_{t-1}}{\text{EPA employees}_{t-1}}$					-0.006
					(0.006)
Constant	x	x	x	x	x
Observations	301	293	309	309	301
R-squared	0.000034	0.0000046	0.000026	0.0038	0.00048

This table reports state-year level regressions to investigate whether the treated group after 1996 can be predicted by changes in state-level enforcement activities from the EPA. It reports regressions in which the dependent variable takes the value of one if the state is in the 11th Circuit after 1996 (included) and zero otherwise. The variable *EPA employees*$_{it}$ is constructed using data from the U.S. Office of Personnel Management's (OPM) Enterprise Human Resources Integration System. We compute the total number of full-time employees working at the EPA at the state level based on the location of the duty stations. The sample spans 1992 to 1999. The symbols *, **, and *** denote statistical significance at the 10%, 5%, and 1% levels, respectively.

Table H.4: Prediction of The Shock with State-Level Variables: enforcement (3/3)

B. EPA total pay

	Dependent variable: Treated group × Post 1996 (included)				
$\log(\frac{\text{EPA Total pay}_{it}}{\text{EPA Total pay}_{it-3}})$	-0.016				
	(0.019)				
$\log(\frac{\text{EPA Total pay}_{it}}{\text{EPA Total pay}_{it=1995}})$		0.062			
		(0.074)			
$\text{EPA Total pay}_{it}$			-0.000		
			(0.000)		
$\log(\text{EPA Total pay}_{it})$				0.005	
				(0.005)	
$\frac{\text{EPA Total pay}_{it}-\text{EPA Total pay}_{t-1}}{\text{EPA Total pay}_{it-1}}$					0.000
					(0.007)
Constant	x	x	x	x	x
Observations	292	284	309	301	300
R-squared	0.00072	0.0064	0.000030	0.0031	0.00000035

This table reports state-year level regressions to investigate whether the treated group after 1996 can be predicted by changes in state-level enforcement activities from the EPA. It reports regressions in which the dependent variable takes the value of one if the state is in the 11th Circuit after 1996 (included) and zero otherwise. The variable *EPA Total Pay*$_{it}$ is constructed using data from the U.S. Office of Personnel Management's (OPM) Enterprise Human Resources Integration System. We compute the total pay of full-time employees working at the EPA at the state level based on the location of the duty stations. The sample spans 1992 to 1999. The symbols *, **, and *** denote statistical significance at the 10%, 5%, and 1% levels, respectively.

Table H.5: Magnitudes: difference

	$\frac{\text{Pollution}_{cit}}{\text{Emp}_{it=1994}}$
11th Circuit$_i$	-0.259
	(0.249)
R2	0.063
Observations	78,500
Legal status$_i \times$Year FE$_t$	x
Chemical FE$_c \times$Year FE$_t$	x
NAICS FE$_i \times$Year FE$_t$	x
Firm-level controls$_{it}$	x

Note: This table reports the simple differences between the 11th Circuit (treated group) and the others (control group) before 1996, the year that the Lender Liability Act was passed. The dependent variable is $\frac{\text{Pollution}_{cit}}{\text{Emp}_{it=1994}}$, which is the total onsite pollution divided by the employment in 1994. 11th Circuit$_i$ takes the value one for plants located within the 11th Circuit and zero otherwise. Chemical FE$_c$ is a chemical fixed effect defined at the CAS registry number level. Year FE$_t$ is a year fixed effect, and NAICS FE$_i$ is an industry fixed effect based on the two-digit NAICS code as defined in the LBD dataset. Legal status FE$_i$ is a fixed effect that groups together firms with the same legal status as defined by the LBD. Firm-level controls$_{it}$ includes 15 time-varying controls defined at the firm level that are commonly used in empirical corporate finance. These controls are the capital intensity, capx, cash flow, cash holding, cost of capital, firm sales, the log of firm size, R&D intensity, the return on asset (ROA), the return on equity (ROE), the tangibility ratio, the net income, Tobin's Q, total asset, total firms' liability. Estimates and sample size have been rounded to four significant digits according to Census Bureau disclosure guidelines. The symbols*, **, and *** denote statistical significance at the 10%, 5%, and 1% levels, respectively.

Table H.6: Placebo on Air Pollution

	Dependent variable: log(air pollution+1)$_{cit}$			
	(1)	(2)	(3)	(4)
Post$_t$ × 11th Circuit$_i$	-0.0306	-0.0223	-0.0291	-0.0289
	(0.0529)	(0.0433)	(0.0409)	(0.0397)
Observations	210,000	210,000	210,000	210,000
R-squared	71.5	76.6	76.6	76.6
Facility FE$_i$× Chemical FE$_c$	x	x	x	x
Year FE$_t$	x			
Legal status$_i$× Year FE$_t$		x	x	x
Chemical FE$_c$× Year FE$_t$		x	x	x
NAICS FE$_i$× Year FE$_t$			x	x
Firm-level controls$_{it}$				x

Note: This table reports the difference-in-differences between the 11th Circuit (treated group) and the others (control group) after 1996, the year that the Lender Liability Act was passed. The dependent variable is log(on-site air pollution+1)$_{cit}$, the log of air pollution plus one. The dataset is at the chemical-year level from 1992 through 1999. Post$_t$ is a variable that takes the value one after 1996 (included) and zero otherwise. 11th Circuit$_i$ takes the value one for plants located within the 11th Circuit and zero otherwise. Chemical FE$_c$ is a chemical fixed effect defined at the CAS registry number level. Year FE$_t$ is a year fixed effect, Facility FE$_i$ is a facility fixed effect, and NAICS FE$_i$ is an industry fixed effect based on the two-digit NAICS code as defined in the LBD dataset. Legal status FE$_i$ is a fixed effect that groups together firms with the same legal status as defined by the LBD. Firm-level controls$_{it}$ includes 15 time-varying controls defined at the firm level that are commonly used in empirical corporate finance. These controls are the capital intensity, capx, cash flow, cash holding, cost of capital, firm sales, the log of firm size, R&D intensity, the return on asset (ROA), the return on equity (ROE), the tangibility ratio, the net income, Tobin's Q, total asset, total firms' liability. Estimates and sample size have been rounded to four significant digits according to Census Bureau disclosure guidelines. The symbols*, **, and *** denote statistical significance at the 10%, 5%, and 1% levels, respectively.

Table H.7: Other Measures of Pollution

Panel A: Inverse hyperbolic sine

Dependent variable: IHS(on-site CERCLA pollution)$_{cit}$				
	(1)	(2)	(3)	(4)
Post$_t$ × 11th Circuit$_i$	0.160***	0.150***	0.148***	0.144***
	(0.0135)	(0.0170)	(0.0181)	(0.0173)
Observations	210,000	210,000	210,000	210,000
R-squared	70.3	72.4	72.4	72.4
Facility FE$_i$× Chemical FE$_c$	x	x	x	x
Year FE$_t$	x			
Legal status$_i$× Year FE$_t$		x	x	x
Chemical FE$_c$× Year FE$_t$		x	x	x
NAICS FE$_i$× Year FE$_t$			x	x
Firm-level controls$_{it}$				x
Mean Dep. Var.	0.7476	0.7476	0.7476	0.7476

Panel B: Pollution per capital

Dependent variable: on-site CERCLA pollution per facility capital$_{cit}$				
	(1)	(2)	(3)	(4)
Post$_t$ × 11th Circuit$_i$	0.654**	0.734**	0.747**	0.704**
	(0.213)	(0.297)	(0.275)	(0.241)
Observations	183,000	183,000	183,000	183,000
R-squared	67.6	67.6	67.6	67.6
Facility FE$_i$× Chemical FE$_c$	x	x	x	x
Year FE$_t$	x			
Legal status$_i$× Year FE$_t$		x	x	x
Chemical FE$_c$× Year FE$_t$		x	x	x
NAICS FE$_i$× Year FE$_t$			x	x
Firm-level controls$_{it}$				x

Note: This table reports the difference-in-differences between the 11th Circuit (treated group) and the others (control group) after 1996, the year that the Lender Liability Act was passed, for alternative measures of pollution. The dependent variable of Panel A is the inverse hyperbolic sine ($f(x) = log(x + \sqrt{1+x^2})$) of the on-site CERCLA release of chemical c at time t. The inverse hyperbolic sine is approximately equal to the natural logarithm of x, but is well defined in 0. The dependent variable for Panel B is the on-site CERCLA release of chemical c at time t divided by the capital of the facility's building and other structures (as defined the CMF/ASM). The dataset is at the establishment-chemical-year level. Post$_t$ is a variable that takes the value one after 1996 (included) and zero otherwise. 11th Circuit$_i$ takes the value one for plants located within the 11th Circuit and zero otherwise. Chemical FE$_c$ is a chemical fixed effect defined at the CAS registry number level. Year FE$_t$ is a year fixed effect, Facility FE$_i$ is a facility fixed effect, and NAICS FE$_i$ is an industry fixed effect based on the two-digit NAICS code as defined in the LBD dataset. Legal status FE$_i$ is a fixed effect that groups together firms with the same legal status as defined by the LBD. Firm-level controls$_{it}$ includes 15 time-varying controls defined at the firm level that are commonly used in empirical corporate finance. These controls are the capital intensity, capx, cash flow, cash holding, cost of capital, firm sales, the log of firm size, R&D intensity, the return on asset (ROA), the return on equity (ROE), the tangibility ratio, the net income, Tobin's Q, total asset, total firms' liability. Estimates and sample size have been rounded to four significant digits according to Census Bureau disclosure guidelines.

Table H.8: Include Adjacent Circuits as Control Group

Panel A: CERCLA on-site pollution (continuous)

Dependent variable: log(on-site CERCLA pollution+1)$_{cit}$				
	(1)	(2)	(3)	(4)
Post$_t$ × 11th Circuit$_i$	0.147**	0.131**	0.131**	0.123**
	(0.02)	(0.028)	(0.026)	(0.025)
Observations	99,000	99,000	99,000	99,000
R-squared	69.8	72.4	72.4	72.5
Facility FE$_i$× Chemical FE$_c$	x	x	x	x
Year FE$_t$	x			
Legal status$_i$× Year FE$_t$		x	x	x
Chemical FE$_c$× Year FE$_t$		x	x	x
NAICS FE$_i$× Year FE$_t$			x	x
Firm-level controls$_{it}$				x

Panel B: CERCLA on-site pollution (discrete)

Dependent variable: 𝟙(on-site CERCLA pollution)$_{cit}$				
	(1)	(2)	(3)	(4)
Post$_t$ × 11th Circuit$_i$	2.079**	2.146**	2.134**	2.072**
	(0.310)	(0.399)	(0.396)	(0.355)
Observations	99,000	99,000	99,000	99,000
R-squared	67.6	69.4	69.4	69.5
Facility FE$_i$× Chemical FE$_c$	x	x	x	x
Year FE$_t$	x			
Legal status$_i$× Year FE$_t$		x	x	x
Chemical FE$_c$× Year FE$_t$		x	x	x
NAICS FE$_i$× Year FE$_t$			x	x
Firm-level controls$_{it}$				x

This table reports the difference-in-differences between the 11th Circuit and the states from adjacent Circuits, namely the 4th Circuit (Maryland, North Carolina, South Carolina, Virginia, and West Virginia); 5th Circuit (Louisiana, Mississippi, and Texas); and 6th Circuit (Kentucky, Michigan, Ohio, and Tennessee). The treatment year is 1996, the year that the Lender Liability Act was passed. The dependent variable of Panel A is log(on-site CERCLA pollution+1)$_{cit}$ the log of the total amount in pounds of on-site pollution +1. The dependent variable of Panel B is 𝟙(on-site CERCLA pollution)$_{cit}$, a dummy variable that takes one if the total amount in pounds of on-site pollution is strictly positive and zero otherwise. The dataset is at the establishment-chemical-year level. Post$_t$ is a variable that takes the value one after 1996 (included) and zero otherwise. 11th Circuit$_i$ takes the value one for plants located within the 11th Circuit and zero otherwise. Chemical FE$_c$ is a chemical fixed effect defined at the CAS registry number level. Year FE$_t$ is a year fixed effect, Facility FE$_i$ is a facility fixed effect, and NAICS FE$_i$ is an industry fixed effect based on the two-digit NAICS code as defined in the LBD dataset. Legal status FE$_i$ is a fixed effect that groups together firms with the same legal status as defined by the LBD. Firm-level controls$_{it}$ includes 15 time-varying controls defined at the firm level that are commonly used in empirical corporate finance. These controls are the capital intensity, capx, cash flow, cash holding, cost of capital, firm sales, the log of firm size, R&D intensity, the return on asset (ROA), the return on equity (ROE), the tangibility ratio, the net income, Tobin's Q, total asset, total firms' liability. Estimates and sample size have been rounded to four significant digits according to Census Bureau disclosure guidelines. The symbols*, **, and *** denote statistical significance at the 10%, 5%, and 1% levels, respectively.

Table H.9: Exclude Adjacent Circuits

Panel A: CERCLA on-site pollution (continuous)

	Dependent variable: log(on-site CERCLA pollution+1)$_{cit}$			
	(1)	(2)	(3)	(4)
Post$_t$ × 11th Circuit$_i$	0.150***	0.144***	0.140***	0.140***
	(0.0180)	(0.0200)	(0.0217)	(0.0212)
Observations	123,000	123,000	123,000	123,000
R-squared	69.7	72.1	72.2	72.2
Facility FE$_i$× Chemical FE$_c$	x	x	x	x
Year FE$_t$	x			
Legal status$_i$× Year FE$_t$		x	x	x
Chemical FE$_c$× Year FE$_t$		x	x	x
NAICS FE$_i$× Year FE$_t$			x	x
Firm-level controls$_{it}$				x

Panel B: CERCLA on-site pollution (discrete)

	Dependent variable: 𝟙(on-site CERCLA pollution)$_{cit}$			
	(1)	(2)	(3)	(4)
Post$_t$ × 11th Circuit$_i$	2.061***	2.254***	2.185***	2.196***
	(0.399)	(0.408)	(0.416)	(0.390)
Observations	123,000	123,000	123,000	123,000
R-squared	68.5	70.2	70.2	70.2
Facility FE$_i$× Chemical FE$_c$	x	x	x	x
Year FE$_t$	x			
Legal status$_i$× Year FE$_t$		x	x	x
Chemical FE$_c$× Year FE$_t$		x	x	x
NAICS FE$_i$× Year FE$_t$			x	x
Firm-level controls$_{it}$				x

This table reports the difference-in-differences between the 11th Circuit and the states that are not from adjacent Circuits. This means that I exclude states from the 4th Circuit (Maryland, North Carolina, South Carolina, Virginia, and West Virginia); 5th Circuit (Louisiana, Mississippi, and Texas); and 6th Circuit (Kentucky, Michigan, Ohio, and Tennessee).. The treatment year is 1996, the year that the Lender Liability Act was passed. The dependent variable of Panel A is log(on-site CERCLA pollution+1)$_{cit}$ the log of the total amount in pound of on-site pollution +1. The dependent variable of Panel B is 𝟙(on-site CERCLA pollution)$_{cit}$, a dummy variable that takes the value 100 if the total amount in pound of on-site pollution is strictly positive and zero otherwise. The dataset is at the establishment-chemical-year level. Post$_t$ is a variable that takes the value one after 1996 (included) and zero otherwise. 11th Circuit$_i$ takes the value one for plants located within the 11th Circuit and zero otherwise. Chemical FE$_c$ is a chemical fixed effect defined at the CAS registry number level. Year FE$_t$ is a year fixed effect, Facility FE$_i$ is a facility fixed effect, and NAICS FE$_i$ is an industry fixed effect based on the two-digit NAICS code as defined in the LBD dataset. Legal status FE$_i$ is a fixed effect that groups together firms with the same legal status as defined by the LBD. Firm-level controls$_{it}$ includes 15 time-varying controls defined at the firm level that are commonly used in empirical corporate finance. These controls are the capital intensity, capx, cash flow, cash holding, cost of capital, firm sales, the log of firm size, R&D intensity, the return on asset (ROA), the return on equity (ROE), the tangibility ratio, the net income, Tobin's Q, total asset, total firms' liability. Estimates and sample size have been rounded to four significant digits according to Census Bureau disclosure guidelines. The symbols*, **, and *** denote statistical significance at the 10%, 5%, and 1% levels, respectively.

Table H.10: Different Time Ranges for the Sample

Panel A: 1993 through 1998

	$\frac{\text{Pollution}_{cit}}{\text{Emp}_{it=1994}}$	Log(Pollution+1$_{cit}$)	$\mathbb{1}$(Pollution)$_{cit}$
	(1)	(2)	(3)
Post$_t$ × 11thCircuit$_i$	0.520***	0.128***	2.090***
	(0.0962)	(0.0146)	(0.287)
Observations	157,000	157,000	157,000
Facility FE$_i$×Chemical FE$_c$	x	x	x
Legal status$_i$×Year FE$_t$		x	x
Chemical FE$_c$×Year FE$_t$		x	x
NAICS FE$_i$×Year FE$_t$			x
Firm-level controls$_{it}$	x	x	x

Panel B: 1994 through 1997

	$\frac{\text{Pollution}_{cit}}{\text{Emp}_{it=1994}}$	Log(Pollution+1$_{cit}$)	$\mathbb{1}$(Pollution)$_{cit}$
	(1)	(2)	(3)
Post$_t$ × 11thCircuit$_i$	0.430***	0.103***	1.884***
	(0.0856)	(0.0138)	(0.251)
Observations	157,000	157,000	157,000
FacilityFE$_i$×ChemicalFE$_c$	x	x	x
Legalstatus$_i$×YearFE$_t$		x	x
ChemicalFE$_c$×YearFE$_t$		x	x
NAICSFE$_i$×YearFE$_t$			x
Firm-levelcontrols$_{it}$	x	x	x

Panel C: 1995 through 1996

	$\frac{\text{Pollution}_{cit}}{\text{Emp}_{it=1994}}$	Log(Pollution+1$_{cit}$)	$\mathbb{1}$(Pollution)$_{cit}$
	(1)	(2)	(3)
Post$_t$ × 11thCircuit$_i$	0.203***	0.0587***	1.083***
	(0.0414)	(0.0169)	(0.267)
Observations	157,000	157,000	157,000
FacilityFE$_i$×ChemicalFE$_c$	x	x	x
Legalstatus$_i$×YearFE$_t$		x	x
ChemicalFE$_c$×YearFE$_t$		x	x
NAICSFE$_i$×YearFE$_t$			x
Firm-levelcontrols$_{it}$	x	x	x

This table reports the difference-in-differences between the 11th Circuit and the states that are not from adjacent Circuits. This means that I exclude states from the 4th Circuit (Maryland, North Carolina, South Carolina, Virginia, and West Virginia); 5th Circuit (Louisiana, Mississippi, and Texas); and 6th Circuit (Kentucky, Michigan, Ohio, and Tennessee).. The treatment year is 1996, the year that the Lender Liability Act was passed. The dependent variable of Panel A is log(on-site CERCLA pollution+1)$_{cit}$ the log of the total amount in pounds of on-site pollution +1. The dependent variable of Panel B is $\mathbb{1}$(on-site CERCLA pollution)$_{cit}$, a dummy variable that takes the value 100 if the total amount in pounds of on-site pollution is strictly positive and zero otherwise. The dataset is at the establishment-chemical-year level. Post$_t$ is a variable that takes the value one after 1996 (included) and zero otherwise. 11th Circuit$_i$ takes the value one for plants located within the 11th Circuit and zero otherwise. Chemical FE$_c$ is a chemical fixed effect defined at the CAS registry number level. Year FE$_t$ is a year fixed effect, Facility FE$_i$ is a facility fixed effect, and NAICS FE$_i$ is an industry fixed effect based on the two-digit NAICS code as defined in the LBD dataset. Legal status FE$_i$ is a fixed effect that groups together firms with the same legal status as defined by the LBD. Firm-level controls$_{it}$ includes 15 time-varying controls defined at the firm level that are commonly used in empirical corporate finance. These controls are the capital intensity, capx, cash flow, cash holding, cost of capital, firm sales, the log of firm size, R&D intensity, the return on asset (ROA), the return on equity (ROE), the tangibility ratio, the net income, Tobin's Q, total asset, total firms' liability. Estimates and sample size have been rounded to four significant digits according to Census Bureau disclosure guidelines. The symbols*, **, and *** denote statistical significance at the 10%, 5%, and 1% levels, respectively.

Table H.11: Reporting Robustness Tests

	$\frac{\text{Pollution}_{cit}}{\text{Emp}_{it=1994}}$	Log(Pollution+1$_{cit}$)	$\mathbb{1}$(Pollution)$_{cit}$
	(1)	(2)	(3)
Post$_t$ × 11thCircuit$_i$	0.466***	0.125***	1.983***
	(0.0603)	(0.0158)	(0.300)
Observations	157,000	157,000	157,000
Facility FE$_i$×Chemical FE$_c$	x	x	x
Legal status$_i$×Year FE$_t$		x	x
Chemical FE$_c$×Year FE$_t$		x	x
NAICSFE$_i$×Year FE$_t$			x
Firm-level controls$_{it}$	x	x	x

This table reports triple difference-in-differences between the 11th Circuit (treated group) and the others (control group) after 1996, the year that the Lender Liability Act was passed. All chemicals that were not reported before 1995 have been removed. The dataset is at the chemical-year level, from 1992 through 1999. The dependent variable of column (1) is $\frac{\text{Pollution}_{cit}}{\text{Emp}_{it=1994}}$, the ratio of on-site CERCLA pollution, in pounds, over employment during the year of 1994. The dependent variable of column (2) is log(on-site CERCLA pollution+1)$_{cit}$, the log of the total amount in pounds of on-site pollution +1. The dependent variable of column (3) is $\mathbb{1}$(on-site CERCLA pollution)$_{cit}$, a dummy variable that takes the value 100 if the total amount in pounds of on-site pollution is strictly positive and zero otherwise. Post$_t$ is a variable that takes the value one after 1996 (included) and zero otherwise. 11th Circuit$_i$ takes the value one for plants located within the 11th Circuit and zero otherwise. Chemical FE$_c$ is a chemical fixed effect defined at the CAS registry number level. Year FE$_t$ is a year fixed effect, Facility FE$_i$ is a facility fixed effect, and NAICS FE$_i$ is an industry fixed effect based on the two-digit NAICS code as defined in the LBD dataset. Legal status FE$_i$ is a fixed effect that groups together firms with the same legal status as defined by the LBD. Firm-level controls$_{it}$ includes 15 time-varying controls defined at the firm level that are commonly used in empirical corporate finance. These controls are the firm sales, capx, capital intensity, cash flow, cash holding, cost of capital, total asset, the log of firm size, the net income, R&D intensity, the return on asset (ROA), the return on equity ROE, tangibility ratio, Tobin's Q and the total firms' liability. The symbols*, **, and *** denote statistical significance at the 10%, 5%, and 1% levels, respectively. Estimates and sample size have been rounded to four significant digits according to Census Bureau disclosure guidelines.

Table H.12: Clustering Robustness Tests

	$\frac{\text{Pollution}_{cit}}{\text{Emp}_{it=1994}}$	Log(Pollution+1$_{cit}$)	$\mathbb{1}$(Pollution)$_{cit}$
	(1)	(2)	(3)
Post$_t$ × 11thCircuit$_i$	0.466***	0.125***	1.983***
	(0.0603)	(0.0158)	(0.300)
Observations	157,000	157,000	157,000
Facility FE$_i$×Chemical FE$_c$	x	x	x
Legal status$_i$×Year FE$_t$		x	x
Chemical FE$_c$×Year FE$_t$		x	x
NAICSFE$_i$×Year FE$_t$			x
Firm-level controls$_{it}$	x	x	x

This table reports different robustness tests to compute the standard errors for the baseline regression. Column (1) reports the coefficients estimated on a sample that takes the average of all variables before and after 1996. Column (2) takes the first difference of the previous before/after comparison, and reports the coefficient of this cross-section. The coefficient estimated in this sample by ordinary least squares is an asymptotically consistent estimator. Column (3) reports the standard errors estimated on the previous sample using the bootstrapped approach. Running the bootstrapped approach on this sample significantly reduces the simulation time. Column (4) clusters at the firm level, while column (5) reports the standard errors clustered at the chemical level. Estimates and sample sizes have been rounded to four significant digits according to Census Bureau disclosure guidelines. The symbols*, **, and *** denote statistical significance at the 10%, 5%, and 1% levels, respectively.

Table H.13: Impact on loans: robustness tests

	Maturity$_{itj}$ (log) (1)	Secured$_{itj}$ (2)	All-In-Drawn Spread$_{itj}$ (log) (3)	All-In-Undrawn Spread$_{itj}$ (log) (4)	Amount$_{itj}$ (log) (5)
Post$_t$ × 11th Circuit$_i$	Absorbed	-0.00337	Absorbed	Absorbed	Absorbed
		(0.0120)			
Post$_t$ × 11th Circuit$_i$ × secured$_{itj}$	0.143***				
	(0.0412)				
Post$_t$ × 11th Circuit$_i$ × High Pollution$_i$			0.140**	-0.276***	-0.363***
			(0.0531)	(0.0612)	(0.0533)
Observations	10,000	10,000	10,000	10,000	10,000
R-squared	45.6	63	45.4	70.6	63.9
Controls$_{itj}$	x	x	x	x	x
Establishment FE$_i$	x	x	x	x	x
Legal status$_i$× YearFE$_t$	x	x	x	x	x
NAICS FE$_i$× Year FE$_t$	x	x	x	x	x
Circuit $_i$× Year FE$_t$	x		x	x	x

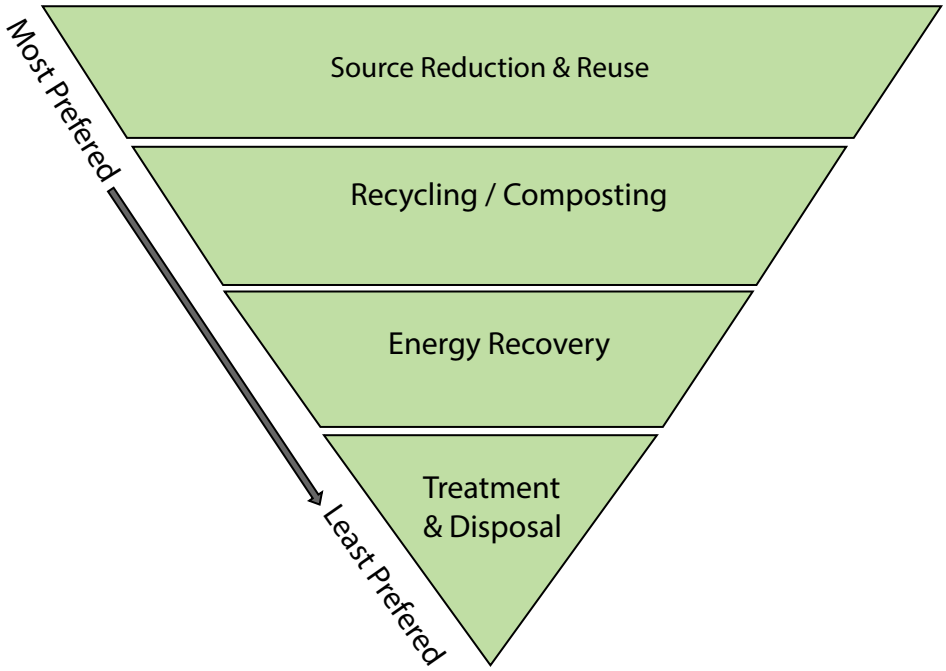
This table reports the difference-in-differences between the 11th Circuit (treated group) and the others (control group) after 1996, the year that the Lender Liability Act was passed. The dependent variable of Column (1) is Maturity$_{itj}$ log, which is the log of the maturity of the loan's facility. The dependent variable of Column (2) is Secured$_{itj}$, which is a dummy variable that takes the value one if the loan facility is secured, and zero otherwise. The dependent variable of Column (3) is AllInDrawn$_{itj}$ (log), which is the log of Loan Spread + Annual Fee. The dependent variable of Column (4) is AllInUndrawn$_{itj}$ (log), which is the log of Commitment Fee + Annual Fee. The dependent variable of Column (5) is Amount$_{itj}$ (log), which is the log of the loan's amount. The dataset is at the loan facility-bank-year level from 1992 through 1999. In cases where a loan facility has multiple lead arrangers, each lender is weighted in the regression according to their share of the loan facility. The definition of lead arrangers follows closely Chakraborty, Goldstein, and MacKinlay (2018). $Post_t$ is a variable that takes the value one after 1996 (included) and zero otherwise. 11th Circuit$_i$ takes the value one for plants located within the 11th Circuit and zero otherwise. Secured$_{itj}$ is a dummy variable that takes the value one if the facility is secured, and 0 otherwise. We include all interaction terms among *Post*$_t$, *11th Circuit*$_i$, and *Secured*$_j$ (or *High Pollution*$_i$), unless already absorbed by the fixed effects. Year FE$_t$ is a year fixed effect, Establishment FE$_i$ is an establishment fixed effect, and NAICS FE$_i$ is an industry fixed effect based on the two-digit NAICS code as defined in the LBD dataset. Legal status FE$_i$ is a fixed effect that groups together firms with the same legal status as defined by the LBD. Firm-level controls$_{it}$ includes time-varying controls defined at the firm level that are commonly used in empirical corporate finance. These controls are the capital intensity, capx, cash flow, cash holding, cost of capital, firm sales, the log of firm size, R&D intensity, the return on asset (ROA), the return on equity (ROE), the tangibility ratio, the net income, Tobin's Q, total asset, total firms' liability, other debt contracts' features, and the loan share of the bank. Table 1 provides additional descriptions of how the variables are defined. Estimates and sample size have been rounded to four significant digits according to Census Bureau disclosure guidelines. The symbols *, **, and *** denote statistical significance at the 10%, 5%, and 1% levels, respectively.

Table H.14: Impact on ROA

		$\frac{\text{Net Income}_{it}}{\text{Total Assets}_{it}}$	
$\text{Post}_t \times$ 11th Circuit_i	p**		p**
Firm $_i \times$ Year FE_t	x		x
Legal $\text{status}_i \times$ Year FE_t	x		x
Chemical $\text{FE}_c \times$ Year FE_t	x		x
NAICS $\text{FE}_i \times$ Year FE_t	x		x
Firm-level controls_{it}			x

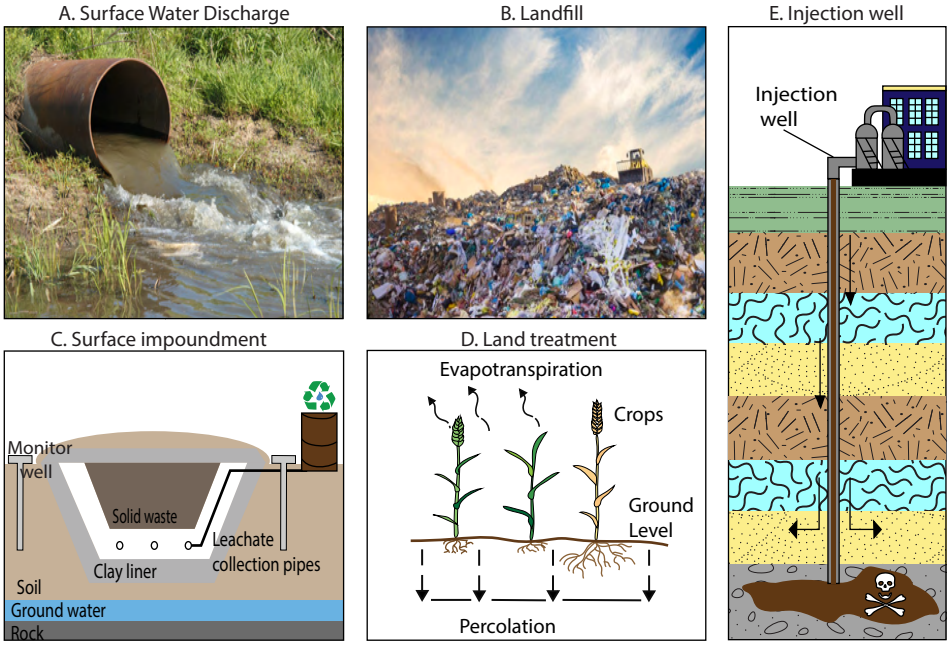
This table reports the difference-in-differences between the 11th Circuit (treated group) and the others (control group) after 1996, the year that the Lender Liability Act was passed. The dependent variable is the return on assets at the firm level, calculated using Compustat. Due to Census Bureau disclosure requirements, we are unable to report the coefficient estimates and instead only disclose the sign and statistical significance of the results. "p" means that the sign is positive. Post_t is a variable that takes the value one after 1996 (included) and zero otherwise. 11th Circuit_i takes the value one for plants located within the 11th Circuit and zero otherwise. Year FE_t is a year fixed effect, Firm FE_i is a firm fixed effect, and NAICS FE_i is an industry fixed effect based on the two-digit NAICS code as defined in the LBD dataset. Legal status FE_i is a fixed effect that groups together firms with the same legal status as defined by the LBD. Firm-level controls_{it} includes 15 time-varying controls defined at the firm level that are commonly used in empirical corporate finance. These controls are the firm sales, capx, capital intensity, cash flow, cash holding, cost of capital, total asset, the log of firm size, the net income, R&D intensity, the return on equity ROE, tangibility ratio, Tobin's Q and the total firms' liability. The symbols*, **, and *** denote statistical significance at the 10%, 5%, and 1% levels, respectively.

Figure H.1: Waste Management Hierarchy (EPA)



This figure represents the Waste Management hierarchy, as defined by the EPA.

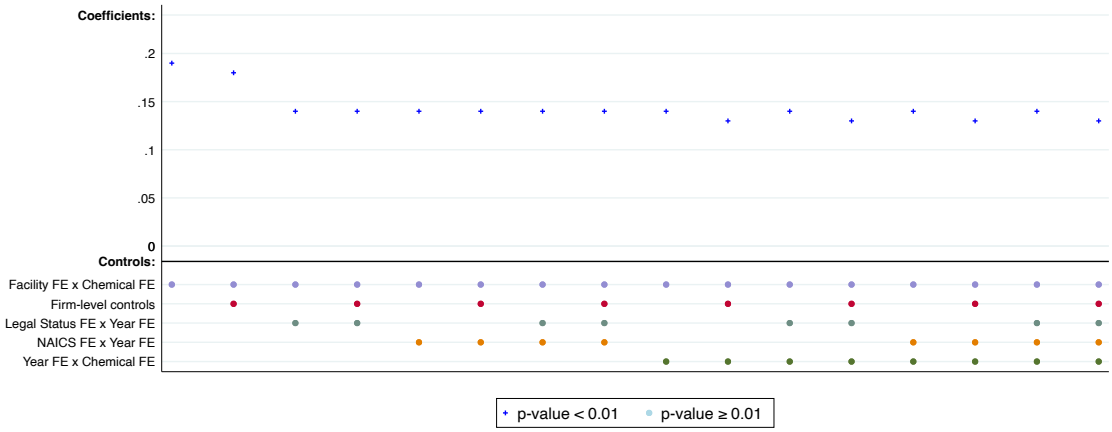
Figure H.2: Types of On-site Pollution



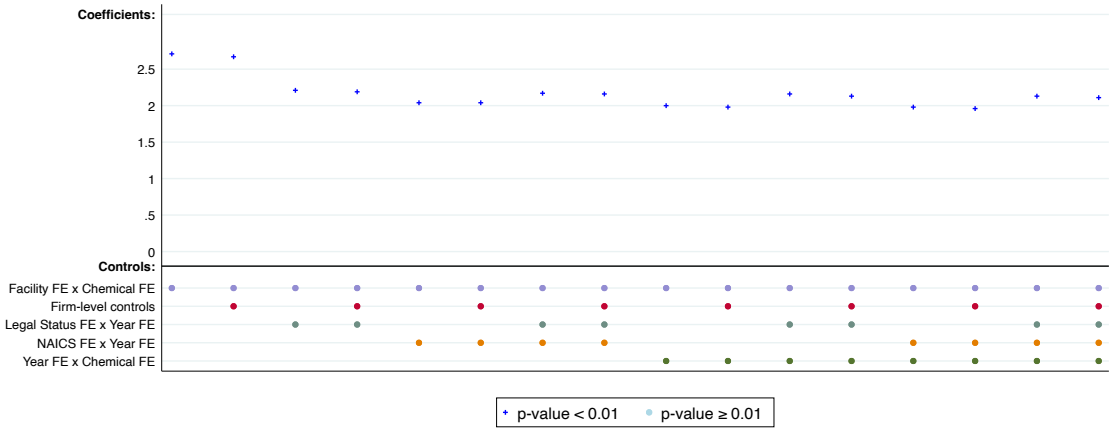
This figure depicts the different ways in which releases occur of on-site toxic pollutants included in the measure of on-site CERCLA pollution, the on-site releases minus air pollution for chemical c, time t, and facility i.

Figure H.3: Specification Curves: Pollution and abatement Outcomes (1/2)

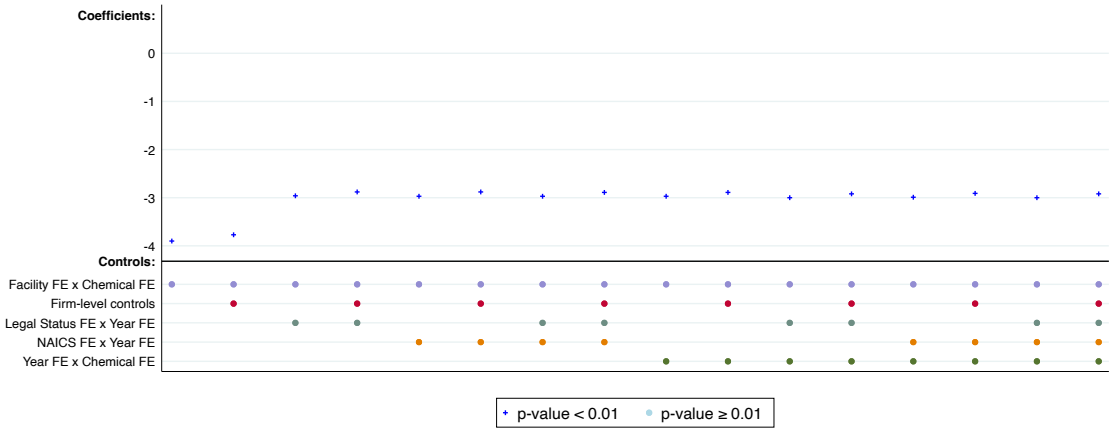
Panel A: on-site releases



Panel B: on-site releases (extensive margin)



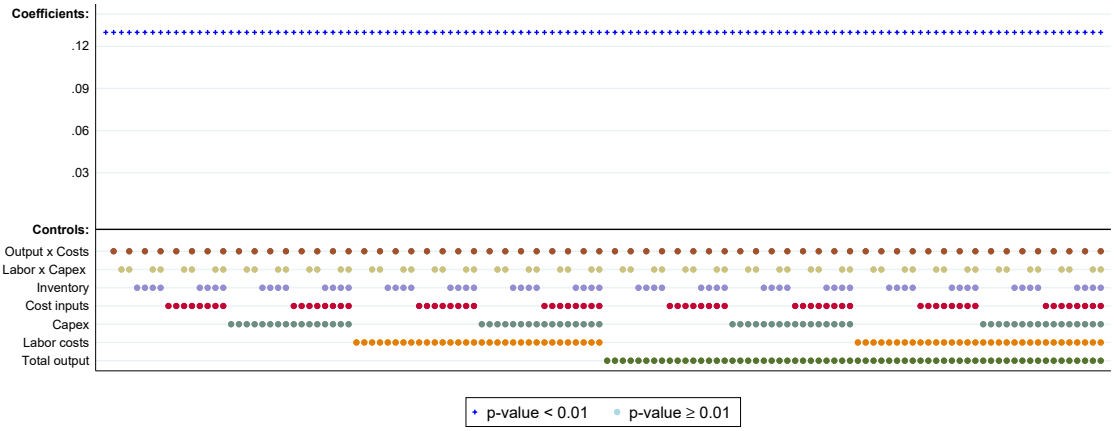
Panel C: Abatement activities



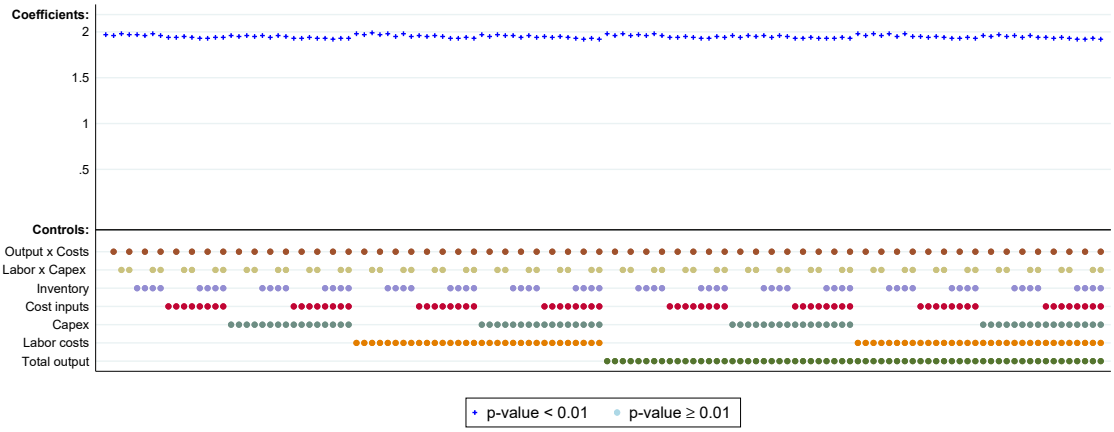
Note: The goal of these figures is to show how the coefficient of $\text{Post}_t \times 11\text{th Circuit}_i$ varies for changes in the controls and fixed effects for the baseline specification of Equation 1. Specifically, the figures of Panels A, B, and C report the coefficient of $\text{Post}_t \times 11\text{th Circuit}_i$ when the dependent variable is $\log(\text{pollution}+1)_{cit}$, $\mathbb{1}(\text{Pollution})_{cit}$, and $\mathbb{1}(\text{Process-related abatement})_{cit}$, respectively. Chemical FE_c is a chemical fixed effect defined at the CAS registry number level. Year FE_t is a year fixed effect, Facility FE_i is a facility fixed effect, and NAICS FE_i is an industry fixed effect based on the two-digit NAICS code as defined in the LBD dataset. Legal status FE_i is a fixed effect that groups together firms with the same legal status as defined by the LBD. Firm-level controls_{it} includes 15 time-varying controls defined at the firm level that are commonly used in empirical corporate finance. These controls are the capital intensity, capx, cash flow, cash holding, cost of capital, firm sales, the log of firm size, R&D intensity, the return on asset (ROA), the return on equity (ROE), the tangibility ratio, the net income, Tobin's Q, total asset, total firms' liability. Table 1 provides additional descriptions of how the variables are defined.

Figure H.4: Specification Curves: Pollution and abatement Outcomes (2/2)

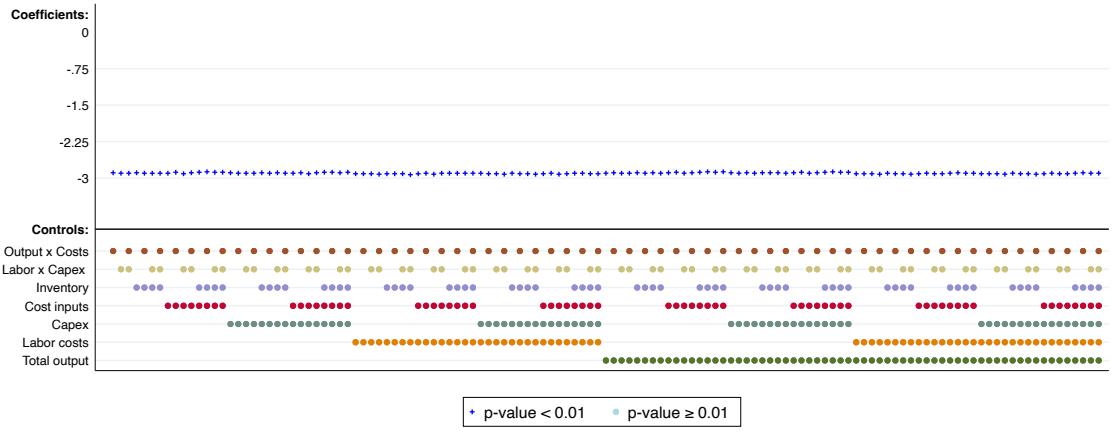
Panel A: on-site releases



Panel B: on-site releases (extensive margin)

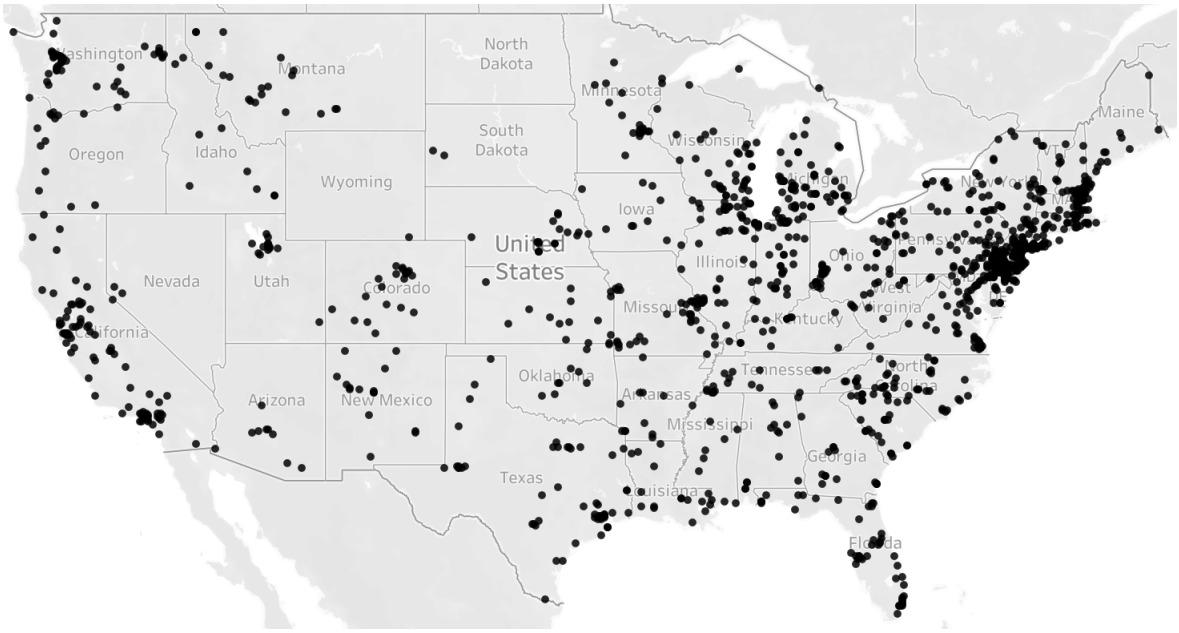


Panel C: Abatement activities



Note: The goal of these figures is to show how the coefficient of $\text{Post}_t \times 11\text{th Circuit}_i$ varies for changes when additional controls are included for the baseline specification of Equation 1. These controls come from the smaller sample of establishments that are also surveyed in the ASM/CMF. Output represents the real output of the facility. Cost inputs include the following variables: cost of fuels, cost of materials and parts, cost of resales, cost of contract work, and cost of purchased electricity. Inventory includes the total value of shipments and work-in-process inventory end. Labor costs regroup the total employment, the total worker hours, and earnings. Capex represents the capital expenditure on new and used buildings and other structures. Labor is the total employment and is interacted with Capex. The figures of Panels A, B, and C report the coefficient of $\text{Post}_t \times 11\text{th Circuit}_i$ when the dependent variable is $\log(\text{pollution+1})_{cit}$, $\mathbb{1}(\text{Pollution})_{cit}$, and $\mathbb{1}(\text{Process-related abatement})_{cit}$, respectively.

Figure H.5: Active Contaminated Sites in the USA



Note: This figure depicts the distribution of brownfields in the Superfund National Priorities List (NPL), when the sites have status information, in April 2021.